

THREE COLOURS FOR THE PLANE OVER A NUMBER FIELD

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ABSTRACT. For a field F of characteristic 0 let $\chi(F^2)$ be the chromatic number of the graph on F^2 that joins x and y when $(x_1 - y_1)^2 + (x_2 - y_2)^2 = 1$. We prove that for every number field F , $\chi(F^2) \leq 3$ if and only if some prime of F above 2 ramifies in $F(i)$ or some prime of F above 3 has residue degree 1. The “if” part is known (colourings through a residue field); the “only if” part extends to all number fields our earlier result that $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for $d \equiv 11 \pmod{12}$. It answers, for three colours, the question whether the chromatic number of the plane over a number field is decided by a single completion, and it classifies the finite extensions K of $\mathbb{Q}$: $\chi(K^2)$ is 2, 3 or at least 4 according to explicit local conditions. We also give an elementary family (every real field containing $\sqrt{a}$ and $\sqrt{b}$ with $a \equiv 2 \pmod{3}$ and $b \equiv 7 \pmod{8}$ needs four colours), new examples such as $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{7})^2) = 4$, and, as far as we know, the first non-integral value of the circular chromatic number of the plane over a number field: $\chi_c(\mathbb{Q}(\sqrt{11})^2) = \chi_c(\mathbb{Q}(\sqrt{35})^2) = 7/2$ (and $\chi_c(\mathbb{Q}^2) = 7/2$), although these planes have chromatic number 4. Below 4, the best upper bound on χ_c that a locally constant colouring of one completion gives is 2, 3 or $7/2$. We show that these are the only values below 4: for every number field F , $\chi_c(F^2)$ is 2, 3, $7/2$ or at least 4, and it is $7/2$ exactly when $\chi(F^2) \geq 4$ and some prime of F above 7 has residue degree 1; below 4 it is the least circular chromatic number of the planes over the completions of F , and it is the circular chromatic number of some finite unit-distance graph in F^2 ; so is the value 4, when it occurs (for example over $\mathbb{Q}(\sqrt{59})$), as at four colours the winding argument fails only on tight squares; over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ we give explicit such graphs, with 1874 and 1657 vertices, certified by checked SAT refutations. The proofs of this classification and of the attainment of 4 have one computer-assisted step, an exact finite computation along the rational rotations whose denominators divide 325; without it we show that $\chi_c(F^2) \geq 56/17$ whenever $\chi(F^2) \geq 4$.

1. INTRODUCTION

Let F be a field of characteristic 0. The *plane over F* is the graph on F^2 in which x and y are adjacent when $(x_1 - y_1)^2 + (x_2 - y_2)^2 = 1$, and $\chi(F^2)$ is its chromatic number. For $F = \mathbb{R}$ this is the Hadwiger–Nelson problem, and $5 \leq \chi(\mathbb{R}^2) \leq 7$: the lower bound is due to de Grey [6] and to Exoo and Ismailescu [8], the upper bound to Isbell (see [26]). Over number fields the question goes back to Woodall ($\chi(\mathbb{Q}^2) = 2$ [29]) and was studied by Johnson [19], Fischer [9], Benda and Perles [1], Moorhouse [23] and Madore [21]. All upper bounds below 7 known to us come from a colouring that is locally constant at one completion F_v : reduction modulo a power of a prime. Writing $L = F(i)$, the unit vectors of F^2 are the elements of norm one of L , and at a place where $x^2 + y^2$ is anisotropic they reduce into a finite group of roots of unity of the residue field. Two such reductions give colourings with at most three colours:

- (a) at a prime of F above 2 that ramifies in $F(i)$, every unit vector is $\equiv 1$ modulo the maximal ideal, which gives a 2-colouring ([22, Theorem A], [15, Proposition 3]; earlier, for special classes of fields, Fischer [9, Thm. 8], Moorhouse [23] and Johnson [19], whose paper we have not seen);

Date: Draft, 6 October 2026.

2020 Mathematics Subject Classification. Primary 05C15, 52C10; Secondary 11R04, 11S15, 05C25.

Key words and phrases. Hadwiger–Nelson problem, chromatic number of the plane, unit-distance graph, number field, local–global principle, circular chromatic number.

These results were obtained with the assistance of an AI system (Claude, by Anthropic), which found the proofs, wrote them and the programs that check them, and helped to draft this note. Separate AI agents, called independent checkers below, refereed the proofs with their own programs. No mathematician has checked the note yet. The author is not a professional mathematician and submits it for independent verification.

- (b) at a prime of F above 3 of residue degree 1, the unit vectors reduce into $\mu_4 \subset \mathbb{F}_9$, and $a + bi \mapsto a + b$ gives a 3-colouring (Madore [21, Cor. 3.4]; for real quadratic fields Moorhouse [23, Lemma 8.2 and Cor. 8.3] and Fischer [9, Thm. 9]).

In [15] we proved that (a) is also necessary for $\chi(F^2) = 2$; we call that result Theorem A, which is why there is no Theorem A here. Here we prove the same at three colours.

Theorem B. *Let F be a number field. Then $\chi(F^2) \leq 3$ if and only if (a) or (b) holds; and $\chi(F^2) \leq 2$ if and only if (a) holds.*

For real quadratic fields $\mathbb{Q}(\sqrt{d})$, d squarefree, (a) holds iff $d \not\equiv 3 \pmod{4}$ and (b) iff $d \not\equiv 2 \pmod{3}$, so Theorem B contains the main result of [16] (Theorem 1 and Corollary 2 there): $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ iff $d \equiv 11 \pmod{12}$. Whether (a) or (b) holds depends only on $F \otimes \mathbb{Q}_2$ and $F \otimes \mathbb{Q}_3$, and is computed from a defining polynomial (the relative discriminant of $F(i)/F$, and the residue degrees above 3). By the theorem of de Bruijn and Erdős [5], $\chi(F^2) \geq 4$ if and only if some finite unit-distance graph in F^2 needs four colours, and Theorem B says that this happens unless one of the completions of F has a locally constant 3-colouring: a local–global principle for three colours. The same proof decides the local fields (Corollary 3): for a finite extension K of $\mathbb{Q}_p$, $\chi(K^2) = 2$ if $p = 2$ and $K(i)/K$ is ramified, $\chi(K^2) = 3$ if $p = 3$ and K has residue degree 1, and $\chi(K^2) \geq 4$ otherwise; for instance the plane over the unramified cubic extension of $\mathbb{Q}_3$ needs at least four colours, although $x^2 + y^2$ is anisotropic there, as over $\mathbb{Q}_3$.

Besides the fields that contain i , whose planes need infinitely many colours, two families of fields were known to need four colours: those containing $\sqrt{d}$ with $d \equiv 11 \pmod{12}$ [16], and those containing $\sqrt{3}$ and $\sqrt{q}$ with $q \equiv 2 \pmod{3}$, by chains of unit rhombi ([13], `notes/local_colourings.md`, Section 11, Theorem 5). They contain $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ (both families), the field of the Moser spindle, whose plane has chromatic number 4 [10, 22], and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ (the second), whose plane has chromatic number 4 [14]. Theorem B adds, for example, $\mathbb{Q}(\sqrt{2}, \sqrt{7})$, $\mathbb{Q}(\sqrt{2}, \sqrt{31})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{55})$, which contain no unit triangle and whose quadratic subfields all have $\chi \leq 3$, and $\mathbb{Q}(2 \cos \frac{2\pi}{7}, \sqrt{7})$, of degree 6, where 3 has residue degree 3. For many fields an explicit proof is short (Proposition 2), and for $\mathbb{Q}(\sqrt{2}, \sqrt{7})$ we give a certificate checked by computer that does not depend on the general argument.

Theorem C. $\chi_c(\mathbb{Q}(\sqrt{11})^2) = \chi_c(\mathbb{Q}(\sqrt{35})^2) = \chi_c(\mathbb{Q}_7^2) = 7/2$, while the three chromatic numbers are 4. Moreover no plane over a number field has circular chromatic number strictly between 2 and 3.

Theorem D. *Let F be a number field with $\chi(F^2) \geq 4$. Then $\chi_c(F^2) \geq 56/17 \approx 3.294$.*

Theorem E. *Let F be a number field with $\chi(F^2) \geq 4$. Then $\chi_c(F^2) \geq 7/2$. So $\chi_c(F^2) \in \{2, 3\} \cup [7/2, \infty]$ for every number field F , and the value $7/2$ is attained.*

Theorem F. *Let F be a number field, and consider the condition*

- (7) *some prime of F above 7 has residue degree 1.*

Then $\chi_c(F^2) = 2$ if (a) holds; $\chi_c(F^2) = 3$ if (b) holds and (a) fails; $\chi_c(F^2) = 7/2$ if (7) holds and (a), (b) fail; and $\chi_c(F^2) \geq 4$ if (a), (b) and (7) all fail.

Theorem E rests on one computer-assisted step (Proposition 8), and Theorem F on another (Proposition 9), which also gives Theorem E again: exact finite computations, whose certificates are checked by programs written separately from those that produced them, and which two independent checkers confirmed by methods of their own. The lower bounds of Theorem C and Proposition 4 are proved in Section 8 with exact certificates checked by computer (the former also follow from Theorem E), and the case $f = 1$, $p \geq 11$ of Proposition 6, used only for Corollary 5, rests on a finite computation of κ_1 for the primes $11 \leq p < 1001$ (run up to 3000); its case $f \geq 3$, which Lemma 19 also uses, is proved by hand. The rest of the proofs are by hand. Theorem D, which Theorem E supersedes, is proved entirely by hand. So below 4 the circular chromatic number of the plane over a number field takes only the values 2, 3 and $7/2$, each given by a locally constant colouring of one completion; more precisely, below 4 it is the least circular chromatic number of the planes over its completions (Corollary 6): a circular local–global

principle below 4 that extends Theorem B. Below 4 the value is also the circular chromatic number of a finite unit-distance graph in F^2 (Corollary 7), and so is the value 4 when $\chi_c(F^2) = 4$ (Theorem G). The proofs give no explicit bound on the size of such a graph (for the value 4 the remark after Lemma 24 replaces the compactness step of Corollary 8, though not the one behind Proposition 10, by a finite construction), but computer searches find explicit ones, certified: for the value 4, over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, ones with 1874 and 1657 vertices (found with base-point periods, Lemma 24); for $7/2$, over $\mathbb{Q}(\sqrt{11})$, one with 155 vertices, and others over $\mathbb{Q}(\sqrt{455})$, $\mathbb{Q}(\sqrt{191})$ and $\mathbb{Q}(\sqrt{911})$; for the value 3, over $\mathbb{Q}(\sqrt{7})$, whose plane has no unit triangle, one with nine vertices, the fewest possible by an exhaustive enumeration (Section 10). For a real quadratic field $\mathbb{Q}(\sqrt{d})$, d squarefree, (a) or (b) holds unless $d \equiv 11 \pmod{12}$. If $d \equiv 11 \pmod{12}$, then $\chi_c = 7/2$ when $(d/7) \neq -1$ (for example $d = 23, 71$); $\chi_c = 4$ when $(d/7) = -1$ and $d \equiv 11 \pmod{24}$ (for example $d = 59, 83$), as then the place above 2 gives $\chi \leq 4$; and only when $(d/7) = -1$ and $d \equiv 23 \pmod{24}$ ($d = 47, 143, 167, 215, \dots$) do we not know whether $\chi_c = 4$, which holds if and only if $\chi = 4$ (for $d = 47$, $4 \leq \chi_c \leq 19/4$; Section 10).

The tools are those of [16]: a characterisation of 3-colourings of Cayley graphs of abelian groups by characters (Theorem W^+ , Section 2), and the exact shape of the characters along the rational rotations whose denominators are powers of 5. These rotations act as a probe: along them, a character that keeps every unit vector in $[1/3, 2/3]$ is, up to a bounded error, either a 2-adic “half” character or a 3-adic character of level one, and a compactness argument turns this into the dichotomy between the places above 2 and those above 3. For Theorem D we show that the probe keeps this shape, with a small error also for type q, when $[1/3, 2/3]$ is replaced by $[\theta, 1 - \theta]$ with $\theta > 17/56$ (Proposition 7). This probe cannot pass $10/3$: for $\theta \leq 3/10$ it has 5-adic characters. For Theorems E and F we add one rational rotation with another prime, $\sigma = (5 + 12i)/13$. Along the rotations $i^a \rho^j \sigma^l$ a finite computation modulo 65 shows that the probe keeps the shape c or q for $\theta > 2/7$, and one modulo 325 that for $\theta > 1/4$ it has only these two shapes and a third, 7-adic, one; a compactness argument at the places above 2, 3 and 7 then proves the theorems.

Sections 2–5 prove Theorem B; Section 6 draws the consequences (circular colourings below 3, fields split above 2 and 3, local fields); Section 7 gives the explicit family; Section 8 proves Theorem C and shows that, below 4, the best upper bound on $\chi_c(F^2)$ given by the restriction to F^2 of a locally constant colouring of a completion F_v^2 is 2, 3 or $7/2$ (Corollary 5); Section 9 proves Theorem D and Section 10 Theorems E and F, with the local fields and finite witnesses; Section 11 lists the computer checks, and Section 12 asks what happens above 4, at four colours, and how small finite witnesses can be. The programs and certificates are in [13].

2. PRELIMINARIES

Characters and colourings. For an abelian group Γ and $S = -S \subseteq \Gamma$, $\text{Cay}(\Gamma, S)$ is the graph on Γ in which $x \sim y$ when $y - x \in S$. A *character* of Γ is a homomorphism $\xi : \Gamma \rightarrow \mathbb{R}/\mathbb{Z}$, and $\|t\|$ is the distance from t to the nearest integer. For integers $p \geq 2q \geq 1$, the circular clique $K_{p/q}$ has vertex set $\mathbb{Z}/p$, with $a \sim b$ when $q \leq (b - a \bmod p) \leq p - q$, and the circular chromatic number $\chi_c(G)$ is the infimum of the p/q such that G has a homomorphism to $K_{p/q}$.

Theorem W^+ ([11, Theorem 1]). *Let $p \geq 2q \geq 2$ with $p < 4q$. Then $\text{Cay}(\Gamma, S)$ has a homomorphism to $K_{p/q}$ if and only if some character ξ of Γ satisfies $\xi(s) \in [q/p, 1 - q/p] + \mathbb{Z}$ for every $s \in S$. In particular $\text{Cay}(\Gamma, S)$ is 3-colourable if and only if some character maps S into $[1/3, 2/3] + \mathbb{Z}$.*

The “if” direction is classical (colour x by $\lfloor p\xi(x) \rfloor$); the “only if” direction is the content of [11], where it is also proved in Lean for every abelian group Γ and finite S . We apply it to the groups $\Gamma = \mathbb{Z}U$ spanned by finite sets U of unit vectors, and in Section 8 to finite groups $O_w/\mathfrak{m}^k$.

The plane as a Cayley graph. Let F be a number field with $i \notin F$, and $L = F(i)$. The map $(x, y) \mapsto x + iy$ identifies F^2 with L , and $x^2 + y^2 = z\bar{z}$ for $z = x + iy$, where $z \mapsto \bar{z}$ is the non-trivial automorphism of L/F . So the unit vectors form the group $T = \{z \in L : z\bar{z} = 1\}$, and the plane over F is $\text{Cay}(L, T)$. As a $\mathbb{Q}(i)$ -vector space L has dimension $n = [F : \mathbb{Q}]$, and $\text{Tr} = \text{Tr}_{L/\mathbb{Q}(i)}$ gives a non-degenerate pairing $L \times L \rightarrow \mathbb{Q}(i)$.

The probe. Let $N = 5^k$, $k \geq 1$, let G_N be the set of $\gamma \in \mathbb{Q}(i)$ with $\gamma\bar{\gamma} = 1$ and $N\gamma \in \mathbb{Z}[i]$ (the rational rotations whose denominator divides N), $I = [1/3, 2/3]$, and

$$S_N = \{c \in \mathbb{C} : \operatorname{Re}(\bar{c}\gamma) \in I + \mathbb{Z} \text{ for every } \gamma \in G_N\}.$$

Let $E_c = \frac{1+i}{2} + \mathbb{Z}[i]$, $E_q = \{\frac{\alpha+\beta i}{3} : \alpha, \beta \in \{1, 2\}\} + \mathbb{Z}[i]$, $E = E_c \cup E_q$, and $r = \sqrt{10}/18$. Elements of E_c have *type c*, those of E_q *type q*.

Proposition 1 ([16, Proposition 9]). *Every $c \in S_N$ can be written $c = N\varepsilon + x$, either with $\varepsilon \in E_c$ and $|x| \leq r$, or with $\varepsilon \in E_q$ and $x = 0$.*

Lemma 1 (exact relations; cf. [16, Lemma 10]). *Let $c_j = N\varepsilon_j + x_j \in S_N$ be as in Proposition 1 ($1 \leq j \leq \ell$), and let $\mu_1, \dots, \mu_\ell \in \mathbb{Z}[i]$ satisfy $\sum_j \mu_j c_j = 0$ and $N > 6r \sum_j |\mu_j|$. Then $\sum_j \mu_j \varepsilon_j = 0$.*

Proof. $\sum_j \mu_j N\varepsilon_j = -\sum_j \mu_j x_j$. The left side lies in $\frac{N}{6}\mathbb{Z}[i]$, because $6E \subseteq \mathbb{Z}[i]$, and the nonzero elements of $\frac{N}{6}\mathbb{Z}[i]$ have modulus at least $\frac{N}{6}$; the right side has modulus at most $r \sum_j |\mu_j| < \frac{N}{6}$. So both sides vanish. $\square$

Only two properties of E are used below. Let $\mathbb{F}_9 = \mathbb{Z}[i]/3$ and $A' = \{\pm 1 \pm i\} \subset \mathbb{F}_9$, the set of non-squares of $\mathbb{F}_9$ (equivalently, of the $y \in \mathbb{F}_9$ with $y^4 = -1$). Then

$$(1) \quad \varepsilon \in E_c \Rightarrow v_{1+i}(\varepsilon) = -1 \text{ and } v_3(\varepsilon) \geq 0; \quad \varepsilon \in E_q \Rightarrow v_{1+i}(\varepsilon) \geq 0 \text{ and } 3\varepsilon \bmod 3 \in A'.$$

In particular every $\varepsilon \in E$ has $v_{1+i}(\varepsilon) \geq -1$ and $v_3(\varepsilon) \geq -1$, with equality in the second exactly for type q.

3. FROM COLOURINGS TO A LINEAR PROBLEM OVER $\mathbb{Q}(i)$

Lemma 2. *Let F be a number field with $i \notin F$, and suppose that F^2 is 3-colourable. Then for every finite $V \subset T$ there is $\beta \in L$ with $\operatorname{Tr}(\beta v) \in E$ for every $v \in V$.*

Proof. We may enlarge V , so we assume that it contains a $\mathbb{Q}(i)$ -basis $b_1, \dots, b_n$ of L . Such a basis exists in T : the $\mathbb{Q}(i)$ -span W of T is a subring of L , as T is a group; being finite-dimensional over $\mathbb{Q}(i)$ and contained in a field, it is a field. For $t \in F$ the element $z = (t+i)/(t-i)$ lies in T , so W contains $(z-1)^{-1}$ and $t = i(z+1)/(z-1)$; hence $W = L$.

For $v \in V$ write $D_v v = \sum_j \mu_{v,j} b_j$ with $D_v \in \mathbb{Z} \setminus \{0\}$ and $\mu_{v,j} \in \mathbb{Z}[i]$, and choose $N = 5^k$ with $N > 6r \max_v (|D_v| + \sum_j |\mu_{v,j}|)$. The set $U = G_N V$ is finite, symmetric ($-1 \in G_N$) and consists of unit vectors, so $\operatorname{Cay}(\mathbb{Z}U, U)$ is a subgraph of $\operatorname{Cay}(L, T)$ and is 3-colourable. By Theorem W⁺ there is a character ξ of $\mathbb{Z}U$ with $\xi(U) \subseteq I + \mathbb{Z}$. Extend ξ to L (the group $\mathbb{R}/\mathbb{Z}$ is divisible) and restrict it to a lattice $\Lambda = \frac{1}{M} \bigoplus_j \mathbb{Z}[i] f_j \supseteq U$, where $f_1, \dots, f_n$ is a $\mathbb{Q}$ -basis of F , so that $L = \bigoplus_j \mathbb{Q}(i) f_j$. A character of $\Lambda \cong \mathbb{Z}^{2n}$ lifts to a real-linear form, and every real-linear form on $\mathbb{C}^n$ is the real part of a $\mathbb{C}$ -linear one; so there is a $\mathbb{Q}(i)$ -linear map $\varphi : L \rightarrow \mathbb{C}$ with $\xi(w) = \operatorname{Re} \varphi(w) \bmod 1$ for $w \in \Lambda$. For $v \in V$ and $\gamma \in G_N$ we get $\xi(\gamma v) = \operatorname{Re}(\gamma \varphi(v))$, so $\varphi(v) \in S_N$, and $\varphi(v) \in S_N$ because G_N is closed under conjugation. Write $\varphi(v) = N\varepsilon_v + x_v$ as in Proposition 1. Applying φ to $D_v v - \sum_j \mu_{v,j} b_j = 0$ and Lemma 1 give $D_v \varepsilon_v = \sum_j \mu_{v,j} \varepsilon_{b_j}$. Let $\beta \in L$ be the element with $\operatorname{Tr}(\beta b_j) = \varepsilon_{b_j}$ for all j . Then $D_v \operatorname{Tr}(\beta v) = \sum_j \mu_{v,j} \varepsilon_{b_j} = D_v \varepsilon_v$, so $\operatorname{Tr}(\beta v) = \varepsilon_v \in E$ for every $v \in V$. $\square$

Lemma 2 is also a practical test: if for some finite $V \subset T$ no such β exists, then F^2 is not 3-colourable. Whether β exists is a finite computation (Section 11).

4. LOCALISATION AT 2 AND 3

For a prime p let $L_p = L \otimes_{\mathbb{Q}} \mathbb{Q}_p = \prod_{w|p} L_w$, and for a place u of F let $L_u = L \otimes_F F_u$ and $T(F_u) = \{z \in L_u : z\bar{z} = 1\}$. Put $T_p = \prod_{u|p} T(F_u)$. Since 2 ramifies and 3 is inert in $\mathbb{Q}(i)$, $\mathbb{Q}(i) \otimes \mathbb{Q}_2 = \mathbb{Q}_2(i)$ and $\mathbb{Q}(i) \otimes \mathbb{Q}_3 = \mathbb{Q}_9$, the unramified quadratic extension of $\mathbb{Q}_3$; the trace extends to $\operatorname{Tr} : L_2 \rightarrow \mathbb{Q}_2(i)$ and $\operatorname{Tr} : L_3 \rightarrow \mathbb{Q}_9$. Let $O_2 = \mathbb{Z}_2[i]$, $O_3 = \mathbb{Z}_9$ (the integers of $\mathbb{Q}_9$), and $\mathcal{A} = \{\frac{\pm 1 \pm i}{3}\} + O_3$. By (1), $E_c \subseteq (1+i)^{-1} O_2^\times \cap O_3$ and $E_q \subseteq O_2 \cap \mathcal{A}$.

Lemma 3 (weak approximation for T). *For every finite set Σ of places of F , T is dense in $\prod_{u \in \Sigma} T(F_u)$.*

Proof. At a place u with $i \notin F_u$, $t \mapsto (t+i)/(t-i)$ is a homeomorphism from F_u onto $T(F_u) \setminus \{1\}$, with inverse $z \mapsto i(z+1)/(z-1)$. At a place with $i \in F_u$, $L_u \cong F_u \times F_u$, $T(F_u) = \{(s, s^{-1})\}$, and $t \mapsto (s, s^{-1})$ with $s = (t+i)/(t-i)$, $i^2 = -1$, is a homeomorphism from $F_u \setminus \{\pm i\}$ onto $T(F_u) \setminus \{1\}$. These maps send F into T , F is dense in $\prod_{u \in \Sigma} F_u$, and $T(F_u)$ has no isolated points. $\square$

Lemma 4. *If F^2 is 3-colourable and $i \notin F$, then one of the following holds:*

- (c) *there is $\beta_2 \in L_2$ with $v_{1+i}(\text{Tr}(\beta_2 z)) = -1$ for every $z \in T_2$;*
- (q) *there is $\beta_3 \in L_3$ with $\text{Tr}(\beta_3 z) \in \mathcal{A}$ for every $z \in T_3$.*

Proof. Fix a $\mathbb{Q}(i)$ -basis $b_1, \dots, b_n \subset T$ of L (proof of Lemma 2), and let

$$K_2 = \{\beta \in L_2 : \text{Tr}(\beta b_j) \in (1+i)^{-1}O_2 \ \forall j\}, \quad K_3 = \{\beta \in L_3 : \text{Tr}(\beta b_j) \in \frac{1}{3}O_3 \ \forall j\}.$$

They are compact, since $\beta \mapsto (\text{Tr}(\beta b_j))_j$ is a linear isomorphism onto $\mathbb{Q}_2(i)^n$, resp. $\mathbb{Q}_3^n$. On the compact space $\Omega = \{c, q\}^T \times K_2 \times K_3$ consider, for $v \in T$, the condition $P_v(\tau, \beta_2, \beta_3)$:

$$\tau(v) = c, \ v_{1+i}(\text{Tr}(\beta_2 v)) = -1, \ \text{Tr}(\beta_3 v) \in O_3; \quad \text{or} \quad \tau(v) = q, \ \text{Tr}(\beta_2 v) \in O_2, \ \text{Tr}(\beta_3 v) \in \mathcal{A}.$$

Each P_v defines a clopen subset of Ω . For a finite $V \supseteq \{b_1, \dots, b_n\}$, Lemma 2 gives $\beta \in L$; its images in L_2 and L_3 lie in K_2 and K_3 , and with $\tau(v)$ the type of $\text{Tr}(\beta v)$ the triple satisfies P_v for every $v \in V$, by (1). So the closed sets $\{P_v \text{ for all } v \in V\}$ have the finite intersection property, and some $(\tau, \beta_2, \beta_3) \in \Omega$ satisfies P_v for every $v \in T$.

For $z \in T_2$ let $a(z)$ be the statement $v_{1+i}(\text{Tr}(\beta_2 z)) = -1$ and $a'(z)$ the statement $\text{Tr}(\beta_2 z) \in O_2$; for $y \in T_3$ let $b(y)$ be $\text{Tr}(\beta_3 y) \in O_3$ and $b'(y)$ be $\text{Tr}(\beta_3 y) \in \mathcal{A}$. They are locally constant, a and a' exclude each other, and so do b and b' . The set

$$Z = \{(z, y) \in T_2 \times T_3 : (a(z) \wedge b(y)) \vee (a'(z) \wedge b'(y))\}$$

is clopen and contains the diagonal image of T , which is dense by Lemma 3; so $Z = T_2 \times T_3$. If $a(z_0)$ holds for some z_0 , then $b(y)$ holds for every y , so b' holds nowhere, and $a(z)$ holds for every z : this is (c). Otherwise a' holds everywhere, and then b' holds everywhere: this is (q). $\square$

Case (c). For $z \in L$ let $g(z) = (1+i) \text{Tr}(\beta_2 z) \in \mathbb{Q}_2(i)$. Fix a representative y_A of each class A of $\mathbb{Q}_2(i)$ modulo O_2 , and colour z by $(g(z) - y_A) \bmod (1+i) \in O_2/(1+i) = \mathbb{F}_2$, where A is the class of $g(z)$. A unit step $v \in T$ adds $g(v) \in O_2^\times$ to $g(z)$: it keeps the class modulo O_2 and changes the residue modulo $1+i$. This is a proper 2-colouring of $L \cong F^2$, so by [15, Theorem 1] some prime of F above 2 ramifies in $F(i)$.

5. CASE (Q): THE PLACES ABOVE 3

Let u be a place of F above 3, $f = f_u$ its residue degree, $q = 3^f$, and $g_u(z) = \text{Tr}_{L_u/\mathbb{Q}_9}(\beta_u z)$, where $\beta_u = (\beta_u)_u$; then $\text{Tr}(\beta_3 y) = \sum_u g_u(y_u)$ for $y = (y_u) \in T_3$.

Even f . Then $\mathbb{F}_9$ lies in the residue field of F_u , so $i \in F_u$, $L_u \cong F_u \times F_u$ and $T(F_u) = \{(s, s^{-1}) : s \in F_u^\times\}$. Write $g_u(s, t) = \ell_1(s) + \ell_2(t)$ with $\mathbb{Q}_3$ -linear $\ell_1, \ell_2 : F_u \rightarrow \mathbb{Q}_9$. In case (q), $g_u(y_u)$ stays in a bounded set when y_u varies and the other components are fixed. If $\ell_1(s_0) \neq 0$, then $g_u(3^{-m}s_0, 3^m s_0^{-1}) = 3^{-m}\ell_1(s_0) + 3^m\ell_2(s_0^{-1})$ is unbounded as $m \rightarrow \infty$; so $\ell_1 = 0$, likewise $\ell_2 = 0$, and $g_u = 0$.

Odd f . Then $L_u = F_u(i) = F_u\mathbb{Q}_9$ is a field, unramified of degree 2 over F_u , with residue field $\mathbb{F}_{q^2}$; its non-trivial automorphism σ acts on $\mathbb{Q}_9$ as the Frobenius ($\sigma(i) = -i$). The group $T(F_u)$ lies in $O_{L_u}^\times$, contains $\mu_4 \subset \mathbb{Q}_9$ and the Teichmüller lifts of $\mu_{q+1} \subset \mathbb{F}_{q^2}^\times$ (as $\zeta\sigma(\zeta) = \zeta^{1+q}$), and reduces onto μ_{q+1} . Since g_u is $\mathbb{Q}_9$ -linear, $g_u(\epsilon z) = \epsilon g_u(z)$ for $\epsilon \in \mu_4$.

Lemma 5 (one place). *In case (q) there are a place $u_0 \mid 3$ of odd residue degree and $\beta \in L_{u_0}$ with $\text{Tr}_{L_{u_0}/\mathbb{Q}_9}(\beta z) \in \mathcal{A}$ for every $z \in T(F_{u_0})$.*

Proof. Only the places of odd residue degree contribute. For such a place u and $z, z' \in T(F_u)$, varying y_u alone gives $g_u(z) - g_u(z') \in \mathcal{A} - \mathcal{A} \subseteq \frac{1}{3}O_3$. With $z' = 1$ and $z = i$ this says $(i-1)g_u(1) \in \frac{1}{3}O_3$, and $i-1$ is a unit at 3; so $g_u(T(F_u)) \subseteq \frac{1}{3}O_3$. Let $Z_u \subseteq \mathbb{F}_9$ be the set of residues of $3g_u(z)$, $z \in T(F_u)$; it is stable under μ_4 , and (q) says that $\sum_u z_u \in \mathcal{A}'$ for every

choice of $z_u \in Z_u$. Suppose two places $u \neq u'$ had nonzero elements $w \in Z_u$ and $w' \in Z_{u'}$. Fix $z_{u''} \in Z_{u''}$ at the other places and put $c = \epsilon' w' + \sum z_{u''}$ with $\epsilon' \in \mu_4$. The four distinct elements $c + \epsilon w$, $\epsilon \in \mu_4$, lie in A' , which has four elements; so they are all of A' , and summing them gives $4c = c = \sum A' = 0$. As this holds for every $\epsilon' \in \mu_4$, $w' = iw'$, so $w' = 0$: a contradiction. Since $0 \notin A'$, exactly one place u_0 has $Z_{u_0} \neq \{0\}$; the other g_u take values in O_3 , and $\beta = \beta_{u_0}$ works, because $\mathcal{A} + O_3 = \mathcal{A}$. $\square$

Lemma 6 (level one). *Let $u \mid 3$ have odd residue degree f , and let $\beta \in L_u$ satisfy $\text{Tr}_{L_u/\mathbb{Q}_3}(\beta z) \in \mathcal{A}$ for every $z \in T(F_u)$. Then $\lambda(x) = 3 \text{Tr}_{L_u/\mathbb{Q}_3}(\beta x) \bmod 3$ is defined on O_{L_u} and vanishes on its maximal ideal $\mathfrak{m}$, and the induced $\mathbb{F}_9$ -linear map $\lambda_1 : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_9$ satisfies $\lambda_1(\mu_{q+1}) \subseteq A'$.*

Proof. The closed additive span R of $T(F_u)$ is a closed subring of O_{L_u} . It contains $\mathbb{Z}_3[\mu_{q+1}]$, which is the ring of integers W of the unramified extension of $\mathbb{Q}_3$ of degree $2f$ (the order of 3 modulo $q+1$ is $2f$). Let π be a uniformiser of F_u . Then $z = (\pi + i)/(\pi - i)$ and $\sigma(z) = z^{-1}$ lie in $T(F_u)$, and $2 - z - \sigma(z) = 4/(\pi^2 + 1)$ is a unit of O_{L_u} in R ; its inverse is a limit of its powers, so $(\pi^2 + 1)/4 \in R$, and $\pi = (z - \sigma(z))(\pi^2 + 1)/(4i) \in R$. Hence $R \supseteq W[\pi] = O_{L_u}$, since π is also a uniformiser of L_u . By continuity $\text{Tr}(\beta O_{L_u}) \subseteq \frac{1}{3}O_3$, so $\lambda : O_{L_u} \rightarrow \mathbb{F}_9$ is defined, $\mathbb{Z}_9$ -linear, and zero on $3O_{L_u} = \mathfrak{m}^e$.

Suppose $\lambda(\mathfrak{m}) \neq 0$, and let $j \geq 1$ be the largest integer with $\lambda(\mathfrak{m}^j) \neq 0$. Identify $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ with $\mathbb{F}_{q^2}$ through $x \mapsto (x/\pi^j) \bmod \mathfrak{m}$; then λ induces a nonzero additive map $\lambda_j : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_9$. For $\eta \in O_{F_u}$ the element $t = (1 + i\pi^j\eta)/(1 - i\pi^j\eta)$ lies in $T(F_u)$ (as $\sigma(t) = t^{-1}$), and $t \equiv 1 + 2i\pi^j\eta \pmod{\mathfrak{m}^{j+1}}$. For $z \in T(F_u)$ with residue $\hat{z} \in \mu_{q+1}$ this gives $\lambda(zt) = \lambda(z) + \lambda_j(2i\hat{z}\hat{\eta})$, where $\hat{\eta}$ is the residue of η . As $\hat{\eta}$ runs over $\mathbb{F}_q$, the second term runs over an additive subgroup Σ_z of $\mathbb{F}_9$, and $\lambda(z) + \Sigma_z \subseteq A'$. But $A' = \{a + bi : a, b \neq 0\}$ contains no coset of a nonzero subgroup of $\mathbb{F}_9$, so $\Sigma_z = 0$: λ_j vanishes on $i\hat{z}\mathbb{F}_q$ for every $\hat{z} \in \mu_{q+1}$. These sets span $\mathbb{F}_{q^2}$ additively, because the $\mathbb{F}_q$ -span of μ_{q+1} is a subfield not contained in $\mathbb{F}_q$; so $\lambda_j = 0$, a contradiction. Hence $\lambda(\mathfrak{m}) = 0$, the induced map λ_1 is $\mathbb{F}_9$ -linear, and $\lambda_1(\mu_{q+1}) \subseteq A'$ because $T(F_u)$ reduces onto μ_{q+1} . $\square$

Lemma 7 (residue fields). *Let f be odd and $q = 3^f$. There is an $\mathbb{F}_9$ -linear map $\lambda_1 : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_9$ with $\lambda_1(\mu_{q+1}) \subseteq A'$ if and only if $f = 1$.*

Proof. For $f = 1$, $\mu_4 \subset \mathbb{F}_9$ and $\lambda_1(x) = (1 + i)x$ works. Let $f \geq 3$. Every $\mathbb{F}_9$ -linear map is $\lambda_1(x) = \text{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_9}(bx) = \sum_{j < f} (bx)^{9^j}$ for some $b \in \mathbb{F}_{q^2}$. In characteristic 3,

$$\lambda_1(z)^4 = \lambda_1(z)^3 \lambda_1(z) = \sum_{j,k < f} b^{e_{jk}} z^{e_{jk}}, \quad e_{jk} = 3 \cdot 9^j + 9^k.$$

Since $A' = \{y : y^4 = -1\}$, if $\lambda_1(\mu_{q+1}) \subseteq A'$ then $\Phi(z) = 1 + \sum_{j,k} b^{e_{jk}} z^{e_{jk}}$ vanishes on μ_{q+1} . On this cyclic group, of order prime to 3, the characters $z \mapsto z^e$ ($e \bmod q+1$) are linearly independent; so for every class $e \bmod q+1$ the sum of the coefficients of the $z^{e_{jk}}$ with $e_{jk} \equiv e$ vanishes (or equals -1 , for $e \equiv 0$).

Modulo $q+1$, $g = -3$ satisfies $g^f = -3^f \equiv 1$ (as f is odd), $9^j \equiv g^{2j}$ and $3 \cdot 9^j \equiv -g^{2j+1}$. Let $e_{j'k'} \equiv e_{jj} = 4 \cdot 9^j$. Multiplying by g^{-2j} gives $4 \equiv -g^\alpha + g^\gamma$ with $\alpha \equiv 2j' + 1 - 2j$ and $\gamma \equiv 2k' - 2j \pmod{f}$, $0 \leq \alpha, \gamma < f$. As integers, $(-3)^\alpha$ and $(-3)^\gamma$ have absolute value at most 3^{f-1} , and $4 + 2 \cdot 3^{f-1} < 3^f + 1$ for $f \geq 3$; so the congruence is the equality $(-3)^\gamma - (-3)^\alpha = 4$. It forces $\gamma = 0$ (otherwise 3 divides the left side, or $(-3)^\gamma = 5$), then $(-3)^\alpha = -3$ and $\alpha = 1$; so $j' = k' = j$, as 2 is invertible modulo f . Hence the class of $4 \cdot 9^j$ contains only e_{jj} , it is not 0 (because $q+1 > 4$ and $3 \nmid q+1$), and its coefficient $b^{4 \cdot 9^j}$ must vanish. So $b = 0$, and then $\Phi = 1 \neq 0$. $\square$

Proof of Theorem B. If $i \in F$, then (a) and (b) fail ($F(i) = F$, and every residue field above 3 contains $\mathbb{F}_9$), and $\chi(F^2) = \infty$ by a theorem of Davies [4] (see [12, Proposition 3]); so the theorem holds. Let $i \notin F$.

Only if. Let F^2 be 3-colourable. By Lemma 4, (c) or (q) holds. In case (c), Section 4 gives (a). In case (q), Lemmas 5 and 6 give a place $u_0 \mid 3$ of odd residue degree f and an $\mathbb{F}_9$ -linear $\lambda_1 : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_9$ with $\lambda_1(\mu_{q+1}) \subseteq A'$, and Lemma 7 gives $f = 1$: this is (b).

If. Under (a) the plane is 2-colourable [22, Theorem A'], [15, Theorem 1]. Under (b), let $u \mid 3$ have residue degree 1 and let w be the place of L above it. Then $i \notin F_u$, $L_w = F_u(i)$ has residue field $\mathbb{F}_9$, and the unit vectors of F_u^2 are units of O_w whose residues lie in μ_4 . A unit step preserves the class of z modulo O_w ; fix a representative x_A of each class A and colour each $z \in A$ by $\ell((z - x_A) \bmod \mathfrak{m}_w)$, where $\ell(a + bi) = a + b$ on $\mathbb{F}_9 = \mathbb{F}_3 + \mathbb{F}_3 i$. Since $\ell(\pm 1) = \pm 1$ and $\ell(\pm i) = \pm 1$, every unit step changes the colour. This colours $F_u^2 \supseteq F^2$ with three colours. Finally $\chi(F^2) \leq 2$ if and only if (a), by [15, Theorem 1]. $\square$

The proof uses Theorem W^+ only through Lemma 2, for the finite sets $G_N V$; the rest is compactness, weak approximation for the norm-one torus, the arithmetic of the places above 2 and 3, and, in case (c), the two-colour theorem of [15].

6. CONSEQUENCES

Corollary 1 (no circular chromatic number between 2 and 3). *Let F be a number field. If F^2 has a homomorphism to a circular clique $K_{p/q}$ with $p/q < 3$, then $\chi(F^2) \leq 2$. Hence $\chi_c(F^2) = 2$, $\chi_c(F^2) = 3$ or $\chi_c(F^2) > 3$, according as $\chi(F^2) = 2, 3$ or at least 4; and F^2 has a homomorphism to a cycle of odd length at least 5 only if it is bipartite.*

Proof. If $i \in F$ there is nothing to prove, so let $i \notin F$. For every finite U as in the proof of Lemma 2, Theorem W^+ gives a character with $\xi(U) \subseteq [q/p, 1 - q/p] + \mathbb{Z}$, and $q/p > 1/3$. In that proof $\varphi(v)$ then satisfies $\operatorname{Re}(\overline{\varphi(v)}) \in [q/p, 1 - q/p] + \mathbb{Z}$ (take $\gamma = 1$), while a point $N\varepsilon$ with $\varepsilon \in E_q$ (by Proposition 1 these are the points of type q) has real part in $\{\frac{1}{3}, \frac{2}{3}\} + \mathbb{Z}$, as $3 \nmid N$. So every ε_v has type c, the conditions P_v in the proof of Lemma 4 can all be taken with $\tau(v) = c$, and (c) holds; so $\chi(F^2) \leq 2$. For the second statement: if $\chi_c(F^2) < 3$, then by the definition of the infimum F^2 maps to some $K_{p/q}$ with $p/q < 3$, so $\chi(F^2) \leq 2$; hence $\chi_c(F^2) = 3$ when $\chi(F^2) = 3$, as $\chi_c \leq \chi$. If $\chi(F^2) \geq 4$, then by the theorem of de Bruijn and Erdős [5] some finite subgraph H has $\chi(H) \geq 4$, so $\chi_c(F^2) \geq \chi_c(H) > \chi(H) - 1 \geq 3$. $\square$

Call a number field F *split above 2 and 3* if $i \in F_v$ for every place v of F above 2 and above 3, that is, if every such place splits in $F(i)$; then (a) and (b) fail. The admissible fields of [15, Section 4], which must moreover satisfy $\sqrt{3} \notin F$ and $i \in F_v$ at the places above 7, 11 and 19, are split above 2 and 3.

Corollary 2. *Every number field F that is split above 2 and 3 has $\chi(F^2) \geq 4$.*

A real quadratic field $\mathbb{Q}(\sqrt{d})$ is split above 2 and 3 exactly when $d \equiv 23 \pmod{24}$, so for real quadratic fields Corollary 2 is contained in [16] (not for imaginary ones: $\mathbb{Q}(\sqrt{-73})$ is split above 2 and 3). Of the biquadratic fields listed as admissible in the account of [13, `notes/local_global.md`, Section 4], $\mathbb{Q}(\sqrt{2}, \sqrt{47})$, $\mathbb{Q}(\sqrt{11}, \sqrt{13})$ and $\mathbb{Q}(\sqrt{10}, \sqrt{38})$ (which contains $\sqrt{95}$) contain some $\sqrt{d}$ with $d \equiv 11 \pmod{12}$; for $\mathbb{Q}(\sqrt{2}, \sqrt{31})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{55})$ the bound is new, as far as we know.

Corollary 3 (local fields). *Let K be a finite extension of $\mathbb{Q}_p$. Then $\chi(K^2) = 2$ if $p = 2$ and $K(i)/K$ is ramified, $\chi(K^2) = 3$ if $p = 3$ and the residue degree of K is 1, and $\chi(K^2) \geq 4$ in every other case.*

Proof. In the first case the 2-colouring of the proof of [15, Proposition 3] colours K^2 itself (that proof colours F_v^2 , and it applies verbatim to every finite extension K of $\mathbb{Q}_2$ with $K(i)/K$ ramified); in the second case the 3-colouring of the proof of Theorem B colours K^2 itself; and $\chi(K^2) \geq 3$ when $p = 3$, because $\mathbb{Q}(\sqrt{7}) \subset \mathbb{Q}_3 \subseteq K$ and $\chi(\mathbb{Q}(\sqrt{7})^2) = 3$. If $p \geq 5$, then $\chi(K^2) \geq \chi(\mathbb{Q}_p^2) \geq 4$ by [16, Corollary 3]. If $i \in K$, then $\chi(K^2) = \infty$ [12, Proposition 3]. There remain the case $p = 2$, $i \notin K$, $K(i)/K$ unramified, and the case $p = 3$ with odd residue degree at least 3. In both we find a number field $F \subset K$ for which (a) and (b) fail; then $\chi(K^2) \geq \chi(F^2) \geq 4$ by Theorem B.

Let $m = [K : \mathbb{Q}_p]$ and let p' be the other element of $\{2, 3\}$. If $p = 2$, then m is even: if m were odd, $\mathbb{Q}_2^\times / \mathbb{Q}_2^{\times 2}$ would inject into $K^\times / K^{\times 2}$, and since -1 and -5 are not squares in $\mathbb{Q}_2$, $i \notin K$ and $K(i) \neq K(\sqrt{5})$, which is the unramified quadratic extension of K ; so $K(i)/K$ would be ramified. Choose monic polynomials over $\mathbb{Q}_p$, $\mathbb{Q}_{p'}$ and $\mathbb{Q}_5$ of a common even degree n :

- over $\mathbb{Q}_p$, $h_K h'$, where h_K is the minimal polynomial of a generator of K and h' is a product of distinct irreducible quadratics, coprime to h_K , whose roots generate $\mathbb{Q}_2(i)$ (if $p = 2$) or $\mathbb{Q}_9$ (if $p = 3$); if $p = 3$ and m is odd, h' also contains one cubic factor, coprime to h_K , generating the unramified cubic extension $\mathbb{Q}_{27}$ of $\mathbb{Q}_3$;
- over $\mathbb{Q}_{p'}$, a product of distinct irreducible quadratics whose roots generate $\mathbb{Q}_2(i)$, resp. $\mathbb{Q}_9$;
- over $\mathbb{Q}_5$, an irreducible polynomial.

By weak approximation choose a monic $h \in \mathbb{Q}[x]$ whose coefficients are so close to the three targets that, by Krasner's lemma, h factors over $\mathbb{Q}_p$, $\mathbb{Q}_{p'}$ and $\mathbb{Q}_5$ with the same degrees and the same fields as the targets. Then $F = \mathbb{Q}[x]/(h)$ is a field (as h is irreducible over $\mathbb{Q}_5$); it embeds in K , through a root of h close to the chosen generator of K ; its places above p are the place that gives K and places that contain i (or, for the cubic factor, have residue degree 3); and its places above p' all contain i . So no place above 2 ramifies in $F(i)$ and no place above 3 has residue degree 1. $\square$

For example $\chi(\mathbb{Q}_{27}^2) \geq 4$, although $i \notin \mathbb{Q}_{27}$ and $x^2 + y^2$ is anisotropic over $\mathbb{Q}_{27}$, as over $\mathbb{Q}_3$; and $\chi(\mathbb{Q}_2(\sqrt{3})^2) \geq 4$, as $\mathbb{Q}_2(\sqrt{3}, i) = \mathbb{Q}_2(\sqrt{3}, \sqrt{-3})$ is unramified over $\mathbb{Q}_2(\sqrt{3})$ (this also follows from $\chi(\mathbb{Q}(\sqrt{11})^2) = 4$, since $\mathbb{Q}_2(\sqrt{11}) = \mathbb{Q}_2(\sqrt{3})$). Corollary 3 extends [16, Corollary 3] ($p \geq 5$) and the values $\chi(\mathbb{Q}_2^2) = 2$ and $\chi(\mathbb{Q}_3^2) = 3$, which follow from Madore's results [21] (see [12, Theorem 2]).

7. AN EXPLICIT FAMILY: TWO SQUARE ROOTS

For many fields the general argument can be replaced by two explicit relations. For $c \geq 1$ put

$$u_c = \frac{(1 + i\sqrt{c})^2}{1 + c} = \frac{(1 - c) + 2\sqrt{c}i}{1 + c}, \quad \text{so that} \quad (1 + c)(u_c + \bar{u}_c) = 2(1 - c).$$

Proposition 2. *Let $F \subset \mathbb{R}$ contain $\sqrt{a}$ and $\sqrt{b}$, where $a \equiv 2 \pmod{3}$ and $b \equiv 7 \pmod{8}$ are positive integers ($a = b$ is allowed), and let $N = 5^k > \frac{4\sqrt{10}}{3} \max(a, b)$. Then no character of the group generated by $U = G_N\{1, u_a, \bar{u}_a, u_b, \bar{u}_b\}$ maps every element of U into $[1/3, 2/3] + \mathbb{Z}$. Hence $\chi(F^2) \geq 4$.*

Proof. As in the proof of Lemma 2, such a character gives numbers $w_v = \varphi(v) \in S_N$ ($v \in \{1, u_a, \bar{u}_a, u_b, \bar{u}_b\}$) that satisfy the two relations exactly. The sums of the absolute values of their coefficients are $4a$ and $4b$, and $6r \cdot 4c = \frac{4\sqrt{10}}{3}c < N$; so Lemma 1 gives, for the types,

$$(1 + a)(\varepsilon_{u_a} + \varepsilon_{\bar{u}_a}) = 2(1 - a)\varepsilon_1, \quad (1 + b)(\varepsilon_{u_b} + \varepsilon_{\bar{u}_b}) = 2(1 - b)\varepsilon_1.$$

At 3. $3 \mid 1 + a$ and $3 \nmid 2(1 - a)$. By (1) the left side of the first relation has $v_3 \geq 0$, hence $v_3(\varepsilon_1) \geq 0$, and ε_1 is not of type q. *At 2.* $v_2(1 + b) \geq 3$ and $v_2(2(1 - b)) = 2$, as $1 - b \equiv 2 \pmod{8}$. By (1) the left side of the second relation has $v_{1+i} \geq 2 \cdot 3 - 1 = 5$, while $v_{1+i}(2(1 - b)\varepsilon_1) = 4 + v_{1+i}(\varepsilon_1) = 3$ if ε_1 has type c. So ε_1 is not of type c either, a contradiction. $\square$

For $a = b = d$ this is the case $d \equiv 23 \pmod{24}$ of [16, Theorem 1], where [16, Theorem 4(1)] needs only $d < 21 \cdot 5^{k-1}$. Proposition 2 covers, for example, $\mathbb{Q}(\sqrt{2}, \sqrt{7})$, $\mathbb{Q}(\sqrt{2}, \sqrt{31})$, $\mathbb{Q}(\sqrt{5}, \sqrt{7})$, $\mathbb{Q}(\sqrt{2}, \sqrt{55})$ and $\mathbb{Q}(\sqrt{3}, \sqrt{5})$ (with $b = 15$); every field containing one of them then has $\chi \geq 4$ as well, as $\chi(F^2)$ can only grow with F . For $\mathbb{Q}(\sqrt{2}, \sqrt{7})$ the bound is sharp: $\sqrt{2} \in \mathbb{Q}_7$, as $2 \equiv 3^2 \pmod{7}$, so $\mathbb{Q}(\sqrt{2}, \sqrt{7})$ embeds in $\mathbb{Q}_7(\sqrt{7})$, whose residue field is $\mathbb{F}_7$, and Proposition 3 gives $\chi_c \leq 7/2$, hence $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{7})^2) = 4$. In terms of Theorem B, $\sqrt{a}$ makes every place above 3 have even residue degree, and $\sqrt{b}$ puts i in every completion above 2.

8. CIRCULAR COLOURINGS

For the plane over $\mathbb{R}$, DeVos, Ebrahimi, Ghebleh, Goddyn, Mohar and Naserasr proved $\chi_c(\mathbb{R}^2) \geq 4$ [7]; since de Grey's theorem gives a finite subgraph with $\chi = 5$, in fact $\chi_c(\mathbb{R}^2) > 4$. For planes over number fields the values 2 and 3 are immediate (a bipartite plane; a plane with $\chi = 3$ containing a unit triangle), and by Corollary 1 the planes with $\chi_c \leq 3$ are exactly those

with $\chi \leq 3$. We know of no earlier non-integral value. Theorem C shows that $7/2$ occurs; its upper bound comes from a residue field.

Proposition 3. *Let v be a place of F whose residue field is $\mathbb{F}_p$, with $p \equiv 3 \pmod{4}$, let $m \geq 1$ be an integer, and let $\lambda : \mathbb{F}_{p^2} \rightarrow \mathbb{F}_p$ be $\mathbb{F}_p$ -linear with $\lambda(\mu_{p+1}) \subseteq \{m, m+1, \dots, p-m\}$. Then F_v^2 , hence F^2 , has a homomorphism to $K_{p/m}$. For $p = 7$, $\lambda(a+bi) = 2a+3b$ takes the values $2, 3, 4, 5$ on $\mu_8 \subset \mathbb{F}_{49}$, so F^2 maps to $K_{7/2}$, and $\chi_c(F^2) \leq 7/2$.*

Proof. As -1 is not a square in $\mathbb{F}_p$, $i \notin F_v$, and $L_w = F_v(i)$ is unramified over F_v with residue field $\mathbb{F}_{p^2}$. A unit vector z has $z\bar{z} = 1$, so it is a unit of O_w , and since the conjugation of L_w/F_v reduces to the Frobenius $x \mapsto x^p$, its residue satisfies $\hat{z}^{p+1} = 1$. A unit step preserves the class modulo O_w ; fix a representative x_A of each class A and colour each $x \in A$ by $\lambda((x - x_A) \bmod \mathfrak{m}_w) \in \mathbb{Z}/p$. A unit step changes the colour by an element of $\{m, \dots, p-m\}$. For $p = 7$, $\mu_8 = \{\pm 1, \pm i, \pm 2 \pm 2i\}$, and $2a+3b$ takes the values $2, 5, 3, 4, 3, 5, 2, 4$ at $1, -1, i, -i, 2+2i, 2-2i, -2+2i, -2-2i$. $\square$

Proof of Theorem C. Upper bounds. 7 splits in $\mathbb{Q}(\sqrt{11})$ ($11 \equiv 2^2 \pmod{7}$) and ramifies in $\mathbb{Q}(\sqrt{35})$, so both fields, and $\mathbb{Q}_7$, have a place with residue field $\mathbb{F}_7$; Proposition 3 applies.

Lower bounds. For $d \in \{11, 35\}$ let $u_n = (n + i\sqrt{d})^2 / (n^2 + d)$, $V = \{1, u_1, \bar{u}_1, u_7, \bar{u}_7, u_{19}, \bar{u}_{19}\}$ and $U = G_{25}V$, a set of 140 unit vectors (70 up to sign). We checked by computer that no character of $\mathbb{Z}U$ maps every element of U into the open interval $(2/7, 5/7) + \mathbb{Z}$. The certificate branches on the integer values of the relations among the vectors (66 relations; 17 744 nodes for $d = 11$ and 14 199 for $d = 35$) and closes every leaf with an exact Farkas vector; two exact checkers that share no code accept it (Section 11). If F^2 , $F = \mathbb{Q}(\sqrt{d})$, mapped to $K_{p/q}$ with $p/q < 7/2$, Theorem W^+ would give a character with values in $[q/p, 1 - q/p] \subset (2/7, 5/7)$ on U . So $\chi_c(F^2) \geq 7/2$. Since $\mathbb{Q}(\sqrt{11}) \subset \mathbb{Q}_7$, also $\chi_c(\mathbb{Q}_7^2) \geq 7/2$.

Chromatic numbers. $K_{7/2} \rightarrow K_4$ gives $\chi \leq 4$, and $\chi \geq \chi_c = 7/2 > 3$ gives $\chi \geq 4$ (for the two quadratic fields this is also [16, Theorem 1], for $\mathbb{Q}_7$ [12, Theorem 2]). The last sentence of the theorem is Corollary 1. $\square$

The lower bound is about the infinite graph $\text{Cay}(\mathbb{Z}U, U)$, to which Theorem W^+ is applied directly, and gives no finite subgraph of $\mathbb{Q}(\sqrt{11})^2$ with circular chromatic number exactly $7/2$; one exists by Corollary 7, and Section 10 gives one with 155 vertices, found by computer. The open interval is essential: the characters that come from the 7-adic colourings take values in the closed interval $[2/7, 5/7]$ on U . The certificates were found with a floating-point mixed-integer solver, which reports $\max_{\xi} \min_{u \in U} \|\xi(u)\| = 2/7$ for both sets; nothing in the proof depends on it.

The circular chromatic number separates planes with the same chromatic number:

Proposition 4. $\chi_c(\mathbb{Q}(\sqrt{59})^2) \geq 53/15 > 7/2$, while $\chi(\mathbb{Q}(\sqrt{59})^2) = 4$.

Proof. For $d = 59$ the same set $U = G_{25}V$ has no character into the open interval $(15/53, 38/53)$: the certificate has 127 181 nodes and 105 761 leaves, and both checkers accept it (Section 11). As in the proof of Theorem C, Theorem W^+ gives $\chi_c \geq 53/15$. Here 7 is inert, so Proposition 3 does not apply. We have $\chi \geq 4$ by [16, Theorem 1], as $59 \equiv 11 \pmod{12}$. And $\chi \leq 4$: let v be the place of $F = \mathbb{Q}(\sqrt{59})$ above 2. As $59 \cdot 3 \equiv 1 \pmod{8}$, $F_v = \mathbb{Q}_2(\sqrt{59}) = \mathbb{Q}_2(\sqrt{3})$, and $L_w = F_v(i) = \mathbb{Q}_2(\sqrt{3}, \sqrt{-3})$ is unramified over F_v , with residue field $\mathbb{F}_4$. So the unit vectors are units of O_w whose residues lie in the group $\mu_3 = \mathbb{F}_4^\times$, and, as in the proof of Proposition 3, colouring each class of L_w modulo O_w by the residue modulo $\mathfrak{m}_w$ is a proper 4-colouring (a bound of Fischer [9]). $\square$

Theorem F gives more: $\chi_c(\mathbb{Q}(\sqrt{59})^2) = 4$. Proposition 4 is an explicit form of the separation: 70 explicit unit vectors already push χ_c above $7/2$.

Corollary 4. $\chi_c(K^2) = 7/2$ for every finite extension K of $\mathbb{Q}_7$ with residue degree 1, and $\chi_c(F^2) = 7/2$ for every number field F that contains $\sqrt{11}$ or $\sqrt{35}$ and has a place above 7 of residue degree 1, for example $\mathbb{Q}(\sqrt{7}, \sqrt{11})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{35})$.

Proof. $K \supseteq \mathbb{Q}_7 \supset \mathbb{Q}(\sqrt{11})$ and $F \supseteq \mathbb{Q}(\sqrt{11})$ or $\mathbb{Q}(\sqrt{35})$ give $\chi_c \geq 7/2$ by Theorem C; the residue field $\mathbb{F}_7$ gives $\chi_c \leq 7/2$ by Proposition 3 (whose proof applies to K itself). $\square$

Locally constant colourings. Below 4, the best upper bounds that colourings through a single completion can give are 2, 3 and $7/2$ (Corollary 5). Let K be a finite extension of $\mathbb{Q}_p$ with $i \notin K$ and $K(i)/K$ unramified (the second condition is automatic for p odd; for $p = 2$ it is the local form of the failure of (a)), let $\mathbb{F}_q$ be the residue field of K ($q = p^f$), w the place of $L = K(i)$, and

$$\kappa_1(K) = \max_{\lambda} \min_{z \in \mu_{q+1}} \|\lambda(z)\|,$$

the maximum over the additive maps $\lambda : \mathbb{F}_{q^2} \rightarrow \frac{1}{p}\mathbb{Z}/\mathbb{Z}$.

Proposition 5. *Suppose K^2 has a locally constant homomorphism to $K_{P/Q}$, $P/Q < 4$. Then $Q/P \leq \kappa_1(K)$. Conversely K^2 maps to $K_{p/m}$ when $m/p = \kappa_1(K) > 0$.*

Proof. The converse is the colouring of Proposition 3, with $\mathbb{F}_q$ in place of $\mathbb{F}_p$. Let π be a uniformiser of K (also one of L) and σ the non-trivial automorphism of L/K . On the compact set O_w the homomorphism is constant on the cosets of some power $\mathfrak{m}^k$ of the maximal ideal, and only its restriction to O_w is used. It induces a homomorphism from $\text{Cay}(O_w/\mathfrak{m}^k, \bar{T})$, where $\bar{T}$ is the image of $T(K)$, to $K_{P/Q}$; by Theorem W^+ there is a character ξ of $O_w/\mathfrak{m}^k$ with $\|\xi(z)\| \geq Q/P > 1/4 \geq 1/(2p)$ for every $z \in T(K)$. We show that ξ vanishes on $\mathfrak{m}$; then ξ is a map λ as in the definition of κ_1 , and $Q/P \leq \kappa_1(K)$, since $T(K)$ reduces onto μ_{q+1} (it contains the Teichmüller lifts of μ_{q+1} , on which σ acts as $x \mapsto x^q$). Suppose not, and let $J \geq 1$ be the largest integer with $\xi(\mathfrak{m}^J) \neq 0$. Through $x \mapsto (x/\pi^J) \bmod \mathfrak{m}$, ξ induces a nonzero additive map $\lambda_J : \mathbb{F}_{q^2} \rightarrow \frac{1}{p}\mathbb{Z}/\mathbb{Z}$. For $x \in O_w$, $t = (1 + \pi^J x)/\sigma(1 + \pi^J x)$ lies in $T(K)$ and $t \equiv 1 + \pi^J(x - \sigma(x)) \pmod{\mathfrak{m}^{J+1}}$, and the residues of $x - \sigma(x)$ fill the $\mathbb{F}_q$ -line $D = \{y - y^q : y \in \mathbb{F}_{q^2}\}$. For $z \in T(K)$ with residue $\hat{z}$, $\xi(zt) = \xi(z) + \lambda_J(\hat{z}d)$, $d \in D$; the values $\lambda_J(\hat{z}D)$ form a subgroup of $\frac{1}{p}\mathbb{Z}/\mathbb{Z}$, and if it were the whole group, $\xi(T(K))$ would contain a coset of $\frac{1}{p}\mathbb{Z}/\mathbb{Z}$ and a point with $\|\cdot\| \leq \frac{1}{2p}$. So λ_J vanishes on $\hat{z}D$ for every $\hat{z} \in \mu_{q+1}$, hence on their additive span, which is D times the $\mathbb{F}_q$ -span of μ_{q+1} , that is $\mathbb{F}_{q^2}$: a contradiction. $\square$

Proposition 6. $\kappa_1(K) > 1/4$ only when $p = 3$ and $f = 1$, where $\kappa_1 = 1/3$, or $p = 7$ and $f = 1$, where $\kappa_1 = 2/7$. In every other case with $i \notin K$ and $K(i)/K$ unramified, $\kappa_1(K) < 1/4$; it is 0 when $p = 2$ or $f \geq 3$.

Proof. For $p = 2$: if $\lambda(\mu_{q+1}) = \{\frac{1}{2}\}$, summing over μ_{q+1} , whose elements add up to 0, gives $\lambda(0) = \frac{q+1}{2} \equiv \frac{1}{2}$, as $q+1$ is odd; so some z has $\lambda(z) = 0$.

Let p be odd; then $p \equiv 3 \pmod{4}$ and f is odd, as $i \notin K$. Let $\psi(x) = e^{2\pi i x/p}$ on $\mathbb{F}_p$, and $S(b) = \sum_{z \in \mu_{q+1}} \psi(\text{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_p}(bz))$ for $b \in \mathbb{F}_{q^2}^\times$. Every additive $\lambda : \mathbb{F}_{q^2} \rightarrow \frac{1}{p}\mathbb{Z}/\mathbb{Z}$ is $z \mapsto \frac{1}{p} \text{Tr}(bz)$ for some b . Writing the indicator function of $\mu_{q+1} = \ker N_{\mathbb{F}_{q^2}/\mathbb{F}_q}$ with the multiplicative characters χ of $\mathbb{F}_q^\times$, and using the Davenport–Hasse relation $G(\chi \circ N, \psi \circ \text{Tr}) = -G(\chi, \psi_q)^2$ for Gauss sums [3] ($\psi_q = \psi \circ \text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}$; see [20, Chapter 5]), one finds

$$S(b) = \frac{1}{q-1} \sum_{\chi} \bar{\chi}(N(b)) G(\chi \circ N, \psi \circ \text{Tr}) = - \sum_{y \in \mathbb{F}_q^\times} \psi_q\left(y + \frac{N(b)}{y}\right),$$

minus a Kloosterman sum; so $|S(b)| \leq 2\sqrt{q}$ by Weil's bound [28]. Hence, for $b \neq 0$, the number of $z \in \mu_{q+1}$ with $\text{Tr}(bz) = 0$ is $\frac{1}{p} \sum_{a \in \mathbb{F}_p} \sum_z \psi(a \text{Tr}(bz)) = \frac{q+1}{p} + \frac{1}{p} \sum_{a \neq 0} S(ab) \geq \frac{q+1}{p} - \frac{p-1}{p} 2\sqrt{q}$, which is positive when $f \geq 3$ (then $\sqrt{q} \geq p^{3/2} > 2(p-1)$); so $\kappa_1 = 0$ for $f \geq 3$.

Let $f = 1$, $p = 4k + 3$, and $\mathcal{I} = \{x : |x| \leq k\} \subset \mathbb{F}_p$, the residues with $\|x/p\| < \frac{1}{4}$. The number of $z \in \mu_{p+1}$ with $\text{Tr}(bz) \in \mathcal{I}$ is

$$\frac{(p+1)|\mathcal{I}|}{p} + \frac{1}{p} \sum_{a \neq 0} \left(\sum_{x \in \mathcal{I}} \psi(-ax) \right) S(ab) \geq \frac{p^2-1}{2p} - \frac{2\sqrt{p}}{p} \sum_{a=1}^{p-1} \frac{1}{|\sin(\pi a/p)|} \geq \frac{p^2-1}{2p} - 2\sqrt{p}(\ln p + 1),$$

because $|\sum_{x \in \mathcal{I}} \psi(-ax)| = |\sin(\pi a(2k+1)/p)/\sin(\pi a/p)| \leq 1/|\sin(\pi a/p)|$, and, by the symmetry $a \leftrightarrow p-a$ and $\sin x \geq 2x/\pi$ on $[0, \pi/2]$, $\sum_{a=1}^{p-1} 1/|\sin(\pi a/p)| \leq p \sum_{a \leq (p-1)/2} 1/a \leq p(\ln p + 1)$. The last expression is positive for $p = 1001$ and increasing for $p \geq 1001$ (its derivative is

$\frac{1}{2} + \frac{1}{2p^2} - \frac{\ln p + 3}{\sqrt{p}} > 0$ there). For the primes $p \equiv 3 \pmod{4}$ with $11 \leq p < 3000$ a direct computation (in $\mathbb{F}_{p^2} = \mathbb{F}_p(i)$, where bz runs over the circle $s^2 + t^2 = N(b)$ and $\text{Tr}(bz) = 2s$) gives $\kappa_1 < \frac{1}{4}$; the largest value is $\frac{4}{19}$. For $p = 3$ and $p = 7$ a direct computation gives $\kappa_1 = \frac{1}{3}$ and $\kappa_1 = \frac{2}{7}$, attained by the maps of the proof of Theorem B and of Proposition 3. $\square$

Corollary 5 (local circular values). *Let K be a finite extension of $\mathbb{Q}_p$. If K^2 has a locally constant homomorphism to $K_{P/Q}$ with $P/Q < 4$, then either $p = 2$ and $K(i)/K$ is ramified, or $(p, f) = (3, 1)$ and $P/Q \geq 3$, or $(p, f) = (7, 1)$ and $P/Q \geq 7/2$.*

Proof. If $i \in K$, then $\chi(K^2) = \infty$ [12, Proposition 3]. Otherwise combine Propositions 5 and 6. $\square$

So, below 4, the best upper bound on $\chi_c(F^2)$ given by the restriction to F^2 of a locally constant colouring of a completion F_v^2 of a number field is 2 (condition (a)), 3 (condition (b)) or $7/2$ (a place above 7 of residue degree 1).

9. A GAP ABOVE 3

By Corollary 1, $\chi_c(F^2)$ is 2, 3 or larger than 3. Theorem D says that in the last case it stays away from 3. Its proof is that of Theorem B with $I = [1/3, 2/3]$ replaced by a slightly wider interval; the new ingredient is the shape of the characters along the probe for such intervals. The proof is entirely by hand; Section 10 reaches $7/2$ with a second prime and one computer-assisted step.

Notation. In this section $N = 5^k$ with $k \geq 1$, $\rho = (3 + 4i)/5$, $h = (1 + i)/2$ and $\Lambda_{\pm} = \frac{2 \pm i}{5} \mathbb{Z}[i]$. For $\frac{1}{4} < \theta \leq \frac{1}{3}$ let S_N^{θ} be defined as S_N with $[\theta, 1 - \theta]$ in place of I , and put

$$s = \frac{1}{2} - \theta, \quad \delta = \frac{1}{3} - \theta, \quad B_s = \{a + bi : a, b \in \mathbb{R}, |a|, |b| \leq s\},$$

so that $s = \frac{1}{6} + \delta$, and $P_k = \{x \in \mathbb{C} : \bar{x}\rho^j \in B_s \text{ for } |j| \leq k\}$. For $\varepsilon \in E_q$ and $|j| \leq k$, Lemma 8 below gives $\eta_j^N(\varepsilon) \in \{\pm 1 \pm i\}$ with $\overline{N\varepsilon}\rho^j \in h + \eta_j^N(\varepsilon)/6 + \mathbb{Z}[i]$; for $1 \leq m \leq k$ let

$$Y_m^N(\varepsilon) = \{y \in \mathbb{C} : \eta_j^N(\varepsilon)/6 + \bar{y}\rho^j \in B_s \text{ for } |j| \leq m\},$$

and $Y_k(\varepsilon) = Y_k^N(\varepsilon)$. These depend on θ , on N , and on ε only modulo $\mathbb{Z}[i]$; when N is fixed we write $\eta_j(\varepsilon)$ for $\eta_j^N(\varepsilon)$. We write $w_1 = \text{Re } w$ and $w_2 = \text{Im } w$.

Lemma 8. (i) $G_N = \{i^a \rho^j : a \in \mathbb{Z}, |j| \leq k\}$, and $c \in S_N^{\theta}$ if and only if $\bar{c}\rho^j \in h + B_s + \mathbb{Z}[i]$ for every $|j| \leq k$.

(ii) If $\varepsilon \in E_c$ and $|j| \leq k$, then $\overline{N\varepsilon}\rho^j \in h + \mathbb{Z}[i]$.

(iii) If $\varepsilon = \frac{\alpha + \beta i}{3} + t \in E_q$ and $|j| \leq k$, then $\overline{N\varepsilon}\rho^j \equiv (-1)^k (-i)^j \frac{\alpha - \beta i}{3} \pmod{\mathbb{Z}[i]}$. This lies in $h + \eta/6 + \mathbb{Z}[i]$ for a unique $\eta \in \{\pm 1 \pm i\}$; moreover $\eta_{j-1}(\varepsilon) = i \eta_j(\varepsilon)$ for $1 - k \leq j \leq k$ (so $\eta_{j-2}(\varepsilon) = -\eta_j(\varepsilon)$), and $\eta_j(\bar{\varepsilon}) = \bar{\eta}_{-j}(\varepsilon)$.

(iv) $N\varepsilon + P_k \subseteq S_N^{\theta}$ for $\varepsilon \in E_c$, and $N\varepsilon + Y_k(\varepsilon) \subseteq S_N^{\theta}$ for $\varepsilon \in E_q$.

Proof. (i) If $\gamma \in G_N$, then $N\gamma \in \mathbb{Z}[i]$ has norm 5^{2k} , so $N\gamma = u(2 + i)^a(2 - i)^b$ with a unit u and $a + b = 2k$, and $\gamma = u(\frac{2+i}{2-i})^{a-k} = u\rho^{a-k}$; conversely $N\rho^{\pm j} = 5^{k-j}(3 \pm 4i)^j \in \mathbb{Z}[i]$ for $0 \leq j \leq k$. Since $\text{Re}(\pm iw) = \mp \text{Im } w$ and $[\theta, 1 - \theta] + \mathbb{Z} = \frac{1}{2} + [-s, s] + \mathbb{Z}$ is symmetric, $\text{Re}(\bar{c}\gamma) \in [\theta, 1 - \theta] + \mathbb{Z}$ for all $\gamma \in G_N$ if and only if both coordinates of every $\bar{c}\rho^j$ lie in $\frac{1}{2} + [-s, s] + \mathbb{Z}$.

(ii) Write $\varepsilon = h + t$. The Gaussian integer $N\rho^j$ has odd norm, so $N\rho^j = 1 + (1 + i)m$ with $m \in \mathbb{Z}[i]$, and $\bar{h}N\rho^j = \bar{h} + m \equiv h$, as $\bar{h} = h - i$; and $\bar{t}N\rho^j \in \mathbb{Z}[i]$.

(iii) $\overline{N\varepsilon}\rho^j \equiv \frac{\alpha - \beta i}{3} N\rho^j$, and $N\rho^j \equiv (-1)^k (-i)^j \pmod{3}$, because $5 \equiv -1$ and $3 \pm 4i \equiv \pm i \pmod{3}$. Modulo $\mathbb{Z}[i]$, $\frac{\alpha - \beta i}{3}$ times a unit has both coordinates in $\{\frac{1}{3}, \frac{2}{3}\}$, so it lies in $h + \frac{\pm 1 \pm i}{6} + \mathbb{Z}[i]$. Since $(-i)^{j-1} = i(-i)^j$ and $ih \equiv h$, $\eta_{j-1} = i\eta_j$. Finally $\overline{N\varepsilon}\rho^j$ is the conjugate of $\overline{N\varepsilon}\rho^{-j}$, and $\bar{h} \equiv h$.

(iv) follows from (i)–(iii). $\square$

Lemma 9. (i) $P_1 = sP$, where P is the 12-gon with vertices $\pm 1 \pm \frac{i}{3}$, $\pm \frac{1}{3} \pm i$ and $\frac{5}{7}(\pm 1 \pm i)$; so $|x| \leq s\sqrt{10}/3$ for $x \in P_k$.

(ii) $|y| \leq \sqrt{2}\delta$ for $y \in Y_m^N(\varepsilon)$, $1 \leq m \leq k$.

Proof. (i) P_1 is defined by the twelve inequalities $\pm(\bar{x}\rho^j)_e \leq s$, $j \in \{0, \pm 1\}$, $e \in \{1, 2\}$, which are homogeneous in (x, s) ; intersecting their boundary lines in pairs gives the vertices. For instance at $x = 1 + \frac{i}{3}$ we have $\bar{x} = 1 - \frac{i}{3}$, $\bar{x}\rho = \frac{13}{15} + \frac{3}{5}i$ and $\bar{x}\rho^{-1} = \frac{1}{3} - i$. Every vertex has modulus at most $\sqrt{10}/3$, and $P_k \subseteq P_1$.

(ii) As $s = \frac{1}{6} + \delta$, the condition $\eta_j/6 + \bar{y}\rho^j \in B_s$ implies $\eta_{j,e}(\bar{y}\rho^j)_e \leq \delta$ for $e = 1, 2$, where $\eta_j = \eta_{j,1} + \eta_{j,2}i$. So $Y_m^N(\varepsilon) \subseteq Y_1^N(\varepsilon) \subseteq \delta X(\varepsilon)$, where $X(\varepsilon)$ is defined by the six inequalities $\eta_{j,e}(\bar{y}\rho^j)_e \leq 1$, $|j| \leq 1$ (for $\delta = 0$ read $\delta X(\varepsilon)$ as $\{0\}$: as $X(\varepsilon)$ is bounded, the same inequalities with right-hand side 0 have only the solution 0). By Lemma 8(iii), $\eta_j^N(\varepsilon) = \eta_j^N(\pm\varepsilon)$ for $|j| \leq 1$; replacing ε by $i\varepsilon$ replaces η_j by $-i\eta_j$ and X by iX , and multiplication by i permutes the four classes of E_q modulo $\mathbb{Z}[i]$ cyclically. So every $X(\varepsilon)$ is a rotation of $X(\frac{1+i}{3})$ at level 5, where $\eta_{-1} = 1 + i$, $\eta_0 = 1 - i$, $\eta_1 = -1 - i$. This is the hexagon with vertices $-\frac{5}{4}$, $-\frac{5}{4}i$, $-\frac{5}{7}(1 + i)$, $1 - \frac{i}{2}$, $1 + i$ and $-\frac{1}{2} + i$, of moduli at most $\sqrt{2}$. $\square$

Lemma 10. Let $0 < s < 11/56$ and $k \geq 1$, and let $x \in \mathbb{C}$ with $|x| \leq s\sqrt{2}$ and $\bar{x}\rho^{k-2} \in B_s$. Then $\bar{x}\rho^k + \nu \notin B_s + \mathbb{Z}[i]$ for every $\nu \in \Lambda_+ \setminus \mathbb{Z}[i]$. The same holds with ρ^{-k} , ρ^{2-k} and Λ_- in place of ρ^k , ρ^{k-2} and Λ_+ .

Proof. Let $u = \bar{x}\rho^k$ and $w = \bar{x}\rho^{k-2} = u\rho^{-2} = u(-7 - 24i)/25$, so that $w_1 = (-7u_1 + 24u_2)/25$ and $w_2 = -(24u_1 + 7u_2)/25$. Suppose $u + \nu \in B_s + \mathbb{Z}[i]$; changing ν by a Gaussian integer we may assume $u + \nu \in B_s$. The non-zero classes of $\Lambda_+/\mathbb{Z}[i]$ are represented by $\pm \frac{2+i}{5}$ and $\pm \frac{-1+2i}{5}$, whose coordinates have absolute value at most $\frac{2}{5}$; every other element of the class has a coordinate of absolute value at least $\frac{3}{5}$, which $u_e + \nu_e \in [-s, s]$ excludes, as $|u_e| \leq |x| \leq s\sqrt{2}$ and $s(1 + \sqrt{2}) < \frac{3}{5}$. If $\nu = \pm \frac{2+i}{5}$, then $\mp u_1 \geq \frac{2}{5} - s$ and $\mp u_2 \geq \frac{1}{5} - s$, so $\pm w_2 = \mp(24u_1 + 7u_2)/25 \geq (11 - 31s)/25$. If $\nu = \pm \frac{-1+2i}{5}$, then $\pm u_1 \geq \frac{1}{5} - s$ and $\mp u_2 \geq \frac{2}{5} - s$, so $\mp w_1 = \pm(7u_1 - 24u_2)/25 \geq (11 - 31s)/25$. In both cases $|w_e| \geq (11 - 31s)/25 > s$, as $56s < 11$, contradicting $w \in B_s$. The second statement follows by complex conjugation, which exchanges ρ and ρ^{-1} , and Λ_+ and Λ_- , and preserves B_s and $\mathbb{Z}[i]$ (apply the first to $\bar{x}$ and $\bar{\nu}$). $\square$

Lemma 11. Let $0 \leq \delta < 5/168$, $k \geq 2$ and $\varepsilon \in E_q$, and let $y \in \mathbb{C}$ with $|y| < \frac{1}{6}$ and $\eta_{k-2}(\varepsilon)/6 + \bar{y}\rho^{k-2} \in B_s$. Then $\overline{N\varepsilon + y}\rho^k + \nu \notin h + B_s + \mathbb{Z}[i]$ for every $\nu \in \Lambda_+ \setminus \mathbb{Z}[i]$. The same holds with ρ^{-k} , ρ^{2-k} , $\eta_{2-k}(\varepsilon)$ and Λ_- in place of ρ^k , ρ^{k-2} , $\eta_{k-2}(\varepsilon)$ and Λ_+ .

Proof. Replacing (ε, y, ν) by $(i\varepsilon, iy, -i\nu)$ multiplies $\overline{N\varepsilon + y}\rho^k + \nu$ by $-i$ and replaces η_j by $-i\eta_j$; it preserves $h + B_s + \mathbb{Z}[i]$ (as $-ih \equiv h$), E_q , Λ_+ and the hypotheses. So we may assume $\eta_k(\varepsilon) = 1 + i$, and then $\eta_{k-2}(\varepsilon) = -1 - i$ (Lemma 8). Let $u = \bar{y}\rho^k$, so that $\bar{y}\rho^{k-2} = u(-7 - 24i)/25$. Suppose the conclusion fails. Then $\frac{1}{6} + \nu_e + u_e \in [-s, s] + \mathbb{Z}$ for $e = 1, 2$, where we take $\nu_e \in \{\pm \frac{1}{5}, \pm \frac{2}{5}\}$. As $|u_e| < \frac{1}{6}$ and $s < \frac{1}{5}$, the values $\nu_e = \frac{1}{5}$ and $\nu_e = \frac{2}{5}$ are impossible ($\frac{1}{6} + \nu_e + u_e$ then lies in $(\frac{1}{5}, \frac{11}{15})$), and $\nu_e = -\frac{2}{5}$ forces $u_e \geq \frac{1}{15} - \delta$. Of the four non-zero classes $\pm(\frac{2}{5}, \frac{1}{5})$, $\pm(-\frac{1}{5}, \frac{2}{5})$ of $\Lambda_+/\mathbb{Z}[i]$ only $\nu = (-\frac{2}{5}, -\frac{1}{5})$ remains, so $u_1 \geq \frac{1}{15} - \delta$. On the other hand the hypothesis at ρ^{k-2} says that both coordinates of $u(-7 - 24i)/25$ are at least $-\delta$: $7u_1 - 24u_2 \leq 25\delta$ and $24u_1 + 7u_2 \leq 25\delta$. Adding 7 times the first to 24 times the second gives $625u_1 \leq 775\delta$, that is $u_1 \leq \frac{31}{25}\delta$. Hence $\frac{1}{15} \leq \frac{56}{25}\delta$, that is $\delta \geq \frac{5}{168}$, a contradiction. The second statement follows by complex conjugation, as in Lemma 10: $\bar{\varepsilon} \in E_q$, $\bar{h} \equiv h$, and $\eta_{k-2}(\bar{\varepsilon}) = \overline{\eta_{2-k}(\varepsilon)}$ by Lemma 8(iii). $\square$

Proposition 7. Let $17/56 < \theta \leq 1/3$, $k \geq 1$ and $N = 5^k$. Then

$$S_N^\theta = \bigcup_{\varepsilon \in E_c} (N\varepsilon + P_k) \cup \bigcup_{\varepsilon \in E_q} (N\varepsilon + Y_k(\varepsilon)).$$

In particular every $c \in S_N^\theta$ is $c = N\varepsilon + x$, either with $\varepsilon \in E_c$ and $|x| \leq s\sqrt{10}/3$, or with $\varepsilon \in E_q$ and $|x| \leq \sqrt{2}\delta$.

For $\theta = 1/3$ we have $s\sqrt{10}/3 = \sqrt{10}/18$ and $\delta = 0$, and this is Proposition 1.

Proof. The inclusion $\supseteq$ is Lemma 8(iv). Note that $\theta > 17/56$ means $s < 11/56$ and $\delta < 5/168$, and that then $s(1 + \sqrt{2}) < \frac{3}{5}$. The sets on the right are invariant under $N\mathbb{Z}[i]$ and under multiplication by i (which maps E_c and E_q to themselves modulo $\mathbb{Z}[i]$), and so is S_N^θ . We use induction on k . For $\lambda \in \mathbb{Z}[i]$ put $\nu_\pm(\lambda) = \frac{N}{5}\bar{\lambda}\rho^{\pm k}$. Since $\frac{N}{5}\rho^k = \frac{(2+i)^{2k}}{5} = \frac{(2+i)^{2k-1}}{2-i}$, we have $\nu_\pm(\lambda) \in \Lambda_\pm$, and $\nu_+(\lambda) \in \mathbb{Z}[i]$ if and only if $2+i$ divides λ ; likewise $\nu_-(\lambda) \in \mathbb{Z}[i]$ if and only if $2-i$ divides λ . So both are Gaussian integers if and only if $5 \mid \lambda$.

The case $k = 1$. Let $c \in S_5^\theta$. The condition at ρ^0 gives $c \in h + B_s + \mathbb{Z}[i]$, so $c = 5h + x + \lambda$ with $x \in B_s$ and $\lambda \in \mathbb{Z}[i]$, which we take modulo 5; put $z = \bar{x}$, so $|z| \leq s\sqrt{2}$. By Lemma 8(ii) the conditions at $\rho^{\pm 1}$ read $z\rho^{\pm 1} + \nu_\pm(\lambda) \in B_s + \mathbb{Z}[i]$, and as in the proof of Lemma 10 the shift can be taken to be the representative of $\nu_\pm(\lambda)$ modulo $\mathbb{Z}[i]$ with coordinates in $[-\frac{2}{5}, \frac{2}{5}]$. If both $\nu_\pm(\lambda)$ are Gaussian integers, then $5 \mid \lambda$ and $x \in P_1$ (as $|z\rho^{\pm 1}| \leq s\sqrt{2} < 1 - s$). If exactly one is, say $\nu_-(\lambda)$, then $z\rho^{-1} \in B_s$, and Lemma 10 with $k = 1$ contradicts the condition at ρ ; the other case is symmetric. In the remaining 16 classes, multiplication by i leaves four orbits, represented by $\lambda = 1, 2, 2+2i$ and $1+i$. Write $(z\rho)_{1,2} = \frac{1}{5}(3z_1 - 4z_2, 4z_1 + 3z_2)$ and $(z\rho^{-1})_{1,2} = \frac{1}{5}(3z_1 + 4z_2, -4z_1 + 3z_2)$.

- $\lambda = 1$: $\nu_\pm \equiv \frac{-2 \mp i}{5}$, so $(z\rho)_1, (z\rho^{-1})_1 \geq \frac{2}{5} - s$. Their sum is $\frac{6}{5}z_1 \leq \frac{6}{5}s$, so $\frac{4}{5} - 2s \leq \frac{6}{5}s$ and $s \geq \frac{1}{4}$.
- $\lambda = 2$: $\nu_\pm \equiv \frac{1 \mp 2i}{5}$, so $(z\rho)_1, (z\rho^{-1})_1 \leq s - \frac{1}{5}$ and $(z\rho)_2 \geq \frac{2}{5} - s$. As $7(z\rho)_1 + 25(z\rho^{-1})_1 - 24(z\rho)_2 = 0$, we get $0 \leq 32(s - \frac{1}{5}) + 24(s - \frac{2}{5})$, so $s \geq \frac{2}{7}$.
- $\lambda = 2 + 2i$: $\nu_+ \equiv \frac{-1+2i}{5}$ and $\nu_- \equiv \frac{-2+i}{5}$, so $(z\rho)_2 \leq s - \frac{2}{5}$, $(z\rho^{-1})_1 \geq \frac{2}{5} - s$ and $(z\rho^{-1})_2 \leq s - \frac{1}{5}$. As $25(z\rho)_2 - 24(z\rho^{-1})_1 + 7(z\rho^{-1})_2 = 0$, we get $0 \leq 49(s - \frac{2}{5}) + 7(s - \frac{1}{5})$, so $s \geq \frac{3}{8}$.
- $\lambda = 1 + i$: then $c = 5\varepsilon + y$ with $\varepsilon = \frac{2+2i}{3} \in E_q$ and $y = x + \frac{1+i}{6}$, so $|y| \leq \sqrt{2}(s + \frac{1}{6}) < 0.52$. By Lemma 8(iii), $\bar{c}\rho^j \in h + \eta_j(\varepsilon)/6 + \bar{y}\rho^j + \mathbb{Z}[i]$, and as $|\bar{y}\rho^j| < 0.52$ and $\frac{1}{6} + 0.52 < 1 - s$, the condition at ρ^j says $\eta_j(\varepsilon)/6 + \bar{y}\rho^j \in B_s$. So $y \in Y_1^5(\varepsilon)$.

The first three orbits are thus empty, as $s < 11/56 < \frac{1}{4}$.

The step from $k-1$ to $k \geq 2$. Let $c \in S_N^\theta$. As $G_{N/5} \subseteq G_N$, $c \in S_{N/5}^\theta$, and by induction $c = \frac{N}{5}\varepsilon' + x$ with $\varepsilon' \in E_c$ and $x \in P_{k-1}$, or with $\varepsilon' \in E_q$ and $x \in Y_{k-1}^{N/5}(\varepsilon')$.

If $\varepsilon' = h + t \in E_c$, then $\frac{N}{5}\varepsilon' = Nh + \frac{N}{5}(t - 2 - 2i)$, so $c = Nh + x + \frac{N}{5}\lambda$ with $\lambda \in \mathbb{Z}[i]$. By Lemma 8(ii) the condition at $\rho^{\pm k}$ reads $\bar{x}\rho^{\pm k} + \nu_\pm(\lambda) \in B_s + \mathbb{Z}[i]$. Here $|x| \leq s\sqrt{10}/3 < s\sqrt{2}$ by Lemma 9, and $\bar{x}\rho^{\pm(k-2)} \in B_s$ as $x \in P_{k-1}$. By Lemma 10 both $\nu_\pm(\lambda)$ are Gaussian integers, so $5 \mid \lambda$ and $c \in Nh + x + N\mathbb{Z}[i]$; and $\bar{x}\rho^{\pm k} \in B_s + \mathbb{Z}[i]$ with $|x| < 1 - s$ gives $x \in P_k$.

If $\varepsilon' = \frac{\alpha' + \beta'i}{3} + t' \in E_q$, let $\alpha, \beta \in \{1, 2\}$ with $\alpha \equiv 2\alpha'$ and $\beta \equiv 2\beta' \pmod{3}$, and $\varepsilon = \frac{\alpha + \beta i}{3}$. Then $\frac{N}{5}\varepsilon' - N\varepsilon \in \frac{N}{5}\mathbb{Z}[i]$, because $\alpha' - 5\alpha \equiv \alpha' - 2\alpha \equiv 0 \pmod{3}$, and likewise for β ; so $c = N\varepsilon + x + \frac{N}{5}\lambda$ with $\lambda \in \mathbb{Z}[i]$. For $|j| \leq k-1$ we have $\frac{N}{5}\rho^j \in \mathbb{Z}[i]$, so $\eta_j^N(\varepsilon) = \eta_j^{N/5}(\varepsilon')$ for $|j| \leq k-1$, and $x \in Y_{k-1}^N(\varepsilon)$. In particular $|x| \leq \sqrt{2}\delta < \frac{1}{6}$, and the hypotheses of Lemma 11 hold at $\rho^{\pm(k-2)}$. The condition at $\rho^{\pm k}$ says $\bar{N\varepsilon + x}\rho^{\pm k} + \nu_\pm(\lambda) \in h + B_s + \mathbb{Z}[i]$, so by Lemma 11 both $\nu_\pm(\lambda)$ are Gaussian integers, $5 \mid \lambda$ and $c \in N\varepsilon + x + N\mathbb{Z}[i]$. Finally $\eta_{\pm k}(\varepsilon)/6 + \bar{x}\rho^{\pm k} \in B_s + \mathbb{Z}[i]$ with $|x| < \frac{1}{2}$ gives $x \in Y_k(\varepsilon)$. The bounds on $|x|$ are Lemma 9. $\square$

Proof of Theorem D. Suppose that F^2 has a homomorphism to $K_{p/q}$ with $p/q < 56/17$; we show that $\chi(F^2) \leq 3$. If $i \in F$ this cannot happen, as then $\chi(F^2) = \infty$ (proof of Theorem B). If $p/q \leq 3$, then $\chi(F^2) \leq \chi(K_{p/q}) \leq 3$. Otherwise $\theta = q/p \in (17/56, 1/3)$, and we follow the proof of Lemma 2 for a finite $V \subset T$. Theorem W⁺ applies, as $p < 4q$, and gives a character with $\xi(U) \subseteq [\theta, 1 - \theta] + \mathbb{Z}$; so $\varphi(v) \in S_N^\theta$ for $v \in V$, and Proposition 7 writes $\varphi(v) = N\varepsilon_v + x_v$ with $\varepsilon_v \in E$ and $|x_v| \leq r_\theta = \max(s\sqrt{10}/3, \sqrt{2}\delta)$ (so $r_{1/3} = r$). Lemma 1 and its proof hold for such points with r_θ in place of r . So, choosing $N > 6r_\theta \max_v(|D_v| + \sum_j |\mu_{v,j}|)$, we obtain $\beta \in L$ with $\text{Tr}(\beta v) \in E$ for every $v \in V$: the conclusion of Lemma 2. The proof of Lemma 4 uses the 3-colouring only through this conclusion, and the rest of the proof of Theorem B only through (c) or (q). So (a) or (b) holds, and $\chi(F^2) \leq 3$ by Theorem B. Finally, if $\chi(F^2) \geq 4$ and

$\chi_c(F^2) < 56/17$, the definition of χ_c as an infimum gives a homomorphism to some $K_{p/q}$ with $p/q < 56/17$, which we have excluded. $\square$

A computer-assisted extension. With exact computations the same method gives $\chi_c(F^2) \geq 10^7/3001611 > 3.3315$ whenever $\chi(F^2) \geq 4$; Theorem E supersedes this, but the computations show where the one-prime probe stops. Let $\theta^* = 0.3001611$. It suffices that for every $k \geq 17$ the conclusion of Proposition 7 holds for θ^* , so that every $c \in S_N^{\theta^*}$ is $N\varepsilon + x$ with $\varepsilon \in E$ and $|x| \leq r_{\theta^*} < 0.22$: then for $p/q < 1/\theta^*$ the argument above applies with $\theta = q/p$, since $S_N^\theta \subseteq S_N^{\theta^*}$, with r_{θ^*} in place of r_θ and $k \geq 17$. We computed $S_N^{\theta^*}$ exactly, as a finite union of rational polygons modulo N , level by level, with three programs written separately (two of them by independent checkers): all three find the same number of components at every level $k \leq 19$, and only the components of Proposition 7 for $k = 17, 18, 19$. For $k \geq 18$ the induction step of Proposition 7 goes through with Lemmas 10 and 11 replaced by the same statements under the hypotheses $x \in P_{k-1}$, resp. $y \in Y_{k-1}^N(\varepsilon)$, for every shift $\nu \in \Lambda_\pm \setminus \mathbb{Z}[i]$ (the programs do not use the reduction to the shifts of the hand proofs). The infimum of the θ for which these statements hold is an exact linear-programming value. It is non-increasing in k : in the coordinates $\bar{x}\rho^k$, resp. $\bar{y}\rho^k$, and since $\eta_{k-m} = i^m\eta_k$ (Lemma 8), the system at level $k+1$ contains the one at level k . All three programs find that it is at most $0.30005568 < \theta^*$ for $k = 12, 13$ and 14 (two of them up to $k = 22$).

Remark. (1) For a statement valid for every k , the bound $17/56$ in Proposition 7 is best possible: for $\theta = 17/56$, S_N^θ has further isolated points for $N = 5, 25, 125$ and 625 (8, 20, 12 and 4 of them modulo $N\mathbb{Z}[i]$), at which $\min_\gamma \|\operatorname{Re}(\bar{c}\gamma)\| = 17/56$. But the constant $56/17$ comes from the first levels only: by the same computations, for $\theta > 333/1106$ the set $S_{5^5}^\theta$ has only the components of Proposition 7, and the threshold of the induction step is $3303/10981 < 333/1106$ for $4 \leq k \leq 6$, hence at most that for $k \geq 6$; so the levels $k \leq 5$ and the threshold at $k = 6$ already give $\chi_c(F^2) \geq 1106/333 \approx 3.3213$.

(2) The method cannot reach beyond $10/3$. For every k the point $c_k = Nh + \frac{\rho^k}{5} - 5^{k-1}(2-i)$ lies in $S_N^{3/10}$, with $\min_\gamma \|\operatorname{Re}(\bar{c}_k\gamma)\| = \frac{3}{10}$: for $-k \leq j < k$, $5^{k-1}(2+i)\rho^j \in \mathbb{Z}[i]$ and $\bar{c}_k\rho^j \equiv h + \rho^{j-k}/5 \pmod{\mathbb{Z}[i]}$, at distance $\frac{1}{5}$ from h ; and $\bar{c}_k\rho^k \equiv h + \frac{1}{5} - \frac{2+i}{5} = h - \frac{1+i}{5}$, because $(2+i)^{2k+1} \equiv 2+i \pmod{5}$. But its distance to NE_c is at least $\sqrt{5} \cdot 5^{k-1} - \frac{1}{5}$, and to NE_q at least $\frac{7}{6}5^{k-1} - \frac{1}{5}$. So for $\theta \leq 3/10$ the probe has characters of a third, 5-adic, kind. Section 10 removes them with a rotation whose denominator is a power of another prime.

(3) The least θ above which $S_{5^k}^\theta$ has only the components of Proposition 7 is $17/56$ for $k \leq 4$, $333/1106$ for $k = 5$, $3303/10981$ for $k = 6$ and about 0.30016093 for $k = 17$. We expect that it tends to $3/10$, as do the thresholds of the induction step (about 0.300036 for $20 \leq k \leq 22$); that would give $\chi_c(F^2) \geq 10/3$ by this method.

10. TWO PRIMES

The probe of Section 9 uses only the rotations $i^a\rho^j$, whose denominators are powers of 5, and for $\theta \leq 3/10$ it has characters of a third kind (Remark (2) at the end of Section 9). The plane over a number field contains every rational rotation. We add one with another prime, $\sigma = \frac{3+2i}{3-2i} = \frac{5+12i}{13}$ (in this section σ is this rotation, not the automorphism of Section 5), and for $k, m \geq 0$ put

$$G(k, m) = \{i^a\rho^j\sigma^l : a \in \mathbb{Z}, |j| \leq k, |l| \leq m\},$$

$$S^\theta(k, m) = \{c \in \mathbb{C} : \operatorname{Re}(\bar{c}\gamma) \in [\theta, 1 - \theta] + \mathbb{Z} \text{ for every } \gamma \in G(k, m)\},$$

so that $S^\theta(k, 0) = S_N^\theta$ for $N = 5^k$. We keep the notation of Section 9, with $\frac{1}{4} < \theta \leq \frac{1}{3}$. If $i \notin F$, then $G(k, m) \subset T$: its elements lie in $\mathbb{Q}(i)$ and have norm one, and the automorphism of L/F restricts to complex conjugation on $\mathbb{Q}(i)$.

For $c \in \mathbb{C}$ and $\gamma \in \mathbb{Q}(i)$, the *strip index* of c at γ is $n_\gamma(c) = \lfloor \operatorname{Re}(\bar{c}\gamma) \rfloor$. For $c \in S^\theta(k, m)$ and $\gamma \in G(k, m)$, $\operatorname{Re}(\bar{c}\gamma) \in [n_\gamma(c) + \theta, n_\gamma(c) + 1 - \theta]$, and the *margins* $\operatorname{Re}(\bar{c}\gamma) - n_\gamma(c)$ and $n_\gamma(c) + 1 - \operatorname{Re}(\bar{c}\gamma)$ are at least θ . The points of $S^\theta(k, m)$ with a given vector of strip indices form

a compact convex polygon, on which the margins are affine functions; two different vectors give disjoint polygons, as some functional takes its values in two different strips. So the connected components of $S^\theta(k, m)$ correspond to the index vectors that occur, and a point of $S^\theta(k, m)$ lies in $S^{\theta'}(k, m)$ for $\theta' < \theta$, with the same strip indices. As $65\gamma \in \mathbb{Z}[i]$ for $\gamma \in G(1, 1)$, $S^\theta(1, 1)$ is invariant under translation by $65\mathbb{Z}[i]$; its *type points* are the points of $65h + 65\mathbb{Z}[i]$ and of $\frac{65}{3}(\alpha + \beta i) + 65\mathbb{Z}[i]$, $\alpha, \beta \in \{1, 2\}$.

Proposition 8 (computer-assisted). *Let $\frac{2}{7} < \theta \leq \frac{1}{3}$ and $c \in S^\theta(1, 1)$. Then there is a type point c_0 such that c and c_0 have the same strip index at every $\gamma \in G(1, 1)$.*

Proof. Let $r_0 = \frac{7}{25} < \frac{2}{7}$; then $S^\theta(1, 1) \subseteq S^{r_0}(1, 1)$. As $\operatorname{Re}(\bar{c} \cdot i\gamma) = -\operatorname{Im}(\bar{c}\gamma)$ and $[\theta, 1 - \theta] + \mathbb{Z}$ is symmetric, only the 18 functionals $\operatorname{Re}(\bar{c}\gamma)$ and $\operatorname{Im}(\bar{c}\gamma)$ with $\gamma = \rho^j \sigma^l$, $|j|, |l| \leq 1$, matter: at a point of $S^{r_0}(1, 1)$ and at a type point no value $\operatorname{Re}(\bar{c}\gamma)$ is an integer, so $n_{-\gamma} = -n_\gamma - 1$ and $n_{i\gamma} = -n_{-i\gamma} - 1$, and the strip indices at these 18 functionals determine those at all 36 rotations. The proof is a finite computation with exact certificates of two kinds (Section 11).

Completeness. After a translation by $65\mathbb{Z}[i]$, c lies in a cell $[a + r_0, a + 1 - r_0] \times [b + r_0, b + 1 - r_0]$ with $0 \leq a, b \leq 64$, which fixes the strip indices at $\gamma = 1$ and $\gamma = -i$. For each of the other 16 functionals, in a fixed order, the possible strip indices are the integers n for which $[n + r_0, n + 1 - r_0]$ meets the range of the functional on the cell. A tree lists, for each cell and each branch, every possible index either as a branch to follow or as closed; a closed branch carries a Farkas certificate: three inequalities valid on the branch (sides of the cell, and sides of the strips chosen so far) and rational multipliers $\lambda \geq 0$ whose combination is a false inequality between rational constants. The open branches end in 13 leaves (15 620 branch nodes, 4 364 closed branches), so every $c \in S^{r_0}(1, 1)$ has, up to $65\mathbb{Z}[i]$, one of 13 explicit index vectors.

The 13 vectors. Five are those of the type points. Each of the other eight comes with three of its margins ψ_1, ψ_2, ψ_3 and rationals $\lambda_1, \lambda_2, \lambda_3 \geq 0$ with $\sum \lambda_i = 1$ and $\sum \lambda_i \psi_i = \frac{2}{7}$ identically (the linear parts cancel). At a point of $S^\theta(1, 1)$ every margin is at least $\theta > \frac{2}{7}$, so no such point has one of these eight vectors. $\square$

The eight vectors excluded at the end are those of the 7-torsion points $\frac{65}{7}(a + bi)$ with $(a, b) \in \{(1, 2), (1, 5), (2, 1), (2, 6), (5, 1), (5, 6), (6, 2), (6, 5)\}$, at which the least margin is exactly $\frac{2}{7}$. On $G(1, 1)$ they give the characters of the 7-adic colourings of Proposition 3 (the map $\lambda(a + bi) = 2a + 3b$ gives $\frac{65}{7}(1 + 5i)$). The bound $\frac{2}{7}$ is optimal for every finite set of rational rotations: their denominators are prime to 7 (as $7 \equiv 3 \pmod{4}$), and if D is a common denominator, $d \equiv D^{-1} \pmod{7}$ and $\lambda(a + bi) = sa + tb$ is a map as in Proposition 3 with values in $\{2, 3, 4, 5\}$ on μ_8 , then at each of these rotations γ the point $Dd(s + ti)/7$ has the value $\lambda(\gamma \bmod 7)/7$, whose distance to $\mathbb{Z}$ is at least $\frac{2}{7}$. Both primes are needed, and so are the mixed rotations: with the primes 13 and 17 the corresponding window keeps 12 other components above $\frac{2}{7}$ (some with largest least margin $\frac{2}{5}$), and the rotations $1, \rho^{\pm 1}, \sigma^{\pm 1}$ alone keep 180 other components (some with largest least margin $\frac{55}{144} > \frac{1}{3}$); the primes 5 and 17 work as well as 5 and 13.

Lemma 12. *Let $\frac{2}{7} < \theta \leq \frac{1}{3}$, $k \geq 1$, $N = 5^k$, and let $C \in \mathbb{C}$ satisfy $\operatorname{Re}(\bar{C}\gamma) \in [\theta, 1 - \theta] + \mathbb{Z}$ for every $\gamma \in G(k, 1)$. Then $C \in N\varepsilon + P_k + N\mathbb{Z}[i]$ for some $\varepsilon \in E_c$, or $C \in N\varepsilon + Y_k(\varepsilon) + N\mathbb{Z}[i]$ for some $\varepsilon \in E_q$. In particular $C = N\varepsilon + x$ with $\varepsilon \in E$ and $|x| \leq r_\theta = \max(s\sqrt{10}/3, \sqrt{2}\delta) < 0.23$.*

Proof. Digits. For $|j| \leq k$ the conditions at ρ^j and $-i\rho^j$ say that both coordinates of $\bar{C}\rho^j$ lie in $\frac{1}{2} + [-s, s] + \mathbb{Z}$, so $\bar{C}\rho^j = h + g_j + n_j$ with $g_j \in B_s$ and $n_j \in \mathbb{Z}[i]$, unique as $s < \frac{1}{2}$; here $n_j = n_{\rho^j}(C) + i n_{-i\rho^j}(C)$. For $-k < j \leq k$ let $\nu_j = g_j - \rho g_{j-1}$. As $\bar{C}\rho^j = \rho \cdot \bar{C}\rho^{j-1}$,

$$\nu_j = (\rho - 1)h + \rho n_{j-1} - n_j,$$

so ν_j depends only on the strip indices of C at ρ^{j-1} , $-i\rho^{j-1}$, ρ^j and $-i\rho^j$. For any $c \in \mathbb{C}$ we define $n_t(c) = n_{\rho^t}(c) + i n_{-i\rho^t}(c)$, $g_t(c) = \bar{c}\rho^t - h - n_t(c)$ and $\nu_t(c) = g_t(c) - \rho g_{t-1}(c) = (\rho - 1)h + \rho n_{t-1}(c) - n_t(c)$ in the same way.

Type points. Let $|t| \leq 1$. If $c_0 \in 65h + 65\mathbb{Z}[i]$, then, as $65\rho^t$ is a Gaussian integer of odd norm, $\bar{c}_0\rho^t \equiv h \pmod{\mathbb{Z}[i]}$ (as in Lemma 8(ii)), and $\nu_0(c_0) = \nu_1(c_0) = 0$. If $c_0 \in \frac{65}{3}(\alpha + \beta i) + 65\mathbb{Z}[i]$,

then $\bar{c}_0 \rho^t \equiv \frac{\alpha - \beta i}{3} u_t \pmod{\mathbb{Z}[i]}$ with $u_t \equiv 65 \rho^t \equiv -(-i)^t \pmod{3}$, as $65 \equiv -1$ and $\rho \equiv -i \pmod{3}$. So $g_t(c_0) = \eta_t/6$, with $\eta_t \in \{\pm 1 \pm i\}$ and $\eta_{t+1} = -i\eta_t$ (as $-ih \equiv h$), and

$$\nu_t = \frac{\eta_t - \rho \eta_{t-1}}{6} = -\frac{(i + \rho) \eta_{t-1}}{6} = -\frac{1 + 3i}{10} \eta_{t-1} \neq 0, \quad \nu_1 = -i\nu_0.$$

Windows. Let $-k < j < k$ and $c = \rho^{-j} C$, so that $\bar{c} = \rho^j \bar{C}$. For $\gamma \in G(1, 1)$ we have $\rho^j \gamma \in G(k, 1)$, so $c \in S^\theta(1, 1)$, and by Proposition 8 c has the strip indices of some type point c_0 . The strip indices of c at ρ^t and $-i\rho^t$ are those of C at ρ^{j+t} and $-i\rho^{j+t}$; so $(\nu_j, \nu_{j+1}) = (\nu_0(c_0), \nu_1(c_0))$, which is $(0, 0)$ or $(\nu, -i\nu)$ with $\nu = -\frac{1+3i}{10}\eta$, $\eta \in \{\pm 1 \pm i\}$.

Pure words. The pairs (ν_j, ν_{j+1}) , $-k < j < k$, cover $\nu_{1-k}, \dots, \nu_k$, and consecutive pairs share a digit. So either all ν_j vanish, or none does and $\nu_{j+1} = -i\nu_j$ for $1 - k \leq j < k$.

Identification. If all digits vanish, then $g_j = \rho^j g_0$ for $|j| \leq k$. Let $x = \bar{g}_0$; then $\bar{x}\rho^j = g_j \in B_s$, so $x \in P_k$, and $\bar{C}\rho^j \equiv h + \bar{x}\rho^j \equiv \overline{Nh + x\rho^j} \pmod{\mathbb{Z}[i]}$ by Lemma 8(ii). Hence $\text{Re}(\bar{w}\gamma) \in \mathbb{Z}$ for $w = C - Nh - x$ and every γ in G_N , so for every γ in the $\mathbb{Z}$ -span of G_N . This span is $\frac{1}{N}\mathbb{Z}[i]$: it is a $\mathbb{Z}[i]$ -module, it contains $\rho^{\pm k} = (2 \pm i)^{2k}/N$, and $(2 + i)^{2k}$ and $(2 - i)^{2k}$ are coprime. With $\gamma = \frac{1}{N}$ and $\frac{i}{N}$ we get $w \in N\mathbb{Z}[i]$.

Otherwise let $\eta_{j-1} = -10\nu_j/(1+3i)$ for $1 - k \leq j \leq k$, and $\eta_k = -i\eta_{k-1}$. By the windows these lie in $\{\pm 1 \pm i\}$, and $\eta_j = -i\eta_{j-1}$; so $\nu_j = (\eta_j - \rho\eta_{j-1})/6$, as above, and $g_j - \eta_j/6 = \rho(g_{j-1} - \eta_{j-1}/6)$. Let $y = g_0 - \eta_0/6$; then $g_j = \eta_j/6 + \bar{y}\rho^j$ for $|j| \leq k$. Choose $\varepsilon \in E_q$ with $\eta_0(\varepsilon) = \eta_0$; the four classes of E_q give the four values, by Lemma 8(iii). As also $\eta_{j+1}(\varepsilon) = -i\eta_j(\varepsilon)$, we get $\eta_j(\varepsilon) = \eta_j$ for $|j| \leq k$. So $y \in Y_k(\varepsilon)$, and $\bar{C}\rho^j \equiv h + \eta_j(\varepsilon)/6 + \bar{y}\rho^j \equiv \overline{N\varepsilon + y\rho^j} \pmod{\mathbb{Z}[i]}$; as before, $C - N\varepsilon - y \in N\mathbb{Z}[i]$.

The bounds are Lemma 9, whose proof holds for every $\theta \in (\frac{1}{4}, \frac{1}{3}]$; as $s < \frac{3}{14}$ and $\delta < \frac{1}{21}$, $r_\theta < 0.23$. $\square$

Proof of Theorem E. Suppose that F^2 has a homomorphism to $K_{p/q}$ with $p/q < 7/2$; we show that $\chi(F^2) \leq 3$. As in the proof of Theorem D, we may assume that $i \notin F$ and $\theta = q/p \in (\frac{2}{7}, \frac{1}{3})$; and $p < 4q$. Let $V \subset T$ be finite and contain a $\mathbb{Q}(i)$ -basis $b_1, \dots, b_n$ of L , write $D_v v = \sum_j \mu_{v,j} b_j$ as in the proof of Lemma 2, and choose $N = 5^k > 6r_\theta \max_v (|D_v| + \sum_j |\mu_{v,j}|)$. Take $U = G(k, 1)V$ in place of $G_N V$: it is finite, symmetric and consists of unit vectors, so Theorem W⁺ gives a character ξ of $\mathbb{Z}U$ with $\xi(U) \subseteq [\theta, 1 - \theta] + \mathbb{Z}$, and the proof of Lemma 2 gives a $\mathbb{Q}(i)$ -linear map $\varphi : L \rightarrow \mathbb{C}$ with $\xi(\gamma v) = \text{Re}(\gamma \varphi(v)) \pmod{1}$ for $v \in V$ and $\gamma \in G(k, 1)$. So $C = \overline{\varphi(v)}$ satisfies the hypothesis of Lemma 12, and $\varphi(v) = N\varepsilon_v + x_v$ with $\varepsilon_v \in E$ (which is closed under conjugation) and $|x_v| \leq r_\theta$. As in the proof of Theorem D, Lemma 1 with r_θ in place of r gives the conclusion of Lemma 2, and the rest of the proof of Theorem B gives (a) or (b); so $\chi(F^2) \leq 3$.

If $\chi(F^2) \geq 4$ and $\chi_c(F^2) < 7/2$, then F^2 has a homomorphism to some $K_{p/q}$ with $p/q < 7/2$, which we have excluded. So $\chi_c(F^2) \geq 7/2$ when $\chi(F^2) \geq 4$; with Corollary 1, $\chi_c(F^2) \in \{2, 3\} \cup [7/2, \infty]$, and the value $7/2$ is attained by Theorem C. $\square$

With [16, Theorem 1], Theorem E gives the lower bounds of Theorem C again; the certificates of Theorem C show more, namely that 70 explicit unit vectors suffice.

The places above 7. Let $\mathbb{F}_{49} = \mathbb{Z}[i]/7$, let $\mu_8 \subset \mathbb{F}_{49}$ be the group of elements of norm one, which are the residues of the rational rotations ($\rho \equiv 2 + 5i$ has order 8, and $\rho^2 \equiv -i$), and let

$$A'_7 = \{e \in \mathbb{F}_{49} : \text{Re}(\bar{e}z) \in \{2, 3, 4, 5\} \text{ for every } z \in \mu_8\},$$

where $\text{Re}(a + bi) = a$ for $a, b \in \mathbb{F}_7$. Elements of $E_7 = \{\varepsilon \in \frac{1}{7}\mathbb{Z}[i] : 7\varepsilon \pmod{7} \in A'_7\}$ have *type 7*; let $E' = E \cup E_7$, so that $42E' \subseteq \mathbb{Z}[i]$. For $\varepsilon \in E_7$ the character $a \mapsto \text{Re}(\bar{\varepsilon}a) \pmod{1}$ of $\mathbb{Z}_{(7)}[i]$, where $\tilde{a} \in \mathbb{Z}[i]$ is congruent to a modulo $7\mathbb{Z}_{(7)}[i]$ (it factors through $\mathbb{F}_{49}$), takes values in $\frac{1}{7}\{2, 3, 4, 5\} + \mathbb{Z}$ at every rational rotation: these are the characters of the 7-adic colourings of Proposition 3. Let $\tilde{\rho} = 2 + 5i$, and for $j \in \mathbb{Z}$ let $\zeta_j(\varepsilon) \in \{\pm \frac{1}{14}, \pm \frac{3}{14}\}^2$ be given by $\bar{\varepsilon}\tilde{\rho}^j \in h + \zeta_j(\varepsilon) + \mathbb{Z}[i]$ (with $\tilde{\rho}^j$ for $j < 0$ read through $\tilde{\rho}^8 \equiv 1 \pmod{7}$).

Lemma 13. (i) $A'_7 = \{e \in \mathbb{F}_{49} : e\bar{e} = -1\}$, that is, A'_7 is the set of the $a + bi$ with $(a, b) \in \{(2, 3), (2, 4), (3, 2), (3, 5), (4, 2), (4, 5), (5, 3), (5, 4)\}$. It is a single μ_8 -orbit, it is closed under conjugation, its elements add up to 0, and $0 \notin A'_7$.

(ii) A'_7 contains no coset of a nonzero additive subgroup of $\mathbb{F}_{49}$.

(iii) $\text{Re}(A'_7) = \{2, 3, 4, 5\}$.

(iv) For $\varepsilon \in E_7$, $\zeta_{j+2}(\varepsilon) = -i\zeta_j(\varepsilon)$, and the digits $\zeta_j(\varepsilon) - \rho\zeta_{j-1}(\varepsilon)$ form a sequence of period 8 in which the digit 0 alternates with digits $u^{\frac{2+i}{5}}$, $u \in \mu_4 \subset \mathbb{C}$; the eight classes of E_7 modulo $\mathbb{Z}[i]$ give its eight shifts, as $\zeta_t(\varepsilon') = \zeta_{t+j}(\varepsilon)$ when $7\varepsilon' \equiv 7\varepsilon\bar{\rho}^j \pmod{7}$ ($\bar{\rho}\bar{\rho} \equiv 1 \pmod{7}$). Two consecutive digits, at given positions, of a sequence of type c (all digits 0), q (the digits of the proof of Lemma 12, all nonzero) or 7 determine its type and its class.

Proof. Write $e = a + bi$ and $z = x + yi$ with $a, b, x, y \in \mathbb{F}_7$. Then $\text{Re}(\bar{e}z) = ax + by$ and $e\bar{e} = a^2 + b^2$, which vanishes only for $e = 0$, as -1 is not a square modulo 7; the nonzero squares modulo 7 are 1, 2 and 4.

(i) The group μ_8 is the conic $x^2 + y^2 = 1$. Let $e \neq 0$ and $t \in \mathbb{F}_7$. The points of the line $ax + by = t$ are $\frac{t}{e\bar{e}}(a, b) + s(-b, a)$, $s \in \mathbb{F}_7$, and such a point lies on the conic if and only if $(e\bar{e})^2 s^2 = e\bar{e} - t^2$. So t is a value of $z \mapsto \text{Re}(\bar{e}z)$ on μ_8 if and only if $e\bar{e} - t^2$ is 0 or a square. Hence $e \in A'_7$ if and only if $e \neq 0$ and neither $e\bar{e}$ nor $e\bar{e} - 1$ is 0 or a square (the values $t = 0$ and $t = \pm 1$), that is, $e\bar{e}$ and $e\bar{e} - 1$ lie in $\{3, 5, 6\}$, which means $e\bar{e} = 6 = -1$. The solutions of $a^2 + b^2 = -1$ have $\{a^2, b^2\} = \{2, 4\}$; they are the eight elements listed. As $(ez)(\overline{ez}) = e\bar{e}$ for $z \in \mu_8$, $A'_7 = e_0\mu_8$ for every $e_0 \in A'_7$. It is closed under conjugation, its elements add up to $e_0 \sum_{z \in \mu_8} z = 0$, as $\mu_8 = -\mu_8$, and $0 \notin A'_7$.

(ii) A nonzero additive subgroup of $\mathbb{F}_{49} \cong \mathbb{F}_7^2$ is a line or all of $\mathbb{F}_{49}$, so each of its cosets contains a set $\{e + sd : s \in \mathbb{F}_7\}$ with $d \neq 0$, which has 7 elements. On it $(e + sd)(\overline{e + sd}) = e\bar{e} + s(\bar{e}d + \bar{e}d) + s^2 d\bar{d}$ is a polynomial of degree 2 in s , as $d\bar{d} \neq 0$; so at most two of these elements lie in A'_7 , and the coset is not contained in it.

(iii) This is read off the list in (i).

(iv) By the list in (i), both coordinates of every element of A'_7 lie in $\{2, 3, 4, 5\}$, and for these c , $\frac{c}{7} - \frac{1}{2} = \frac{2c-7}{14} \in \{\pm\frac{1}{14}, \pm\frac{3}{14}\}$; as $7\bar{\varepsilon}\bar{\rho}^j \pmod{7}$ lies in A'_7 , which is closed under conjugation and stable under μ_8 , $\zeta_j(\varepsilon)$ is defined. As $\bar{\rho}^2 = -21 + 20i \equiv -i \pmod{7}$ and $7\bar{\varepsilon} \in \mathbb{Z}[i]$, $\bar{\varepsilon}\bar{\rho}^{j+2} \equiv -i\bar{\varepsilon}\bar{\rho}^j \pmod{\mathbb{Z}[i]}$; as $-ih \equiv h \pmod{\mathbb{Z}[i]}$ and $\{\pm\frac{1}{14}, \pm\frac{3}{14}\}^2$ is stable under multiplication by $-i$, $\zeta_{j+2}(\varepsilon) = -i\zeta_j(\varepsilon)$. So the digits $\nu_j = \zeta_j(\varepsilon) - \rho\zeta_{j-1}(\varepsilon)$ satisfy $\nu_{j+2} = -i\nu_j$. For $7\varepsilon \equiv 2 + 3i$ we have $7\bar{\varepsilon} \equiv 2 + 4i$, $(2 + 4i)\bar{\rho} \equiv 5 + 4i$ and $(2 + 4i)\bar{\rho}^2 \equiv 4 + 5i \pmod{7}$, so $\zeta_0 = \frac{-3+i}{14}$, $\zeta_1 = \frac{3+i}{14}$ and $\zeta_2 = \frac{1+3i}{14}$; as $\rho\zeta_0 = -\frac{13+9i}{70}$ and $\rho\zeta_1 = \frac{1+3i}{14}$, we get $\nu_1 = \frac{2+i}{5}$ and $\nu_2 = 0$. Hence $\nu_{2m} = 0$ and $\nu_{2m+1} = (-i)^m \frac{2+i}{5}$ for every m , a sequence of period 8 of the stated form. If $7\varepsilon' \equiv 7\varepsilon\bar{\rho}^j \pmod{7}$, then $\bar{\varepsilon}'\bar{\rho}^t \equiv \bar{\varepsilon}\bar{\rho}^{t+j} \pmod{\mathbb{Z}[i]}$, so $\zeta_t(\varepsilon') = \zeta_{t+j}(\varepsilon)$; as $\bar{\rho} \equiv \bar{\rho}^{-1}$ generates μ_8 and A'_7 is one μ_8 -orbit, the eight classes of E_7 give the eight shifts $j = 0, \dots, 7$. Finally, two consecutive digits are both 0 for type c, both nonzero for type q, and one 0 and one nonzero for type 7. For type q the class is the sequence (η_t) , with $\eta_{t+1} = -i\eta_t$, and one digit $\nu_j = -\frac{1+3i}{10}\eta_{j-1}$ determines it, as in the proof of Lemma 12. For type 7, the position of the nonzero digit $(-i)^m \frac{2+i}{5}$ and m modulo 4 determine the shift modulo 8, hence the class. $\square$

The set $S^\theta(2, 1)$ is invariant under $325\mathbb{Z}[i]$, and its *type points* are the points of $325h + 325\mathbb{Z}[i]$ (type c), of $\frac{325}{3}(\alpha + \beta i) + 325\mathbb{Z}[i]$ with $\alpha, \beta \in \{1, 2\}$ (type q), and of $\frac{325}{7}(a + bi) + 325\mathbb{Z}[i]$, $0 \leq a, b \leq 6$, with $3(a + bi) \pmod{7} \in A'_7$ (type 7; eight classes). As $325 \equiv 3 \pmod{7}$, a point of type 7 has at ρ^t , $|t| \leq 2$, the values of the character of an element of E_7 ; its least margin is $\frac{2}{7}$. Strip indices are floors, so they are defined at these points also for $\theta > \frac{2}{7}$.

Proposition 9 (computer-assisted). *Let $\frac{1}{4} < \theta \leq \frac{1}{3}$ and $c \in S^\theta(2, 1)$. Then there is a type point c_0 , of type c, q or 7, such that c and c_0 have the same strip index at every $\gamma \in G(2, 1)$. If $\theta > \frac{2}{7}$, then c_0 has type c or q.*

Proof. As for Proposition 8, with $r_0 = \frac{249}{1000} < \frac{1}{4}$ and the 30 functionals of the 15 rotations $\rho^j \sigma^l$, $|j| \leq 2$, $|l| \leq 1$: the tree over the 325^2 cells has 550 474 branch nodes, 199 920 closed branches

and 29 leaves. Of the 29 index vectors, five are those of the points of type c and q, eight those of the points of type 7, and each of the other 16 has a certificate $\lambda \geq 0$, $\sum \lambda_i = 1$, $\sum \lambda_i \psi_i = \frac{1}{4}$ on three margins, which excludes it for $\theta > \frac{1}{4}$. Each of the eight vectors of type 7 has a certificate of the same kind, with three positive multipliers and $\sum \lambda_i \psi_i = \frac{2}{7}$ identically, which excludes it for $\theta > \frac{2}{7}$; these certificates use margins at the rotations $\rho^{-2}\sigma^l$, outside $G(1, 1)$. $\square$

The 16 excluded vectors are 2-adic: their largest least margin $\frac{1}{4}$ is attained at 4- and 8-torsion points such as $\frac{325}{4}(1+i)$ and $\frac{325}{8}(2+i)$. So $\frac{1}{4}$ is a limit of the method, consistent with $\chi_c(\mathbb{Q}(\sqrt{59})^2) = 4$. The window $G(1, 1)$ does not suffice below $\frac{2}{7}$: for $\theta = 0.2501$ it has other components, with largest least margins $\frac{11}{42}$ and $\frac{25}{96}$.

Lemma 14. *Let $\frac{1}{4} < \theta \leq \frac{1}{3}$, let k be a positive multiple of 6 (so that $N = 5^k \equiv 1 \pmod{7}$), and let $C \in \mathbb{C}$ satisfy $\text{Re}(\bar{C}\gamma) \in [\theta, 1 - \theta] + \mathbb{Z}$ for every $\gamma \in G(k, 1)$. Then $C \in N\varepsilon + P_k + N\mathbb{Z}[i]$ for some $\varepsilon \in E_c$, or $C \in N\varepsilon + Y_k(\varepsilon) + N\mathbb{Z}[i]$ for some $\varepsilon \in E_q$, or $C \in N\varepsilon + Y_k^7(\varepsilon) + N\mathbb{Z}[i]$ for some $\varepsilon \in E_7$, where*

$$Y_k^7(\varepsilon) = \{x \in \mathbb{C} : \zeta_j(\varepsilon) + \bar{x}\rho^j \in B_s \text{ for } |j| \leq k\};$$

the third case does not occur if $\theta > \frac{2}{7}$. In every case $C = N\varepsilon + x$ with $\varepsilon \in E'$ and $|x| \leq r' = (\frac{1}{4} + \frac{3}{14})\sqrt{2} < 0.66$.

Proof. Define g_j , n_j and the digits ν_j as in the proof of Lemma 12. For $|j| \leq k - 2$ the point $c = \rho^{-j}C$ lies in $S^\theta(2, 1)$, as $\rho^j G(2, 1) \subseteq G(k, 1)$; by Proposition 9 the digits $\nu_{j-1}, \nu_j, \nu_{j+1}, \nu_{j+2}$ of C are those of a type point at $t = -1, 0, 1, 2$. These are 0 for type c, as $325\rho^t \in \mathbb{Z}[i]$ has odd norm; $(\nu, -i\nu, -\nu, i\nu)$ with $\nu \neq 0$ for type q, as in the proof of Lemma 12, now with $325 \equiv 1 \pmod{3}$; and four consecutive digits of the sequence of Lemma 13(iv) for type 7, as the values of such a point at ρ^t are those of a character of type 7. The windows cover $\nu_{1-k}, \dots, \nu_k$, and consecutive windows share three digits, which are, in each window, at given positions. By Lemma 13(iv) two of them determine the type and the class of each window: if the window at j has type 7 and class ε_j (its digits are those of ε_j at the positions $-1, 0, 1, 2$), then the window at $j+1$ has type 7 and the class ε_{j+1} with $7\varepsilon_{j+1} \equiv 7\varepsilon_j \bar{\rho} \pmod{7}$, and likewise for the types c and q. So all digits of C are of one type: all 0, all of type q, or $\nu_j = \zeta_j(\varepsilon) - \rho\zeta_{j-1}(\varepsilon)$ for $1 - k \leq j \leq k$ and one class $\varepsilon \in E_7$ (the class of the window at 0); by Proposition 9, the last case does not occur if $\theta > \frac{2}{7}$. In the first two cases the identification is that of Lemma 12. In the third, $N\rho^j \in \mathbb{Z}[i]$ and $N\rho^j \equiv \bar{\rho}^j \pmod{7}$ for $|j| \leq k$, as $N \equiv 1 \pmod{7}$; so $\bar{N}\varepsilon\rho^j = \bar{\varepsilon}N\rho^j \in h + \zeta_j(\varepsilon) + \mathbb{Z}[i]$. As the digits of C are $\zeta_j(\varepsilon) - \rho\zeta_{j-1}(\varepsilon)$, $g_j - \zeta_j(\varepsilon) = \rho(g_{j-1} - \zeta_{j-1}(\varepsilon))$, and with $x = g_0 - \zeta_0(\varepsilon)$ we get $g_j = \zeta_j(\varepsilon) + \bar{x}\rho^j$ for $|j| \leq k$. So $x \in Y_k^7(\varepsilon)$, $\bar{C}\rho^j \equiv N\varepsilon + \bar{x}\rho^j \pmod{\mathbb{Z}[i]}$, and $C - N\varepsilon - x \in N\mathbb{Z}[i]$ as before. Finally $|x|$ is at most $s\sqrt{10}/3$, $\sqrt{2}\delta$, resp. $|g_0| + |\zeta_0(\varepsilon)| \leq \sqrt{2}(s + \frac{3}{14})$, each less than r' as $s < \frac{1}{4}$. $\square$

Lemma 15. *Let F be a number field with $i \notin F$, and suppose that F^2 has a homomorphism to $K_{p/q}$ with $3 \leq p/q < 4$. Then for every finite $V \subset T$ there is $\beta \in L$ with $\text{Tr}(\beta v) \in E'$ for every $v \in V$.*

Proof. As in the proof of Theorem E, with $\theta = q/p \in (\frac{1}{4}, \frac{1}{3}]$, $6 \mid k$, $N > 42r' \max_v(|D_v| + \sum_j |\mu_{v,j}|)$ and Lemma 14 in place of Lemma 12; E' is closed under conjugation by Lemma 13(i), and Lemma 1 holds with 42 in place of 6, as its proof uses only $42E' \subseteq \mathbb{Z}[i]$ and the bound on $|x|$. $\square$

Let $L_7 = L \otimes_{\mathbb{Q}} \mathbb{Q}_7 = \prod_{u|7} L_u$ and $T_7 = \prod_{u|7} T(F_u)$. As 7 is inert in $\mathbb{Q}(i)$, $\mathbb{Q}(i) \otimes \mathbb{Q}_7 = \mathbb{Q}_{49}$, the unramified quadratic extension of $\mathbb{Q}_7$, and the trace extends to $\text{Tr} : L_7 \rightarrow \mathbb{Q}_{49}$. Let $O_7 = \mathbb{Z}_{49}$ and $\mathcal{A}_7 = \{\frac{e}{7} : e \in O_7, e \bmod 7 \in \mathcal{A}'_7\}$; then $E \subseteq O_7$ and $E_7 \subseteq O_2 \cap O_3 \cap \mathcal{A}_7$.

Lemma 16. *Under the hypotheses of Lemma 15, (c) or (q) of Lemma 4 holds, or*

(7') there is $\beta_7 \in L_7$ with $\text{Tr}(\beta_7 z) \in \mathcal{A}_7$ for every $z \in T_7$.

Proof. As the proof of Lemma 4, with a third prime. Let $K_7 = \{\beta \in L_7 : \text{Tr}(\beta b_j) \in \frac{1}{7}O_7 \forall j\}$, which is compact. On $\{c, q, 7\}^T \times K_2 \times K_3 \times K_7$ let $P_v(\tau, \beta_2, \beta_3, \beta_7)$ be the condition

$$\begin{aligned} \tau(v) = c, \quad v_{1+i}(\text{Tr}(\beta_2 v)) = -1, \quad \text{Tr}(\beta_3 v) \in O_3, \quad \text{Tr}(\beta_7 v) \in O_7; \text{ or} \\ \tau(v) = q, \quad \text{Tr}(\beta_2 v) \in O_2, \quad \text{Tr}(\beta_3 v) \in \mathcal{A}, \quad \text{Tr}(\beta_7 v) \in O_7; \text{ or} \\ \tau(v) = 7, \quad \text{Tr}(\beta_2 v) \in O_2, \quad \text{Tr}(\beta_3 v) \in O_3, \quad \text{Tr}(\beta_7 v) \in \mathcal{A}_7. \end{aligned}$$

These conditions are clopen, Lemma 15 and the local properties of E_c , E_q and E_7 give the finite intersection property, and so some $(\tau, \beta_2, \beta_3, \beta_7)$ satisfies P_v for every $v \in T$. With a, a', b, b' as in the proof of Lemma 4, let $d(z_7)$ and $d'(z_7)$, for $z_7 \in T_7$, be the statements $\text{Tr}(\beta_7 z_7) \in O_7$ and $\text{Tr}(\beta_7 z_7) \in \mathcal{A}_7$; they exclude each other, as $0 \notin \mathcal{A}'_7$. The set of $(z_2, z_3, z_7) \in T_2 \times T_3 \times T_7$ with $(a(z_2) \wedge b(z_3) \wedge d(z_7)) \vee (a'(z_2) \wedge b'(z_3) \wedge d(z_7)) \vee (a'(z_2) \wedge b(z_3) \wedge d'(z_7))$ is clopen and contains the diagonal image of T , which is dense by Lemma 3; so it is everything. If $a(z_0)$ holds for some z_0 , then b and d hold everywhere, so b' and d' hold nowhere, and a holds everywhere: this is (c). Otherwise a' holds everywhere. If $b'(y_0)$ holds for some y_0 , then d holds everywhere, so d' holds nowhere, and b' holds everywhere: this is (q). Otherwise b holds everywhere, and then d' holds everywhere: this is (7'). $\square$

In case (7') let u be a place of F above 7, f its residue degree, $q = 7^f$, and $g_u(z) = \text{Tr}_{L_u/\mathbb{Q}_{49}}(\beta_u z)$, where $\beta_7 = (\beta_u)_u$; then $\text{Tr}(\beta_7 y) = \sum_u g_u(y_u)$ for $y \in T_7$. If f is even, then $i \in F_u$ and $g_u = 0$, exactly as for even f in Section 5 (with 7^m in place of 3^m). If f is odd, $L_u = F_u(i) = F_u\mathbb{Q}_{49}$ is a field, unramified of degree 2 over F_u , with residue field $\mathbb{F}_{q^2}$; its non-trivial automorphism acts on $\mathbb{Q}_{49}$ as the Frobenius; $T(F_u)$ lies in $O_{L_u}^\times$, contains the Teichmüller lifts of μ_{q+1} , among them $\mu_8 \subset \mathbb{Q}_{49}$ (as $8 \mid q+1$), and reduces onto μ_{q+1} ; and $g_u(\omega z) = \omega g_u(z)$ for $\omega \in \mu_8$.

Lemma 17 (one place). *In case (7') there are a place $u_0 \mid 7$ of odd residue degree and $\beta \in L_{u_0}$ with $\text{Tr}_{L_{u_0}/\mathbb{Q}_{49}}(\beta z) \in \mathcal{A}_7$ for every $z \in T(F_{u_0})$.*

Proof. As the proof of Lemma 5. For a place u of odd residue degree and $z, z' \in T(F_u)$, varying y_u alone gives $g_u(z) - g_u(z') \in \mathcal{A}_7 - \mathcal{A}_7 \subseteq \frac{1}{7}O_7$; with $z' = 1$ and $z = i$, as $i - 1$ is a unit at 7, $g_u(T(F_u)) \subseteq \frac{1}{7}O_7$. Let $Z_u \subseteq \mathbb{F}_{49}$ be the set of residues of $7g_u(z)$, $z \in T(F_u)$; it is stable under μ_8 , and $\sum_u z_u \in \mathcal{A}'_7$ for every choice of $z_u \in Z_u$. Suppose two places $u \neq u'$ had nonzero elements $w \in Z_u$ and $w' \in Z_{u'}$. Fix $z_{u''} \in Z_{u''}$ at the other places and put $e_0 = \omega' w' + \sum z_{u''}$ with $\omega' \in \mu_8$. The eight distinct elements $e_0 + \omega w$, $\omega \in \mu_8$, lie in $\mathcal{A}'_7$, which has eight elements; so they are all of $\mathcal{A}'_7$, and summing them gives $8e_0 = e_0 = \sum \mathcal{A}'_7 = 0$ (Lemma 13(i)). As this holds for every $\omega' \in \mu_8$, $w' = 0$: a contradiction. Since $0 \notin \mathcal{A}'_7$, exactly one place u_0 has $Z_{u_0} \neq \{0\}$, and it has odd residue degree; the other g_u take values in O_7 , and $\beta = \beta_{u_0}$ works, because $\mathcal{A}_7 + O_7 = \mathcal{A}_7$. $\square$

Lemma 18 (level one). *Let $u \mid 7$ have odd residue degree f , $q = 7^f$, and let $\beta \in L_u$ satisfy $\text{Tr}_{L_u/\mathbb{Q}_{49}}(\beta z) \in \mathcal{A}_7$ for every $z \in T(F_u)$. Then $\lambda(x) = 7 \text{Tr}_{L_u/\mathbb{Q}_{49}}(\beta x) \bmod 7$ is defined on O_{L_u} and vanishes on its maximal ideal, and the induced $\mathbb{F}_{49}$ -linear map $\lambda_1 : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_{49}$ satisfies $\lambda_1(\mu_{q+1}) \subseteq \mathcal{A}'_7$.*

Proof. The proof of Lemma 6 applies with 7, $\mathbb{Q}_{49}$, $\mathbb{F}_{49}$ and $\mathcal{A}'_7$ in place of 3, $\mathbb{Q}_9$, $\mathbb{F}_9$ and $\mathcal{A}'$ (there σ is the automorphism of L_u/F_u): it uses that the prime is odd, that the order of 7 modulo $q+1$ is $2f$, and, at the end, Lemma 13(ii) in place of the fact that $\mathcal{A}'$ contains no coset of a nonzero subgroup of $\mathbb{F}_9$. $\square$

Lemma 19 (residue fields). *Let f be odd and $q = 7^f$. If some $\mathbb{F}_{49}$ -linear map $\lambda_1 : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_{49}$ satisfies $\lambda_1(\mu_{q+1}) \subseteq \mathcal{A}'_7$, then $f = 1$.*

Proof. By Lemma 13(iii), $\text{Re} \circ \lambda_1 : \mathbb{F}_{q^2} \rightarrow \mathbb{F}_7$ is additive and takes values in $\{2, 3, 4, 5\}$ on μ_{q+1} ; read in $\frac{1}{7}\mathbb{Z}/\mathbb{Z}$, it stays at distance at least $\frac{2}{7}$ from 0 there. So $\kappa_1 > 0$ for the unramified extension of $\mathbb{Q}_7$ of degree f , and Proposition 6 gives $f = 1$. $\square$

Proof of Theorem F. If (a) holds, $\chi_c(F^2) = \chi(F^2) = 2$. If (b) holds and (a) fails, $\chi(F^2) = 3$ by Theorem B, so $\chi_c(F^2) = 3$ by Corollary 1. If (7) holds and (a), (b) fail, $\chi(F^2) \geq 4$ by Theorem B,

so $\chi_c(F^2) \geq 7/2$ by Theorem E (which also follows from Proposition 9; Remark (1) below), and $\chi_c(F^2) \leq 7/2$ by Proposition 3, as a place above 7 of residue degree 1 has residue field $\mathbb{F}_7$.

It remains to show that if F^2 has a homomorphism to $K_{p/q}$ with $p/q < 4$, then (a), (b) or (7) holds; then $\chi_c(F^2) \geq 4$ when all three fail. If $i \in F$ there is no such homomorphism, as $\chi(F^2) = \infty$. If $p/q \leq 3$, then $\chi(F^2) \leq 3$, and (a) or (b) holds by Theorem B. Otherwise $3 < p/q < 4$, and by Lemmas 15 and 16 we are in case (c), (q) or (7'). Cases (c) and (q) give (a) or (b) as in the proof of Theorem B. In case (7'), Lemmas 17, 18 and 19 give a place above 7 of residue degree 1: this is (7). $\square$

Corollary 6 (local fields). *Let K be a finite extension of $\mathbb{Q}_p$. Then $\chi_c(K^2) = 2$ if $p = 2$ and $K(i)/K$ is ramified; $\chi_c(K^2) = 3$ if $p = 3$ and K has residue degree 1; $\chi_c(K^2) = 7/2$ if $p = 7$ and K has residue degree 1; and $\chi_c(K^2) \geq 4$ in every other case. Consequently, for every number field F ,*

$$\min(\chi_c(F^2), 4) = \min\left(4, \min_v \chi_c(F_v^2)\right),$$

the inner minimum over all places v of F : below 4, the circular chromatic number of the plane over a number field is the least circular chromatic number of the planes over its completions.

Proof. In the first three cases the upper bounds are the colourings of Corollary 3 (for $p = 2$ and $p = 3$) and of Proposition 3, which colour K^2 itself. The lower bounds: a bipartite plane with an edge has $\chi_c = 2$; for $p = 3$, $K \supseteq \mathbb{Q}_3 \supset \mathbb{Q}(\sqrt{7})$ and $\chi_c(\mathbb{Q}(\sqrt{7})^2) = 3$ by Corollary 1, as $\chi(\mathbb{Q}(\sqrt{7})^2) = 3$; for $p = 7$, Corollary 4. In the other cases, if $i \in K$ then $\chi(K^2) = \infty$. Otherwise we find, as in the proof of Corollary 3, a number field $F \subset K$ for which (a), (b) and (7) fail; then $\chi_c(K^2) \geq \chi_c(F^2) \geq 4$ by Theorem F. The construction there works with $\mathbb{Q}_7$ added to the primes p, p' (and with 7 in the role of p when $p = 7$): over $\mathbb{Q}_7$ take a product of distinct irreducible quadratics whose roots generate $\mathbb{Q}_{49}$, so that the places of F above 7 contain i ; over $\mathbb{Q}_p$ take $h_K h'$ as there, where for $p = 7$ the factor h' consists of such quadratics and, if $[K : \mathbb{Q}_7]$ is odd, one cubic factor generating the unramified cubic extension of $\mathbb{Q}_7$; and choose the auxiliary prime of the irreducibility condition outside $\{2, 3, 7, p\}$. If $p \notin \{2, 3, 7\}$, take over $\mathbb{Q}_p$ the polynomial h_K times distinct linear factors coprime to it, to adjust the degree. Then every place of F above 2 contains i or is the place giving K , where $K(i)/K$ is unramified; every place above 3 contains i or has residue degree at least 3; and every place above 7 contains i or has odd residue degree at least 3.

For the last statement, $F^2 \subseteq F_v^2$ gives $\chi_c(F^2) \leq \chi_c(F_v^2)$ for every v . If $\chi_c(F^2) < 4$, Theorem F gives (a), (b) or (7), with $\chi_c(F^2) = 2, 3$ or $7/2$ accordingly, and the completion at the corresponding place has the same value by the first part. $\square$

For example $\chi_c(\mathbb{Q}_{19}^2)$ and $\chi_c(\mathbb{Q}(\sqrt{47})^2)$ lie in $[4, 19/4]$: the lower bound is Corollary 6, resp. Theorem F, and the upper bound is Proposition 3 at 19 with $m = 4$, as $\lambda(a + bi) = 4a + 5b$ takes its values in $\{4, \dots, 15\}$ on $\mu_{20} \subset \mathbb{F}_{361}$ (and $\kappa_1 = 4/19$; as $47 \equiv 3^2 \pmod{19}$, $\sqrt{47} \in \mathbb{Q}_{19}$). The same holds for $\mathbb{Q}(\sqrt{5}, \sqrt{7})$. So if $\chi(\mathbb{Q}(\sqrt{47})^2) = 5$, which is open [16, Corollary 2], its circular chromatic number lies in $(4, 19/4]$ and is not an integer.

Remark. (1) Theorem E also follows from Proposition 9: for $\theta > \frac{2}{7}$ it leaves only the types c and q, so Lemma 14 (with $6 \mid k$) gives the conclusion of Lemma 12, and the proof of Theorem E goes through. The certificates that exclude the vectors of type 7 there use rotations outside $G(1, 1)$, so they are not those of Proposition 8. Hence Theorem F rests on Proposition 9 alone, and Theorem E on either proposition.

(2) A second prime is needed. Besides the 5-adic points of Remark (2) in Section 9, the one-prime probe has torsion characters $a \mapsto \text{Re}(\bar{c}\tilde{a})$ of $\mathbb{Z}[\frac{1}{5}][i]$, where $c \in \frac{1}{M}\mathbb{Z}[i]$, $5 \nmid M$, and $\tilde{a}$ is the residue of a modulo M . For $M \leq 400$, those other than the characters of type c and q that keep every $i^a \rho^j$ at margin greater than $\frac{2}{7}$ have order 41 (largest least margin $\frac{12}{41}$, at $c = \frac{12+12i}{41}$; ρ has order 5 modulo 41) or 76 (largest least margin $\frac{11}{38}$). For $M \leq 300$ prime to 65 (and $M \leq 400$, by a second program), no character of this kind other than those of type c and q keeps every element of $\langle i, \rho, \sigma \rangle$ at margin greater than $\frac{2}{7}$; and for $k \leq 5$ every translate $c_k + Nm$ of the points c_k of that Remark has least margin at most $\frac{41}{250}$ on $G(k, 1)$.

Finite witnesses. By compactness, $\chi_c(F^2)$ is the supremum of $\chi_c(H)$ over the finite subgraphs H of F^2 : if every finite subgraph has a homomorphism to $K_{p/q}$, so does F^2 . Below 4 the supremum is attained (Corollary 7), and at 4 as well (Theorem G). In this subsection s denotes an element of a connection set, not the number $\frac{1}{2} - \theta$. For a homomorphism c of a graph to $K_{p/q}$, the *tight digraph* D_c has an arc $x \rightarrow y$ for every edge xy with $c(y) - c(x) \equiv q \pmod{p}$, and a *tight cycle* of c is a directed cycle of D_c . For a finite set S in an abelian group let $\kappa(S) = \max_{\xi} \min_{s \in S} \|\xi(s)\|$, over the characters ξ of $\mathbb{Z}S$ (a maximum, as the dual group is compact); if $\chi_c(\text{Cay}(\mathbb{Z}S, S)) < 4$, then $\chi_c(\text{Cay}(\mathbb{Z}S, S)) = 1/\kappa(S)$ [11, Corollary 2]. The next lemma is the easy half of a characterisation due to Guichard [17] (cf. [30]); we include its short proof.

Lemma 20. *Let H be a finite graph with a homomorphism to $K_{p/q}$. If every homomorphism of H to $K_{p/q}$ has a tight cycle, then $\chi_c(H) = p/q$.*

Proof. Suppose that $\chi_c(H) < p/q$. By definition H has a homomorphism c' to some $K_{p'/q'}$ with $p'/q' < p/q$. For $x \in V(H)$ let $g(x) = \frac{p}{p'}c'(x) \in [0, p]$, reading $c'(x)$ in $\{0, \dots, p' - 1\}$, and $c(x) = \lfloor g(x) \rfloor \pmod{p}$. For an edge xy let $D \in [\frac{pq'}{p'}, p - \frac{pq'}{p'}]$ be the lift of $g(y) - g(x)$ modulo p ; as $\frac{pq'}{p'} > q$, we have $q < D < p - q$. The integer $\lfloor g(x) + D \rfloor - \lfloor g(x) \rfloor$ is $\lfloor D \rfloor$ or $\lceil D \rceil$, so it lies in $[q, p - q]$, and it is congruent to $c(y) - c(x)$; so c is a homomorphism to $K_{p/q}$, and the arc $x \rightarrow y$ is tight exactly when this integer is q . Along a closed walk these integers have the same sum as the lifts D , as the floors telescope. On a tight cycle of length m the first sum is mq , while the second is larger than mq . So c has no tight cycle. $\square$

Lemma 21. *Let Γ be an abelian group, $S = -S \subseteq \Gamma$ finite with $\Gamma = \mathbb{Z}S$, $p < 4q$, and c a homomorphism of $\text{Cay}(\Gamma, S)$ to $K_{p/q}$. For $x \in \Gamma$ and $s \in S$ let $\ell(x, s) \in [q, p - q]$ be the integer congruent to $c(x + s) - c(x)$ modulo p , and $a(s) = M_x \ell(x, s)$ for an invariant mean M on Γ . Then $a(-s) = p - a(s)$, and $s_1 + \dots + s_m \mapsto \frac{1}{p} \sum_k a(s_k) \pmod{1}$ ($s_k \in S$) is a well-defined character ξ of Γ , with $\xi(s) = a(s)/p \in [\frac{q}{p}, 1 - \frac{q}{p}]$ for $s \in S$. If $u_1, \dots, u_r \in S$ and positive integers $n_1, \dots, n_r$ satisfy $\sum_i n_i u_i = 0$ and $a(u_i) = q$ for every i , then every closed walk in $\text{Cay}(\Gamma, S)$ with n_i steps u_i is a closed walk in D_c , and so c has a tight cycle.*

Proof. The first part is the proof of Theorem W⁺ in [11]. For $s, t \in S$ the integers $\ell(x, s) + \ell(x + s, t)$ and $\ell(x, t) + \ell(x + t, s)$ are congruent modulo p and lie in $[2q, 2p - 2q]$, an interval of length $2p - 4q < p$; so they are equal. For a closed walk W with steps $s_1, \dots, s_m$ started at x , with $x_k = x + s_1 + \dots + s_k$, the sum $\Lambda(W, x) = \sum_k \ell(x_{k-1}, s_k)$ is a multiple of p , and for $t \in S$ the identity gives $\Lambda(W, x + t) - \Lambda(W, x) = \sum_k (\ell(x_k, t) - \ell(x_{k-1}, t)) = 0$. As S generates Γ , $\Lambda(W, x)$ does not depend on x , so by invariance of M it equals $\sum_k a(s_k)$. As $\ell(x + s, -s) = p - \ell(x, s)$, invariance also gives $a(-s) = p - a(s)$. So ξ is well defined: two expressions of the same element give a closed walk. For the walk of the statement, $\Lambda(W, x) = q \sum_i n_i$ for every x , while each of its $\sum_i n_i$ terms is at least q ; so every term is q . A closed walk in D_c contains a directed cycle. $\square$

Lemma 22 ([11, Corollary 16]). *Let Γ be an abelian group, $S = -S \subseteq \Gamma$ finite with $\Gamma = \mathbb{Z}S$, and suppose that $\chi_c(\text{Cay}(\Gamma, S)) = p/q$ with $2 < p/q < 4$. Then every homomorphism of $\text{Cay}(\Gamma, S)$ to $K_{p/q}$ has a tight cycle, and some finite subgraph of $\text{Cay}(\Gamma, S)$ has circular chromatic number p/q .*

Proof. Here $\kappa(S) = q/p$, so $\text{Cay}(\Gamma, S)$ has a homomorphism to $K_{p/q}$. Let c be one, and a, ξ as in Lemma 21; let $S_q = \{s \in S : a(s) = q\}$. Suppose that no nonzero nonnegative integers n_t give $\sum_{t \in S_q} n_t t = 0$. Then the images $\bar{t}$ of the elements of S_q in $\Gamma \otimes \mathbb{R}$ have no nonzero nonnegative real relation: in a basis of $\Gamma \otimes \mathbb{Q}$ the cone of such relations is cut out by rational equations, so if it were nonzero it would contain a nonzero rational, hence integral, point (n_t) , and then $\sum n_t t$ would be a torsion element, killed by some $m \geq 1$. By Gordan's theorem there is a linear form on $\Gamma \otimes \mathbb{R}$, hence a homomorphism $h : \Gamma \rightarrow \mathbb{R}$, with $h(t) > 0$ for every $t \in S_q$. For $\varepsilon > 0$, $\xi_\varepsilon = \xi + \varepsilon h \pmod{1}$ is a character of Γ , and for $s \in S$: if $a(s) = q$, then $\xi_\varepsilon(s) = \frac{q}{p} + \varepsilon h(s)$; if $a(s) = p - q$, then $-s \in S_q$ and $\xi_\varepsilon(s) = 1 - \frac{q}{p} - \varepsilon h(-s)$; otherwise $\xi(s)$ lies in the open interval $(\frac{q}{p}, 1 - \frac{q}{p})$, which is nonempty as $p > 2q$. As S is finite, for small ε all values $\xi_\varepsilon(s)$ lie in that

open interval, so $\kappa(S) > q/p$: a contradiction. So such n_t exist, and Lemma 21, applied to the t with $n_t > 0$, gives a tight cycle of c .

If every finite subgraph of $\text{Cay}(\Gamma, S)$ had a homomorphism to $K_{p/q}$ without tight cycles, so would $\text{Cay}(\Gamma, S)$: for each finite $H \subset \Gamma$ the maps $\Gamma \rightarrow \mathbb{Z}/p$ whose restriction to H is such a homomorphism of the subgraph induced on H form a nonempty closed subset of the compact space $(\mathbb{Z}/p)^\Gamma$; these subsets have the finite intersection property, and a tight cycle is finite. So some finite subgraph H has a tight cycle for every homomorphism to $K_{p/q}$, and $\chi_c(H) = p/q$ by Lemma 20, as $\chi_c(H) \leq p/q$. $\square$

Corollary 7 (finite witnesses). *If $\chi_c(F^2) < 4$, then some finite subgraph of F^2 has circular chromatic number $\chi_c(F^2)$. If $\chi_c(F^2) \in \{3, 7/2\}$, there is a finite set $U \subset T$ with $\chi_c(\text{Cay}(\mathbb{Z}U, U)) = \chi_c(F^2)$, and the subgraph can be taken in $\text{Cay}(\mathbb{Z}U, U)$.*

Proof. By Theorem F the value is 2, 3 or $7/2$, and $i \notin F$, as otherwise $\chi(F^2) = \infty$. For 2 an edge is a witness. Fix a $\mathbb{Q}(i)$ -basis $b_1, \dots, b_n \subset T$ of L as in the proof of Lemma 4. It is enough to find a finite $U \subset T$ with $\chi_c(\text{Cay}(\mathbb{Z}U, U)) = \chi_c(F^2)$, as Lemma 22 then gives the witness inside $\text{Cay}(\mathbb{Z}U, U) \subseteq F^2$.

Let $\chi_c(F^2) = 7/2$, so that (7) holds and (a) and (b) fail. If for every finite $V \supseteq \{b_j\}$ there were $\beta \in L$ with $\text{Tr}(\beta v) \in E$ for every $v \in V$, the proof of Lemma 4 (the images of β lie in K_2 and K_3 , and the types of the $\text{Tr}(\beta v)$ give τ) and the proof of Theorem B would give (a) or (b). So there is a finite $V_0 \supseteq \{b_j\}$ for which no such β exists. Let k , with $6 \mid k$, and $N = 5^k$ be as in the proof of Lemma 15 for V_0 (they do not depend on θ), and $U_0 = G(k, 1)V_0$. If $\kappa(U_0) > \frac{2}{7}$, take θ with $\frac{2}{7} < \theta \leq \min(\kappa(U_0), \frac{1}{3})$ and a character ξ of $\mathbb{Z}U_0$ with $\xi(U_0) \subseteq [\theta, 1 - \theta] + \mathbb{Z}$. The proof of Lemma 15, in which the third case of Lemma 14 does not occur as $\theta > \frac{2}{7}$, gives $\beta \in L$ with $\text{Tr}(\beta v) \in E$ for every $v \in V_0$: impossible. So $\kappa(U_0) \leq \frac{2}{7}$. The homomorphism of F^2 to $K_{7/2}$ of Proposition 3 restricts to $\text{Cay}(\mathbb{Z}U_0, U_0)$, so by Theorem W⁺, $\kappa(U_0) \geq \frac{2}{7}$. Hence $\kappa(U_0) = \frac{2}{7}$ and $\chi_c(\text{Cay}(\mathbb{Z}U_0, U_0)) = \frac{7}{2}$.

Let $\chi_c(F^2) = 3$, so that (b) holds and (a) fails (Corollary 1, Theorem B). As in the proof of Corollary 1, if for every finite $V \supseteq \{b_j\}$ there were $\beta \in L$ with $\text{Tr}(\beta v) \in E_c$ for every $v \in V$, the conditions P_v of the proof of Lemma 4 could all be taken with $\tau(v) = c$, and (c), hence (a), would hold. So there is a finite $V_0 \supseteq \{b_j\}$ for which no such β exists. Let N be as in the proof of Lemma 2 for V_0 , and $U_0 = G_N V_0$. If $\kappa(U_0) > \frac{1}{3}$, a character with $\xi(U_0) \subseteq [\theta, 1 - \theta] + \mathbb{Z}$, $\frac{1}{3} < \theta \leq \kappa(U_0)$, gives as in the proof of Corollary 1 points $\varphi(v) = N\varepsilon_v + x_v$ with every ε_v of type c (a point of type q would have real part in $\{\frac{1}{3}, \frac{2}{3}\} + \mathbb{Z}$, as $3 \nmid N$), and then β with $\text{Tr}(\beta v) = \varepsilon_v \in E_c$ for every $v \in V_0$: impossible. So $\kappa(U_0) \leq \frac{1}{3}$; as F^2 is 3-colourable, $\kappa(U_0) \geq \frac{1}{3}$ by Theorem W⁺, and $\chi_c(\text{Cay}(\mathbb{Z}U_0, U_0)) = 3$. $\square$

The proof uses Proposition 9 only for $\theta > \frac{2}{7}$, through Lemma 14, and not the certificates of value $\frac{2}{7}$ themselves. For $\mathbb{Q}(\sqrt{11})$ and $\mathbb{Q}(\sqrt{35})$ one can take for U the 140 vectors of Theorem C, whose certificates give $\kappa(U) \leq \frac{2}{7}$, while Proposition 3 gives $\kappa(U) \geq \frac{2}{7}$: some finite subgraph of the Cayley graph of these 140 explicit unit vectors has circular chromatic number $7/2$. A finite extension K of $\mathbb{Q}_p$ with $\chi_c(K^2) < 4$ (Corollary 6) has finite witnesses too: an edge for the value 2, and for the values 3 and $7/2$ those that the corollary gives for $\mathbb{Q}(\sqrt{7})$, resp. $\mathbb{Q}(\sqrt{11})$, since then $p = 3$ and $K \supseteq \mathbb{Q}_3 \supset \mathbb{Q}(\sqrt{7})$, where $\chi_c(\mathbb{Q}(\sqrt{7})^2) = 3$ by Theorem B and Corollary 1, resp. $p = 7$ and $K \supseteq \mathbb{Q}_7 \supset \mathbb{Q}(\sqrt{11})$. The proof, by compactness, gives no bound on the size of the subgraph. When $\sqrt{3} \in F$ and $\chi_c(F^2) = 3$, a unit triangle is a witness; over $\mathbb{Q}(\sqrt{7})$, which has none, there is one with nine vertices (below). The Moser spindle, a unit-distance graph on 7 vertices over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$, maps to $K_{7/2}$ (the common vertex of its two rhombi coloured 0, the far tips 6 and 1, and the others 2, 4, 3, 5), and its independence number is 2, so $\chi_c \geq \chi_f \geq 7/2$; its circular chromatic number is $7/2$. But it contains unit triangles, which $\mathbb{Q}(\sqrt{11})^2$ does not, and $\chi_c(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) \geq 4$ by Theorem F: (a) and (b) fail, as $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$ [10, 22], and (7) fails, as 7 is inert in $\mathbb{Q}(\sqrt{3})$. By SAT computations, not certified, the 76-vertex unit-distance graph over $\mathbb{Q}(\sqrt{11})$ with no 3-colouring in [13] (data/quadratic_planes/q11.json) has circular chromatic number $16/5$, and the unit-distance graph induced on the union of its images under

the nine rotations $\rho^j \sigma^l$, $|j|, |l| \leq 1$ (628 vertices and 1 506 edges; the nine copies alone have 1 494) has circular chromatic number in $(16/5, 13/4]$. Exactly, the ball of radius 2 around 0 in the Cayley graph of the 140 vectors (9 941 vertices, 19 600 edges) is bipartite, and the unit-distance graph it induces (216 more edges) has circular chromatic number $5/2$: it contains a 5-cycle and no triangle, and maps to $K_{5/2}$ (`twoprime/finite_ball.py`).

Witnesses for $7/2$, found by computer: let A be the vertex set of the 76-vertex graph. The unit-distance graph on $A + A = \{a + a' : a, a' \in A\}$ (2 237 vertices, 11 300 edges) maps to $K_{7/2}$ (by a 7-adic colouring), and every homomorphism of it to $K_{7/2}$ has a tight cycle, already among 180 listed cycles of lengths 14 to 42; so its circular chromatic number is $7/2$ by Lemma 20. A colouring-guided growth gives smaller ones. Starting from A , a SAT solver looks for a homomorphism to $K_{7/2}$ in which every listed directed cycle has a non-tight arc; the tight cycles of its answer are added to the list, and when its tight digraph has no directed cycle, the points $x + u$ (x a vertex, u a unit vector with the denominator D of the starting graph, here $D = 30$) whose neighbours leave no colour (or, if there are none, those whose neighbours leave one colour) are added to the graph, at most 200 in a round, those with most neighbours first. This stops after 52 rounds, at 653 vertices; deleting vertices while the property persists leaves a unit-distance graph H_{11} with 170 vertices and 468 edges, and $\chi_c(H_{11}) = 7/2$. It is vertex-critical: for every vertex v a homomorphism c of $H_{11} - v$ to $K_{7/2}$ whose tight digraph has no directed cycle is stored, so $\chi_c(H_{11} - v) < 7/2$ (the other half of the characterisation of [17], explicitly: if a graph G has N vertices and pos numbers them $0, \dots, N - 1$ along a topological order of the tight digraph of a homomorphism $c: G \rightarrow K_{7/2}$, then $Nc + \text{pos}$ is a homomorphism of G to $K_{7N/(2N+1)}$). The same growth with at most 100 new points per round stops at 573 vertices and deletion leaves another vertex-critical witness with 170 vertices; deleting vertices, in a random order, from the unit-distance graph on the union of their vertex sets (205 points) leaves vertex-critical witnesses with 157 and 161 vertices (for two orders), and the same from the union of these two (170 points) leaves a vertex-critical witness with 155 vertices and 404 edges, the smallest we know (`witness_q11b.json.gz`). Started from the 4-chromatic graphs of [13] over $\mathbb{Q}(\sqrt{455})$ and $\mathbb{Q}(\sqrt{191})$ (71 and 96 vertices; $D = 780$ and 240), the same procedure gives vertex-critical witnesses with 175 and 293 vertices (434 and 803 edges); over $\mathbb{Q}(\sqrt{911})$ (327 vertices, $D = 1560$) the growth stops at 873 vertices, and deleting vertices with the cores of refutations (by `kissat`, extracted by `drat-trim`, and then by `CaDiCaL`) leaves a vertex-critical witness with 324 vertices and 866 edges. So $\chi_c = 7/2$ for these planes without Proposition 9: the witnesses give $\chi_c \geq 7/2$, and Proposition 3 gives $\chi_c \leq 7/2$, as 7 ramifies in $\mathbb{Q}(\sqrt{455})$ and splits in $\mathbb{Q}(\sqrt{191})$ and $\mathbb{Q}(\sqrt{911})$ (in agreement with Theorem F, as (a) and (b) fail for $d \equiv 11 \pmod{12}$).

Every refutation is certified: two programs that share no code check the graphs, the colourings and the cycles and write the formula with different variable numberings and arc indicators, and the DRAT proofs of both are verified, one of them also, in LRAT form, by the formally verified checker `cake_lpr`. A referee checked the witnesses with 170, 175 and 293 vertices again with programs of its own: a third encoding (renamed variables, two-sided arc indicators), cycle lists rebuilt from scratch, `cake_lpr` on the stored proofs, and the explicit homomorphisms $Nc + \text{pos}$ of the graphs $H - v$ (`data/number_fields/circular/finite_witness/`); those with 155 and 324 vertices (over $\mathbb{Q}(\sqrt{11})$ and $\mathbb{Q}(\sqrt{911})$), found afterwards, were checked by the two programs above and, by us, with the referee's programs, with the same results. As $\sqrt{11} \in \mathbb{Q}_7$, H_{11} is also an explicit witness for $\mathbb{Q}_7$ and for every finite extension K of $\mathbb{Q}_7$ with $\chi_c(K^2) < 4$.

The same procedure with K_3 in place of $K_{7/2}$ (an arc is now tight when the colour increases by 1 modulo 3) gives a witness for the value 3 in a plane without unit triangles. Over $\mathbb{Q}(\sqrt{7})$, where $\chi_c = 3$, it starts from 0, the 204 unit vectors with denominator 160 and two more points that close a 5-cycle, stops at 607 vertices, and vertex deletion leaves nine points; translated by

$(1, 0)$, they are

$$\begin{aligned} P_0 &= (-1, 0), & P_1 &= \left(-\frac{3+\sqrt{7}}{8}, -\frac{5+\sqrt{7}}{8}\right), & P_2 &= \left(\frac{1}{4}, -\frac{\sqrt{7}}{4}\right), & P_3 &= \left(\frac{\sqrt{7}}{4}, \frac{1}{4}\right), \\ P_4 &= \left(-\frac{1}{4}, \frac{\sqrt{7}}{4}\right), & P_5 &= \left(\frac{3-\sqrt{7}}{8}, \frac{\sqrt{7}-5}{8}\right), & P_6 &= (1, 0), & P_7 &= (0, 0), & S &= (0, 1). \end{aligned}$$

The pairs at distance 1 among them form the 8-cycle $P_0P_1\cdots P_7$, the chords P_0P_4 , P_1P_5 , P_2P_6 and the path P_3SP_7 : their unit-distance graph H_7 is the Wagner graph, the Möbius ladder on 8 vertices, with one of its four chords subdivided. It contains the hexagon $P_0P_4P_3P_2P_6P_7$, the paths $P_0P_1P_2$, $P_4P_5P_6$ and P_3SP_7 that join its opposite vertices, and one more edge, P_1P_5 . The subgraph $M = H_7 - P_1P_5$, a hexagon $v_0\cdots v_5$ whose three long diagonals v_jv_{j+3} are subdivided by vertices m_j (here $v_0, \dots, v_5 = P_0, P_4, P_3, P_2, P_6, P_7$ and $m_0, m_1, m_2 = P_1, P_5, S$), already has a tight cycle in every proper 3-colouring c . Indeed, for an arc $a \rightarrow b$ let $\delta(a, b) = \pm 1$ with $c(b) - c(a) \equiv \delta(a, b) \pmod{3}$; along a closed walk the sum of δ is divisible by 3, so it is ± 3 on a 5-cycle, and on a 6-cycle it is 0 or ± 6 , and ± 6 makes the cycle tight in one direction. Let Z_k be the 5-cycle $v_kv_{k+1}v_{k+2}v_{k+3}m_k$ (indices of v modulo 6, of m modulo 3). As 1-chains, $C_k = Z_{k+1} - Z_k$ is the 6-cycle $v_kv_mv_{k+3}v_{k+4}m_{k+1}v_{k+1}$ and $Z_0 + Z_3$ is the hexagon (each cycle read in the order written); C_{k+3} is C_k reversed, and $Z_3 - Z_0 = C_0 + C_1 + C_2$. If none of C_0, C_1, C_2 is tight in either direction, their sums are 0, the sums over Z_0 and Z_3 are equal, to ± 3 , and the hexagon, with sum ± 6 , is tight. (In H_7 the cycle $C_0 = P_0P_1P_2P_6P_5P_4$ has the chord P_1P_5 and is never tight; the cycles stored with H_7 are C_1, C_2 and the hexagon, in both directions.) As H_7 is 3-colourable, $\chi_c(M) = \chi_c(H_7) = 3$ by Lemma 20. Equivalently, H_7 has no homomorphism to $K_{8/3}$, which is the Wagner graph itself (the vertices $0, 3, 6, 1, 4, 7, 2, 5$ of $K_{8/3}$ form an 8-cycle, and its other edges join opposite vertices): the circular chromatic number of a graph with 9 vertices is a fraction with numerator at most 9 [27, 2, 30], and $8/3$ is the largest such fraction below 3. The Wagner graph has $\chi_c = 8/3$, H_7 is vertex-critical, and nine vertices are the fewest possible in a plane without unit triangles: every triangle-free graph with at most 8 vertices maps to $K_{8/3}$. More precisely, of the 1897 triangle-free graphs with 9 vertices, up to isomorphism, exactly three have $\chi_c = 3$: M , $H_7 = M + m_0m_1$ and $M + m_0m_1 + m_1m_2$. All three are unit-distance graphs over $\mathbb{Q}(\sqrt{7})$, and over $\mathbb{Q}(\sqrt{31})$. Without unit triangles, nine points with the twelve unit pairs of M span M or M plus one or two edges between the m_j , as any other pair at distance 1 would close a triangle, and a direct search for equilateral hexagons whose long diagonals are sums of two unit vectors (`hexagon_search.py`) finds all three over both fields. For example, the points $v_0, \dots, v_5 = (\frac{\sqrt{7}}{4}, \frac{1}{4}), (\frac{\sqrt{7}}{4} - 1, \frac{1}{4}), (-\frac{3}{4}, -\frac{\sqrt{7}}{4}), (0, 0), (-1, 0), (-\frac{1}{4}, \frac{\sqrt{7}}{4})$ and $m_0, m_1, m_2 = (0, 1), (\frac{\sqrt{7}}{4} - 1, -\frac{3}{4}), (-\frac{1}{2} - \frac{\sqrt{7}}{4}, \frac{1}{4})$ span M itself (`witness_q7m.json.gz`): a witness with the fewest vertices and, among those, the fewest edges. The growth over $\mathbb{Q}(\sqrt{31})$ gives nine points that span $M + m_0m_1 + m_1m_2$ (`witness_q31.json.gz`). These facts are checked by enumeration: `check_small.py` checks the points of the three stored witnesses, all 3^9 colour maps, the absence of a homomorphism to $K_{8/3}$ and the vertex-criticality; `small_triangle_free.py` runs over the 4682270 triangle-free graphs on 8 labelled vertices, and `nine_vertices.py` over the triangle-free graphs on 9 vertices up to isomorphism, with two separate tests of $\chi_c = 3$; certified SAT refutations confirm the tight cycles once more. As $\sqrt{7} \in \mathbb{Q}_3$, H_7 is also an explicit witness for $\mathbb{Q}_3$ and for every finite extension K of $\mathbb{Q}_3$ with $\chi_c(K^2) < 4$.

Finite witnesses at four. At $p = 4q$ the proof of Lemma 21 breaks in one place only, and what breaks it is itself a tight cycle. For a homomorphism c of $\text{Cay}(\Gamma, S)$ to $K_4 = K_{4/1}$, with ℓ as in Lemma 21, a *tight square* is a tight cycle $x \rightarrow x + s \rightarrow x + s + t \rightarrow x + t \rightarrow x$ with $s, t \in S$, that is, $\ell(x, s) = \ell(x + s, t) = 1$ and $\ell(x, t) = \ell(x + t, s) = 3$; its four vertices have the colours $c(x), c(x) + 1, c(x) + 2, c(x) + 3$, so they are distinct.

Lemma 23. *Let Γ be an abelian group, $S = -S \subseteq \Gamma \setminus \{0\}$ with $\Gamma = \mathbb{Z}S$, and c a homomorphism of $\text{Cay}(\Gamma, S)$ to K_4 without tight squares. With $a(s) = M_x \ell(x, s)$ as in Lemma 21, the conclusions of Lemma 21 hold with $p = 4$ and $q = 1$: $\xi(s) = a(s)/4$ defines a character of Γ with $\xi(S) \subseteq [\frac{1}{4}, \frac{3}{4}]$, and if $u_1, \dots, u_r \in S$ and positive integers n_i satisfy $\sum_i n_i u_i = 0$ and $a(u_i) = 1$, then c has a*

tight cycle. Conversely, if a character ξ maps S into $[\frac{1}{4}, \frac{3}{4}]$, then $x \mapsto \lfloor 4\tilde{\xi}(x) \rfloor \bmod 4$, where $\tilde{\xi}(x) \in [0, 1)$ is the lift of $\xi(x)$, is a homomorphism of $\text{Cay}(\Gamma, S)$ to K_4 without tight squares.

Proof. For $x \in \Gamma$ and $s, t \in S$ the integers $A = \ell(x, s) + \ell(x + s, t)$ and $B = \ell(x, t) + \ell(x + t, s)$ are congruent modulo 4 and lie in $[2, 6]$. So $A = B$ or $\{A, B\} = \{2, 6\}$; exchanging s and t exchanges A and B , and $A = 2, B = 6$ says exactly, as $\ell(y + u, -u) = 4 - \ell(y, u)$, that $x \rightarrow x + s \rightarrow x + s + t \rightarrow x + t \rightarrow x$ is a tight square. So $A = B$ for all x, s, t , and this is all that the proof of Lemma 21 uses of $p < 4q$; the rest of that proof applies word for word. Conversely, for $s \in S$ let $\sigma \in [\frac{1}{4}, \frac{3}{4}]$ be the lift of $\xi(s)$. As $\xi(x + s) - \xi(x) - \sigma \in \mathbb{Z}$, the integer congruent to $c(x + s) - c(x)$ that the lemma calls $\ell(x, s)$ is $\lfloor 4\tilde{\xi}(x) + 4\sigma \rfloor - \lfloor 4\tilde{\xi}(x) \rfloor \in \{\lfloor 4\sigma \rfloor, \lfloor 4\sigma \rfloor + 1\}$ (it is 3 if $4\sigma = 3$), which lies in $[1, 3]$; so c is a homomorphism to K_4 , while a tight square would need $\ell(x, s) = 1$ and $\ell(x + t, s) = 3$. $\square$

So at four colours Theorem W^+ holds for homomorphisms without tight squares: $\text{Cay}(\Gamma, S)$ has a homomorphism to K_4 without tight squares if and only if some character maps S into $[\frac{1}{4}, \frac{3}{4}]$. Finiteness of S is not used, and the same proof works for $K_{4q/q}$ for every q . In the example $K_4 = \text{Cay}((\mathbb{Z}/2)^2, (\mathbb{Z}/2)^2 \setminus \{0\})$ of [11] every 4-colouring is a tight square. A tight 4-cycle need not be a square: for $\Gamma = \mathbb{Z}/8, S = \{\pm 1, \pm 2\}$, $c(2k) = k$ and $c(2k + 1) = k + 2$, the cycle $0 \rightarrow 2 \rightarrow 4 \rightarrow 6 \rightarrow 0$ is tight, there is no tight square, and $a/4$ is the character $x \mapsto 5x/8$.

Corollary 8. *Let Γ, S be as in Lemma 23, with S finite, $\text{Cay}(\Gamma, S)$ 4-colourable and $\kappa(S) \leq \frac{1}{4}$. Then every homomorphism of $\text{Cay}(\Gamma, S)$ to K_4 has a tight cycle, and some finite subgraph of $\text{Cay}(\Gamma, S)$ has circular chromatic number 4. If $\kappa(S) < \frac{1}{4}$, every such homomorphism has a tight square.*

Proof. Let c be a homomorphism without tight cycles, ξ and a as in Lemma 23, and $S_1 = \{s \in S : a(s) = 1\}$, which is disjoint from $-S_1$ as $a(-s) = 4 - a(s)$. If nonnegative integers n_t , not all 0, gave $\sum_{t \in S_1} n_t t = 0$, Lemma 23 would give a tight cycle. Otherwise, as in the proof of Lemma 22 (where a nonnegative real relation among the images of S_1 in $\Gamma \otimes \mathbb{R}$ is replaced by an integral one, and torsion by a multiple), there is a homomorphism $h : \Gamma \rightarrow \mathbb{R}$ with $h > 0$ on S_1 (any h if S_1 is empty), and for small $\varepsilon > 0$ the character $\xi + \varepsilon h$ maps S into $(\frac{1}{4}, \frac{3}{4}) + \mathbb{Z}$: if $a(s) = 1$ it gives $\frac{1}{4} + \varepsilon h(s)$, if $a(s) = 3$ then $-s \in S_1$ and it gives $\frac{3}{4} - \varepsilon h(-s)$, and otherwise $\xi(s)$ is already inside. So $\kappa(S) > \frac{1}{4}$, a contradiction. The rest is the compactness argument of Lemma 22, with Lemma 20 for $p/q = 4$. If $\kappa(S) < \frac{1}{4}$, Lemma 23 itself excludes homomorphisms without tight squares. $\square$

Conversely, if a finite subgraph H of F^2 has $\chi_c(H) = 4$, then $\kappa(U_H) \leq \frac{1}{4}$ for the set U_H of its edge vectors and their negatives: each component of H lies in a coset of $\mathbb{Z}U_H$, and if $\kappa(U_H) > \frac{1}{4}$, then for a rational $q/p \in (\frac{1}{4}, \kappa(U_H)]$ a character with $\xi(U_H) \subseteq [q/p, 1 - q/p]$ gives a homomorphism of H to $K_{p/q}$ (the *if* direction of Theorem W^+ holds for every $p/q \geq 2$), so $\chi_c(H) \leq p/q < 4$. So, when $\chi(F^2) \leq 4$, some finite subgraph of F^2 has circular chromatic number 4 if and only if some finite $U \subset T$ has $\kappa(U) \leq \frac{1}{4}$.

Proposition 10. *Let F be a number field with $i \notin F$ for which (a), (b) and (7) fail. Then some finite $U \subset T$ has $\kappa(U) \leq \frac{1}{4}$: no character of $\mathbb{Z}U$ maps every element of U into the open interval $(\frac{1}{4}, \frac{3}{4})$.*

Proof. As in the proof of Corollary 7 for $7/2$. If for every finite $V \supseteq \{b_j\}$ there were $\beta \in L$ with $\text{Tr}(\beta v) \in E'$ for every $v \in V$, the proof of Lemma 16, which uses Lemma 15 only through this conclusion (such a β lies in $K_2 \times K_3 \times K_7$, and the types of the $\text{Tr}(\beta v)$ give τ), and Lemmas 17–19 would give (c), (q) or (7'), hence (a), (b) or (7). So some finite $V_0 \supseteq \{b_j\}$ has no such β . Let k be the least multiple of 6 with $5^k > 42r' \max_{v \in V_0} (|D_v| + \sum_j |\mu_{v,j}|)$ (in the fixed basis; r' does not depend on θ), $N = 5^k$, and $U_0 = G(k, 1)V_0$, a finite symmetric subset of T . If $\kappa(U_0) > \frac{1}{4}$, take θ with $\frac{1}{4} < \theta \leq \min(\kappa(U_0), \frac{1}{3})$ and a character ξ of $\mathbb{Z}U_0$ with $\xi(U_0) \subseteq [\theta, 1 - \theta] + \mathbb{Z}$; the proof of Lemma 15, which uses the homomorphism only through such a character (Proposition 9 excludes its sixteen extra vectors for every $\theta > \frac{1}{4}$, and Lemma 14 needs only $\frac{1}{2} - \theta < \frac{1}{4}$), gives β with $\text{Tr}(\beta v) \in E'$ for every $v \in V_0$: impossible. So $\kappa(U_0) \leq \frac{1}{4}$. $\square$

For $i \notin F$ the converse holds too: if (a), (b) or (7) holds, the corresponding colouring and Theorem W^+ give $\kappa(U) \geq \frac{2}{7}$ for every finite $U \subset T$.

Theorem G. *For every number field F with $\chi_c(F^2) \leq 4$, some finite unit-distance graph in F^2 has circular chromatic number $\chi_c(F^2)$. In particular the value 4 is attained by finite unit-distance graphs over $\mathbb{Q}(\sqrt{59})$, $\mathbb{Q}(\sqrt{83})$ and $\mathbb{Q}(\sqrt{3}, \sqrt{11})$.*

Proof. Below 4 this is Corollary 7. If $\chi_c(F^2) = 4$, every finite subgraph has $\chi_c \leq 4$, hence $\chi \leq 4$, and $\chi(F^2) \leq 4$ by the theorem of de Bruijn and Erdős; so $i \notin F$, and (a), (b) and (7) fail by Theorem F, as otherwise $\chi_c(F^2) < 4$. Proposition 10 gives a finite $U \subset T$ with $\kappa(U) \leq \frac{1}{4}$, $\text{Cay}(\mathbb{Z}U, U) \subseteq F^2$ is 4-colourable, and Corollary 8 gives a finite subgraph H with $\chi_c(H) = 4$; the unit-distance graph induced on its vertices also has $\chi_c = 4$, as $4 = \chi_c(H) \leq \chi_c \leq \chi_c(F^2) = 4$. For $\mathbb{Q}(\sqrt{59})$ and $\mathbb{Q}(\sqrt{83})$, $\chi_c = 4$ by the discussion after Theorem F; for $\mathbb{Q}(\sqrt{3}, \sqrt{11})$, $\chi = 4$ and (7) fails, as 7 is inert in $\mathbb{Q}(\sqrt{3})$. $\square$

An independent checker refereed this subsection with two programs of its own, which test Lemma 23, Corollary 8 and the four-colour form of Theorem W^+ on every abelian group of order at most 16 (`data/number_fields/circular/at_four/indep/`). The proof gives no explicit bound on the size of the witness (the remark after Lemma 24 replaces the compactness step of Corollary 8 by a finite construction; the set U of Proposition 10 still comes from compactness); over $\mathbb{Q}(\sqrt{59})$ we know none. Colouring-guided growths over $\mathbb{Q}(\sqrt{59})$ with the unit vectors of denominator 210, 1050 and 2730 stopped at their limits; for 210 this was bound to happen, as the character $\frac{(89a+59b+89c+59e)}{118} \bmod 1$ of the points $(\frac{a+b\sqrt{59}}{210}, \frac{c+e\sqrt{59}}{210})$ keeps all 108 unit vectors with denominator 210 at distance at least $\frac{15}{59}$ from $\mathbb{Z}$, so that every graph built from them maps to $K_{59/15}$. Over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ an explicit set at $\frac{1}{4}$ is small: with R_{60} the rotation by 60° , R_A the rotation with cosine $\frac{5}{6}$ and sine $\frac{\sqrt{11}}{6}$ (the angle of the Moser spindle) and R_G the one with cosine $\frac{11}{14}$ and sine $\frac{5\sqrt{3}}{14}$, the 27 unit vectors $R_{60}^j R_A^k R_G^l(1, 0)$, $j, k, l \in \{-1, 0, 1\}$, have $\kappa = \frac{1}{4}$: a character of order 12 keeps them at distance at least $\frac{1}{4}$ from $\mathbb{Z}$, and an exact certificate (12073 nodes, checked by two programs; `data/number_fields/circular/at_four/`) shows that none maps them into $(\frac{1}{4}, \frac{3}{4})$. So Corollary 8 gives a finite subgraph with $\chi_c = 4$ of the Cayley graph of these 54 vectors (which is 4-colourable, as $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$) directly, without Theorem F. A construction based on the following lemma found one, with 1874 vertices (the graph H_4° below).

Lemma 24 (base-point periods). *Let G be a graph on a finite set of points whose edges join x and $x+s$ with $s \in U \cup -U$, for a finite set U of nonzero vectors with $U \cap (-U) = \emptyset$, and let c be a proper 4-colouring of G without tight cycles. For a closed walk W in G let $\Lambda(W)$ be the sum, over its arcs $x \rightarrow y$, of the representative in $\{1, 2, 3\}$ of $c(y) - c(x)$, and $\text{per}(W) = \Lambda(W)/4 - N(W)$, where $N(W)$ is the number of steps of W in $-U$. Then $\text{per}(W) \in \mathbb{Z}$; per is additive under concatenation and $\text{per}(W^{-1}) = -\text{per}(W)$; it does not change when a backtrack is inserted or deleted, or when two consecutive steps $s \neq \pm t$ are exchanged inside a square of G ; and if W has P steps in U and N in $-U$, $P + N \geq 1$, then $\frac{P-3N}{4} < \text{per}(W) < \frac{3P-N}{4}$.*

Proof. The colour differences along W sum to 0 modulo 4, so $4 \mid \Lambda(W)$. Λ and N are additive; reversing W replaces each representative δ by $4 - \delta$ and exchanges the steps in U and $-U$, so, with $|W| = P + N$, $\text{per}(W^{-1}) = (4|W| - \Lambda(W))/4 - P = -\text{per}(W)$. A backtrack adds $\delta + (4 - \delta) = 4$ to Λ and one step in $-U$. The paths $x \rightarrow x+s \rightarrow x+s+t$ and $x \rightarrow x+t \rightarrow x+s+t$ have sums of representatives in $\{2, \dots, 6\}$ that agree modulo 4; if they differed, one would be 2 and the other 6, and the square $x \rightarrow x+s \rightarrow x+s+t \rightarrow x+t \rightarrow x$, or its reverse, would be a tight cycle (its four points are distinct as $s, t \neq 0$ and $s \neq \pm t$). Finally $|W| \leq \Lambda(W) \leq 3|W|$; if $\Lambda(W) = |W|$, every arc of W is tight, and the closed walk W contains a directed cycle of tight arcs, of length at least 3 as an arc and its reverse are not both tight; if $\Lambda(W) = 3|W|$, the same holds for W^{-1} . So $|W| < \Lambda(W) < 3|W|$, which is the last claim. $\square$

For $\rho = (n_u)$ in the relation lattice $\{n \in \mathbb{Z}^U : \sum_u n_u u = 0\}$ let $P(\rho) = \sum_{n_u > 0} n_u$ and $N(\rho) = \sum_{n_u < 0} |n_u|$; we call $P(\rho) + N(\rho)$ the *length* of ρ and the open interval $(\frac{P(\rho)-3N(\rho)}{4}, \frac{3P(\rho)-N(\rho)}{4})$

its *range*, and we let W_ρ be the walk from a point x_0 with $|n_u|$ steps $\text{sign}(n_u)u$, in a fixed order of U . Suppose that G contains W_ρ for each ρ in a finite set $\mathcal{S}$ of relations, and that each $\rho \in \mathcal{S}$ is joined to 0 by chain steps $\rho' = \rho'' \pm r_j$, where $r_1, \dots, r_m$ is a $\mathbb{Z}$ -basis of the relation lattice, each realised inside G by turning $W_{\rho''}W_{r_j}^{\pm 1}$ into $W_{\rho'}$ with such moves. Then $\text{per}(W_\rho) = p(\rho)$ for an integer homomorphism p on the relation lattice (the empty walk has period 0), with $p(\rho)$ in the range of ρ for every $\rho \neq 0$ met; if no such p exists, every proper 4-colouring of G has a tight cycle. A character ξ of $\mathbb{Z}U$ whose values have representatives $f(u) \in (\frac{1}{4}, \frac{3}{4})$ would give such a p , namely $p(\rho) = \sum_u n_u f(u)$, an integer as $\sum_u n_u \xi(u) = 0$; so an infeasible system on finitely many relations certifies $\kappa(U) \leq \frac{1}{4}$. For the 27 vectors above, the relations with $\sum_u n_u^2 \leq 12$ and 41 of length 8 found by a counterexample search (each R_A^k times a relation among the vectors $\zeta^j g^l$, with $\zeta = e^{i\pi/3}$ and $g = \frac{11+5\sqrt{-3}}{14}$ the rotation R_G , each with a coefficient ± 3 , such as $3\bar{\zeta} + \zeta + g - 3\bar{\zeta}g = 0$) already leave no such p (an integer-programming computation, which only guided the construction). The walks of 256 of them, tied to a basis by bubble sorts, span 13 230 points, and clausal cores of SAT refutations, taken with all unit pairs as edges, reduced these to the graph H_4 of `data/number_fields/circular/finite_witness/witness_q3_11.json.gz`: 1874 points of $\mathbb{Z}U \subset \mathbb{Q}(\sqrt{3}, \sqrt{11})^2$, with all 8085 pairs at distance 1 as edges (198 of them in directions other than the 54 vectors). It has a proper 4-colouring, and every proper 4-colouring has a tight cycle among 4992 listed ones, of lengths 4, 8 and 12, by a SAT refutation that `drat-trim` checks and that an independent encoding, written from scratch by a referee without repository code, confirms with the verified checker `cake_lpr`; Lemma 20 gives $\chi_c(H_4) \geq 4$. So $\chi_c(H_4) = \chi(H_4) = 4$, and the value 4 of Theorem G is attained by an explicit graph over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$. Its subgraph H_4^o on the same points, with only the 7887 pairs that differ by one of the 54 vectors as edges, is an explicit finite subgraph of the Cayley graph of these vectors with $\chi_c = 4$, as in Corollary 8: with the listed cycles it has proper 4-colourings without tight ones, but the tight cycles of such colourings (751, in three rounds) and clausal cores again leave 3389 cycles, of lengths 4 and 8, that every proper 4-colouring meets in a tight one, by a SAT refutation checked by `drat-trim` and `cake_lpr` and confirmed by a referee's own encoding (`data/number_fields/circular/finite_witness/witness_q3_11_cayley.json.gz`, `data/number_fields/circular/finite_witness/indep_W4c/`). Over $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, whose plane is the field $\mathbb{Q}(\zeta_{24})$ with $\zeta_{24} = e^{i\pi/12}$, the same construction starts from the 60 unit vectors $\zeta_{24}^j w^l$ ($j \in \mathbb{Z}/24$, $|l| \leq 2$, one of each pair $\pm u$), with $w = \frac{1+2\sqrt{-2}}{3}$; already the relations with $\sum_u n_u^2 \leq 8$ leave no p (again an integer-programming computation that only guided the construction), and it gives an induced unit-distance graph H_4' with 1657 vertices and 6238 edges (`data/number_fields/circular/finite_witness/witness_q2_3.json.gz`; 39 of the edges are in directions other than the 120 vectors), with $\chi_c(H_4') = \chi(H_4') = 4$ by a SAT refutation checked in the same two ways (the referee's for this graph has two encodings of its own); its subgraph in the Cayley graph of the 120 vectors (6199 edges, 5264 cycles) also has $\chi_c = 4$, checked in the same way (`data/number_fields/circular/finite_witness/witness_q2_3_cayley.json.gz`). As $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) \leq 4$ (a residue colouring at the prime above 2; [14]), this gives $\chi_c(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) = 4$ with an explicit finite graph, and H_4' reproves $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) \geq 4$. An independent referee checked Lemma 24 and its use for $\mathbb{Q}(\sqrt{3}, \sqrt{11})$, with a numerical test on random small instances (`data/number_fields/circular/finite_witness/indep_W4/`). Over $\mathbb{Q}(\sqrt{59})$, for 50 of the 150 unit vectors with denominator 1050 (one of each pair $\pm u$), the same counterexample search stopped with a feasible system of 2926 relations, each of length at most 33.

Remark (Corollary 8 made effective, for S without elements of order 2). Conversely, let U be a finite set of nonzero vectors with $U \cap (-U) = \emptyset$, as in Lemma 24, and $S = U \cup -U$ (for a finite symmetric S such a U exists if and only if no element of S has order 2; in F^2 always). If $\kappa(U) \leq \frac{1}{4}$, with κ taken over all characters of $\mathbb{Z}U$, some finite $\mathcal{S}$ leaves no such p , and the construction gives a finite subgraph as in Corollary 8 effectively (though the set Π below can be very large), without compactness. The least and the largest value of $\sum_u n_u b_u$ on the box $[\frac{1}{4}, \frac{3}{4}]^U$ are $\frac{P(\rho) - 3N(\rho)}{4}$ and $\frac{3P(\rho) - N(\rho)}{4}$, the ends of the range of ρ . Let A be the matrix with rows $r_1, \dots, r_m$, and Π the finite set of $p \in \mathbb{Z}^m$ with p_j in the range of r_j for every j . For $p \in \Pi$ no $f \in (\frac{1}{4}, \frac{3}{4})^U$ has $Af = p$:

otherwise $\sum_u n_u u \mapsto \sum_u n_u f(u) \bmod 1$ would be a well-defined character of $\mathbb{Z}U$ (as $\sum_u n_u f(u)$ is an integer for every relation) with representatives in $(\frac{1}{4}, \frac{3}{4})$. So the affine space $\{f : Af = p\}$ misses the open box; a linear form that separates them is constant on it, so it is $\langle yA, \cdot \rangle$ with $yA \neq 0$, and replacing y by $-y$ if necessary, $\langle y, p \rangle \geq \max_b \langle yA, b \rangle$ over the closed box. The set C_p of all $y \in \mathbb{R}^m$ with this property is $\{y : \langle y, p - Ab \rangle \geq 0 \text{ for every vertex } b \in \{\frac{1}{4}, \frac{3}{4}\}^U\}$, a polyhedral cone defined over $\mathbb{Q}$; it is not $\{0\}$, so it contains a nonzero integer vector y , and $yA \neq 0$ as the r_j are linearly independent. Then $\rho_p = yA$ is a nonzero relation with $p(\rho_p) = \langle y, p \rangle \geq \frac{3P(\rho_p) - N(\rho_p)}{4}$, outside its range, and $\mathcal{S} = \{r_1, \dots, r_m\} \cup \{\rho_p : p \in \Pi\}$ leaves no p . Let X consist of 0, all vertices of the walks W_ρ and W_{r_j} along chains from 0 to the elements of $\mathcal{S}$ (with $x_0 = 0$), and all fourth corners added by the swaps, and let G be the subgraph of $\text{Cay}(\mathbb{Z}U, U \cup -U)$ induced by X . By Lemma 24 and the paragraph after it, every proper 4-colouring of G has a tight cycle; so $\chi_c(G) = 4$ by Lemma 20 if G is 4-colourable (for instance if $\text{Cay}(\mathbb{Z}U, U \cup -U)$ is), and $\chi_c(G) > \chi(G) - 1 \geq 4$ otherwise. As a character with values in $(\frac{1}{4}, \frac{3}{4})$ gives a homomorphism of $\text{Cay}(\mathbb{Z}U, U \cup -U)$ to some $K_{a/b}$ with $a/b < 4$ (as in the converse after Corollary 8), some finite subgraph of $\text{Cay}(\mathbb{Z}U, U \cup -U)$ has $\chi_c \geq 4$ if and only if $\kappa(U) \leq \frac{1}{4}$, and then this construction finds one. The referee checked the remark by computer on 15 sets D of positive integers with $\kappa(D) \leq \frac{1}{4}$, as distance graphs on $\mathbb{Z}$ (`data/number_fields/circular/finite_witness/indep_W4/remark_check.py`): $\mathcal{S}$ leaves no p , and a SAT encoding that does not use Lemma 24 shows that every proper 4-colouring of G has a tight cycle; in 10 of the cases the chains of the basis alone do not suffice.

11. COMPUTER CHECKS

The programs are in the folders `data/number_fields/three_colours/` and `data/number_fields/circular/` of [13], with tests in `tests/test_three_colours.py`, `tests/test_gap_above_three.py` and `tests/test_two_primes.py`.

Lemma 2 as a test. Let $V = \{v_1, \dots, v_m\} \subset T$ contain a $\mathbb{Q}(i)$ -basis $b_1, \dots, b_n$ of L . As β runs over L , the vectors $z = (6 \text{Tr}(\beta v_l))_{l \leq m}$ fill the column space of the matrix $(6 \text{Tr}(b_j v_l))_{l,j}$ over $\mathbb{Q}(i)$, and those with every $\text{Tr}(\beta v_l) \in E$ lie in the lattice $\mathcal{M}$ of its points in $\mathbb{Z}[i]^m = \mathbb{Z}^{2m}$; the types are conditions on z modulo 2 and 3; a program enumerates $\mathcal{M}/2\mathcal{M} \times \mathcal{M}/3\mathcal{M}$ and decides whether some β exists. A second implementation in PARI/GP [24] uses another lattice routine, and an independent checker wrote a third, which agreed on all 78 of its trials. The real quadratic cases of [16] tested ($d = 11, 23, 35, 47, 59$) come out infeasible; $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{7})$ stay feasible; $\mathbb{Q}(\sqrt{2}, \sqrt{7})$ is infeasible already with five vectors, 1, $(7 + 4\sqrt{2}i)/9$, $(23 + 10\sqrt{2}i)/27$, $(1 + 3\sqrt{7}i)/8$ and $(9 + 5\sqrt{7}i)/16$; the sextic field $\mathbb{Q}(2 \cos \frac{2\pi}{7}, \sqrt{7})$ is infeasible with 44 vectors, the obstruction being that of Lemma 7 at its two places above 3 of residue degree 3; $\mathbb{Q}(\sqrt{3}, \sqrt{7})$ stays feasible. An independent checker tested 23 random quartic and sextic fields: every field predicted to be 2- or 3-colourable stayed feasible, and infeasibility was found for all three quartic fields and four of the six sextic fields predicted to need four colours (for the other two the random search found no obstruction; the test is one-sided). The obstructions appear only when V contains vectors that are large at the places above 2 and 3 that split in $F(i)$, as the proof of Lemma 4 suggests.

Exact certificates. The 50 vectors $G_{25}\{1, u_2, \bar{u}_2, u_7, \bar{u}_7\}$ of Section 7 (one per pair $\pm u$), and the 70 vectors $G_{125}V$ for the five vectors above, admit no character into $[1/3, 2/3]$: branch-and-bound trees over the integer values of the relations, with exact Farkas vectors at the leaves (1579 and 11357 nodes), accepted by a separate exact checker, which also verifies that every vector has length 1 and rejects corrupted copies. With Theorem W^+ this proves $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{7})^2) \geq 4$ without Proposition 1 or Sections 3–5; a place above 7 with residue field $\mathbb{F}_7$ gives $\chi \leq 4$. The certificates of Theorem C are of the same kind, for the open interval $(2/7, 5/7)$: a branch covers all integers strictly inside the range of its relation (relations must be nonzero), and a leaf is closed by a Farkas vector valid on the open box. They have 12850 leaves ($d = 11$) and 10241 leaves ($d = 35$), and two exact checkers that share no code accept them; the second one also rebuilds $U = G_{25}V$ from d . Both reject corrupted copies. The certificate of Proposition 4 ($d = 59$, the interval $(15/53, 38/53)$) has 127181 nodes and 105761 leaves; both checkers accept it.

The lemmas. Lemma 7 was checked exhaustively for $f = 1, 3, 5$, and by the independent checkers with their own programs (for $f = 3$ over all 3^{12} $\mathbb{F}_3$ -linear maps); the uniqueness step

for odd $f \leq 79$; Lemma 6 on the finite models $\mathbb{F}_{q^2}[\pi]/(\pi^e)$ ($f = 1$, $e \leq 7$, and $f = 3$, $e \leq 2$); Lemma 5 over all μ_4 -stable subsets of $\mathbb{F}_9$. For the criterion, a PARI/GP script computes (a) and (b) from a defining polynomial; for squarefree $|d| < 200$ it reproduces the rule for quadratic fields, and it agrees with the known values $\chi(\mathbb{Q}(\sqrt{3}, \sqrt{11})^2) = 4$ [10, 22] and $\chi(\mathbb{Q}(\sqrt{2}, \sqrt{3})^2) = 4$ [14, Theorem 1]. For Section 8, two programs written separately compute κ_1 for every (p, f) quoted there, and the bound $|S(b)| \leq 2\sqrt{q}$ and the identity of $S(b)$ with a Kloosterman sum were checked numerically for small (p, f) ; Proposition 5 was checked at level 2, on the finite models $O_w/\mathfrak{m}^2$ for $\mathbb{Q}_7$ and $\mathbb{Q}_{27}$ (`level2.py`).

Theorem D. The programs are in `data/number_fields/circular/probe/`; two independent checkers wrote their own implementations, which share no code with them or with each other, in its subfolders `indep/` and `indep2/`. All use exact rational arithmetic and compute S_N^θ level by level as a union of convex rational polygons: each piece of level $k - 1$, translated by the 25 elements of $\frac{N}{5}\{0, \dots, 4\}^2$, is cut along the strips where the coordinates of $\bar{c}\rho^{\pm k}$ lie in $[\theta, 1 - \theta] + \mathbb{Z}$; a direct cell-by-cell computation agrees with this lifting for $k \leq 3$. Linear-programming values (the thresholds of the induction step, the least s in the base case, the maximum of $\min_\gamma \|\operatorname{Re}(\bar{c}\gamma)\|$ over a component) are floating-point optima certified by exact primal and dual solutions. The implementations agree on everything quoted in Section 9: the base case (least s equal to 0, 11/56 for the eight classes with exactly one integral $\nu_\pm$, and 1/4, 2/7, 3/8, 1/6 for the four orbits with the shifts of the proof; with all shifts the third value is 1/4), the polygons P and $X(\varepsilon)$, the sets S_N^θ for small k at several values of θ in $(17/56, 1/3]$, the extra points at $\theta = 17/56$, the thresholds of the induction step, and the computations at 333/1106 and at θ^* (the same number of components at every level $k \leq 19$). One of them also tested membership in S_N^θ exactly at 350 000 random points ($N = 5, 25$), finding none outside the components of Proposition 7; the 5-adic points c_k of the Remark were checked exactly for $k \leq 15$.

Theorems E and F. The programs are in `data/number_fields/circular/twoprime/`. Propositions 8 and 9 were found by computing $S^\theta(1, 1)$ and $S^\theta(2, 1)$ exactly, by lifting (along the rotations ρ^j and then σ^l , and in the other order) and by a direct enumeration over all cells. Their certificates (the two trees, with 15 620 and 550 474 branch nodes, and the type points and dual certificates of the leaves) are accepted by checkers that share no code with the programs that made them: they rebuild the rotations from $2 \pm i$ and $3 \pm 2i$, recompute the possible strip indices from the corners of the cells, take about one and 45 seconds, and reject corrupted copies. Two independent checkers confirmed the propositions with methods of their own, which share no code with these programs. For Proposition 8: exact clipping of every cell by every strip; the enumeration of all 1 734 260 vertices of the arrangement of strip boundaries in $[0, 65)^2$; and the largest least margin of each component, by exact primal and dual linear programming ($\frac{2}{7}$ for the eight components of type 7, $\frac{1}{3}$ and $\frac{1}{2}$ for types q and c). Both methods find 13 components for $\theta = \frac{7}{25}$ and $\theta = \frac{2}{7}$, and 5 for $\theta = 0.2857143$, $\frac{1429}{5000}$, $\frac{3}{10}$ and $\frac{1}{3}$. For Proposition 9: the enumeration of the vertices of the arrangement in $[0, 325)^2$ (in C with 128-bit integers, each point rechecked with exact fractions), and lifting by rows; both give the same 29 index vectors for $\theta = \frac{249}{1000}$ and the same 13 for $\theta = \frac{1}{4} + 10^{-7}$. The eight dual certificates of value $\frac{2}{7}$ for the vectors of type 7 in Proposition 9 were added after the checkers' reports, by a program of our own (an exact search among the margins that are active at the 7-adic points), and are checked by a further program that rebuilds $325\rho^j\sigma^l$ as products of $2 \pm i$ and $3 \pm 2i$; an independent checker had found the same value $\frac{2}{7}$ by exact linear programming. For Lemma 14, an independent checker lifted $S^\theta(k, 1)$ exactly to the levels $k = 6$ and $k = 12$ and found only the components of the stated form (and, at $k = 5$, components of type 7 not of that form, so the hypothesis $6 \mid k$ is needed). The conclusion of Lemma 12 was tested exactly at every level $k \leq 7$ for $\theta = \frac{3}{10}$ and $k \leq 6$ for $\theta = \frac{201}{700}$: modulo $13 \cdot 5^k$ there are exactly five components, all of the stated form, while for $\theta = \frac{7}{25}$ the test finds the eight components of type 7 at each level $k \leq 3$. Lemma 13, proved above by hand, was also checked by two programs written separately, and its proof by an independent checker with a program of its own that checks (i)–(iv) exhaustively (`indep_L13/`); Lemma 19 directly for $f = 3$ (no $\mathbb{F}_{49}$ -linear map $\mathbb{F}_{7^6} \rightarrow \mathbb{F}_{49}$ maps μ_{344} into A_7'), and Lemma 18 on the finite models $\mathbb{F}_{q^2}[\pi]/(\pi^e)$, $e \leq 3$. The torsion characters of the last Remark were found by two programs; for orders up

to 400, one of them counts each character at its exact order. For Corollary 7, which needs no computation beyond those of Theorem F, two independent checkers tested Lemmas 20–22 with programs of their own: on 291 small graphs and 766 tight relations; on 115 random finite sets S in $\mathbb{Z}$, $\mathbb{Z}^2$ and $\mathbb{Z} \times \mathbb{Z}/m$ with $2 < \chi_c < 4$, at all of whose 409 optimal points (vertices of the optimal set, and midpoints and centroids of them that stay optimal) the tight elements have a nonnegative relation (`indep_FW2/test_kappa_relations.py` with arguments 1 and 180); and by SAT on finite pieces of 13 such Cayley graphs, all of which have finite witnesses (for instance the segment $\{0, \dots, 18\}$ of the distance graph with distances 3, 4, 9, 12, of circular chromatic number $7/2$). One of them clipped $S^{2/7}(2, 1)$ exactly: 13 components, one of type c, four of type q, and the eight points of type 7, each a component by itself. As consistency checks, a program (`tight_walks.py`) finds that each of the eight certificates of type 7 has three positive multipliers, with margins exactly $\frac{2}{7}$ at its 7-adic point and integer relations among the tight rotations (multiplicities 63, 52, 25 or 13, 14, 15); the same holds for the tight elements of the 7-adic characters on the 140 vectors of Theorem C; and the program checks the Moser spindle. The explicit finite witnesses of Section 10, their certificates and the records of every check are in `data/number_fields/circular/finite_witness/verification.txt`.

12. QUESTIONS

- (1) *Above four.* By Theorem F and Corollary 6, below 4 the value of $\chi_c(F^2)$ is the least value of $\chi_c(F_v^2)$ over the completions, and it is given by a locally constant homomorphism of one completion to a circular clique. Is this true above 4? There Theorem W^+ is not available: a homomorphism of a Cayley graph to $K_{p/q}$ with $p/q \geq 4$ need not come from a character. In particular, is there a number field F with $4 < \chi_c(F^2) < 5$? Every F with $\chi(F^2) = 5$ and a place of residue field $\mathbb{F}_{19}$ would be one, as $\kappa_1 = 4/19$ there gives $\chi_c \leq 19/4$; for example $\mathbb{Q}(\sqrt{47})$ if $\chi(\mathbb{Q}(\sqrt{47})^2) = 5$. The 5-chromatic unit-distance graphs we know (those of de Grey [6], Exoo and Ismailescu [8], Heule [18] and Parts [25], and one over $\mathbb{Q}(\sqrt{-3}, \sqrt{-7}, \sqrt{-11})$ in [13], `notes/local_colourings.md`, Section 5) lie over fields that contain $\sqrt{3}$, or $\sqrt{-3}$ and $\sqrt{-7}$, and 3 and -7 are not squares modulo 19.
- (2) *Four colours.* Is $\chi(F^2)$, in general, the least number of colours of a locally constant colouring of one completion? (For real fields this is [15, Question 8], which also allows Borel colourings at the real places.) Theorem B answers this at three colours. Among real quadratic fields, ordered by d , the first open case is $\mathbb{Q}(\sqrt{47})$, where $4 \leq \chi \leq 5$ ([16, Corollary 2] and [23]); by Theorem F, $\chi_c(\mathbb{Q}(\sqrt{47})^2) \geq 4$, so $\chi_c(\mathbb{Q}(\sqrt{47})^2) = 4$ if and only if $\chi(\mathbb{Q}(\sqrt{47})^2) = 4$ (if $\chi_c(F^2) = 4$, every finite subgraph has $\chi_c \leq 4$, hence $\chi \leq 4$, and $\chi(F^2) \leq 4$ by the theorem of de Bruijn and Erdős).
- (3) *Finite witnesses.* By Corollary 7 some finite unit-distance graph in $\mathbb{Q}(\sqrt{11})^2$ has circular chromatic number $7/2$; the proof, by compactness, gives no bound on its size, and the smallest we know has 155 vertices and is vertex-critical (Section 10). How small can it be? (For the value 3 in a plane without unit triangles, nine vertices are needed, and they suffice over $\mathbb{Q}(\sqrt{7})$ and $\mathbb{Q}(\sqrt{31})$; Section 10. Over other such planes the least number is open; over $\mathbb{Q}(\sqrt{15})$ the smallest witness we found has 13 vertices (`data/number_fields/circular/finite_witness/witness_q15.json.gz`), and none of the three nine-vertex graphs appeared with the denominators we tried.) When $\chi_c(F^2) = 4$ the value is attained as well (Theorem G), and the proof gives no explicit bound on the size of a witness (the remark after Lemma 24 makes only its last step, Corollary 8, effective); over $\mathbb{Q}(\sqrt{3}, \sqrt{11})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ we found ones with 1 874 and 1 657 vertices (with Lemma 24 and the paragraph after it), and over $\mathbb{Q}(\sqrt{59})$ we know none: how small can one be? By the converse after Corollary 8 the set U of its edge vectors has $\kappa(U) \leq \frac{1}{4}$, so they cannot all have denominator 210 (Section 10).

ACKNOWLEDGEMENTS

The author thanks the authors of the works cited, whose questions and results this note builds on.

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